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The Click-Based MNL Model: A Framework for Modeling Click Data in Assortment Optimization

(Authors' names blinded for peer review)

We introduce the click-based MNL choice model, a framework for capturing customer purchasing decisions in e-commerce settings. Specifically, we augment the classical Multinomial Logit choice model by assuming that customers only consider the items they have clicked on before they proceed to compare their random utilities. In this context, we study the resulting assortment optimization problem, where the objective is to select a subset of products, made available for purchase, to maximize the expected revenue. Our main algorithmic contribution comes in the form of a polynomial-time approximation scheme (PTAS) for this problem, showing that the optimal expected revenue can be efficiently approached within any degree of accuracy. In the course of establishing this result, we develop several technical ideas, including enumeration schemes and stochastic inequalities, which may be of broader interest. Using data from Alibaba's online marketplace, we fit click-based MNL and latent class MNL models to historical sales and click data in a setting where the online platform recommends a personalized six-product displays to users. We propose an estimation methodology for the click-based MNL model that leverages clickstream data and machine learning classification algorithms. Our results suggest that clickstream data is valuable in predicting choices and that our model can outperform standard logit-based models in certain settings.

Key words: Multinomial Logit model, consideration sets, clickstream data, approximation algorithms

1. Introduction

Assortment optimization is a fundamental computational challenge faced by online platforms, where the set of products displayed to customers can be seamlessly varied and personalized at essentially no cost. Existing prediction models and decision support tools in this context generally take advantage of historical transaction data to infer customers' preferences, as each purchase transaction reveals a choice made by the corresponding customer amongst the available product alternatives. Given this probabilistic prior about customer preferences, the selection of an assortment to maximize expected revenue can be framed as a combinatorial optimization problem, which has received a great deal of attention in the operations research and revenue management literatures.

In recent years, online platforms have increasingly collected data about different forms of customer-product interactions. The primary source of such data is *clickstream* information, corresponding to the customers' search and click behavior. Empirical studies in the marketing literature suggest that clickstream data is valuable to predict customer purchases, brand loyalty, and churn rates (Laudon and Traver 2013). For example, by taking into account the path of webpages browsed by users of an online bookseller, Montgomery et al. (2004) show that the accuracy with which they can predict purchase conversions is increased to 40%, up from a mere 7% without integrating this piece of information. Nevertheless, although discrete-choice models are extensively being employed in operations management, the value added by clickstream data to platforms' decision-making processes still remains unquantified. In particular, leveraging these data sources within the framework of assortment optimization is an apparent blind spot of the existing literature, from both modeling and computational standpoints.

This paper aims to augment standard assortment optimization models by incorporating click signals in e-commerce applications. Our main modeling idea is to assume that these clicks coincide with the subset of alternatives considered by a customer during a product screening phase preceding her final choice decision. This point of view leads us to introduce a choice model that unfolds in two stages. First, the customer forms her consideration set according to product-dependent click propensities. Then, in the second stage, she makes a purchasing decision from among the products in this restricted consideration set according to the classical Multinomial Logit (MNL) choice model. This approach is inspired by the well-established literature in quantitative marketing around two-stage choice models (Howard and Sheth 1969), which were empirically validated by several works such as those of Hauser (1978), Jeuland (1979) and Crompton and Ankomah (1993), and which have recently found applications in operations management (Jagabathula and Rusmevichientong 2016). However, we depart from **the existing operations management literature in explicitly viewing the consideration set as the subset of products a customer clicks on, rather than viewing it as an unobserved latent variable in the choice-making process**. A clear advantage of this approach is that, because e-commerce platforms fully observe customers' clicks, estimating the distribution of consideration sets becomes straightforward. Before describing our choice model in formal terms, we briefly review the most relevant papers on assortment optimization, with a focus on consideration set based choice models.

Brief review of existing literature. Starting with the seminal work of Talluri and van Ryzin (2004), a growing line of research has explored assortment optimization problems under various choice models and operational constraints. When customer choice is governed by the MNL model (Luce 1959, McFadden 1974), and when there are no restrictions on the set of products to be offered, Talluri and van Ryzin (2004) show that there exists an optimal assortment consisting of all

products priced above a certain threshold. Nonetheless, the simple structure of the MNL model is often too restrictive, particularly when the customer population exhibits heterogeneous preferences. For example, a common approach for modeling general substitution patterns is to segment the customers into heterogeneous classes. Since the segment of each customer is not observable, the latter approach gives rise to probabilistic mixture models, such as the latent class MNL model (see, e.g., Rusmevichientong et al. (2014), Désir et al. (2022), Feldman and Topaloglu (2015)).

Another popular approach for capturing customers' choice heterogeneity in the random-utility framework is based on the notion of consideration sets. The crucial observation is that customers may be led to disregard certain products based on prominent features such as price, ratings and reviews, before comparing their random utilities. For example, they may only consider purchasing products of a certain quality that are priced below some threshold (Jagabathula and Rusmevichientong 2016). A rich recent literature has focused on various consideration set-based assortment optimization problems (Davis et al. 2015, Gallego et al. 2020, Aouad and Segev 2021, Gallego and Li 2017, Wang and Sahin 2017, Chu et al. 2020, Derakhshan et al. 2022). In these settings, the consideration sets are not directly observable; their distribution can only be probabilistically inferred from the final purchasing actions. Moreover, in order to develop polynomial-time algorithms for assortment optimization, these papers often place additional structural restrictions either on the consideration sets or on the relative rankings over products' utilities. For example, Feldman and Topaloglu (2018) study an MNL-based model with nested consideration sets. The papers by Honhon et al. (2012), Aouad et al. (2021), and Gallego and Li (2017) require the assumption of a unique ranking, meaning that all customers share a common fixed ranking over product utilities. Such assumptions imply that only a tiny fraction of all possible consideration sets or rankings have a non-zero likelihood, and thus, click data often cannot be used to estimate the model parameters.¹

Click-based MNL modeling approach. Our main modeling idea is to assume that the clicks observed by e-commerce platforms explicitly describe the customers' consideration sets, a key distinctive feature in relation to the existing literature. Specifically, we define each customer's consideration set precisely as the set of products she has clicked on. This assumption is quite natural, considering that customers click on products either to directly purchase them or to collect additional product-specific information. Following this main assumption, the customers' purchasing process unfolds in a two-step process. In the first step, the customer independently decides whether or not to click on each of the offered products according to independent Bernoulli random outcomes,

¹ Consider a simple setting, where we have two products and only two sales data points. For both data points, both products were clicked on. However, the first data point specifies that product 1 was purchased, and the second data point indicates that product 2 was purchased. In this case, there is no universal ranking and distribution over consideration sets that assign a non-zero likelihood for these data.

thereby forming a random consideration set. Here, any possible consideration set has a positive likelihood, which is consistent with the lack of explicit combinatorial structure in real-world click-stream data. In the second step, the customer makes a purchasing decision by ranking the utilities of the products she has clicked on; such ranking decisions are captured via a standard MNL choice model. The resulting choice model, called the *click-based MNL model*, is formally described next.

1.1. Problem formulation

We consider a setting where a retailer has access to a collection of n items. Each item $i \in [n]$ is associated with a selling price of r_i , an MNL-based preference weight of w_i , and a consideration probability of λ_i . Given the assortment $S \subseteq [n]$, the representative customer makes a purchasing decision in two stages:

1. *Generating a consideration set.* First, a random subset $C_S \subseteq S$ is generated, by independently picking each item $i \in S$ with probability λ_i . We refer to C_S as the (random) consideration set induced by the assortment S , whose total weight is denoted by $w(C_S) = \sum_{i \in C_S} w_i$.

2. *Picking from the consideration set.* For any given realization of the consideration set C_S , a representative customer makes a purchase from among this subset according to the MNL choice model. Namely, each item $i \in C_S$ is chosen with probability $\frac{w_i}{1+w(C_S)}$, and the no-purchase option is chosen with the residual probability, $\frac{1}{1+w(C_S)}$.

In order to define the expected revenue function, let $\pi(i, S)$ be the probability of picking item i , given that the assortment S is offered. Based on the preceding discussion, this purchase probability is clearly zero when $i \notin S$. In the more interesting scenario when $i \in S$, the latter probability can be written as

$$\pi(i, S) = \Pr[i \in C_S] \cdot \mathbb{E} \left[\frac{w_i}{1 + w(C_S)} \middle| i \in C_S \right] = \lambda_i \cdot \mathbb{E} \left[\frac{w_i}{1 + w_i + w(C_S^{-i})} \right], \quad (1)$$

where the expectation above is taken over the randomness in generating the consideration set C_S , with the shorthand notation $C_S^{-i} = C_S \setminus \{i\}$. The objective is to identify an assortment $S \subseteq [n]$ that maximizes the resulting expected revenue $\mathcal{R}(S) = \sum_{i \in S} r_i \cdot \pi(i, S)$. This assortment optimization formulation is termed the *click-based MNL assortment problem*, for short.

Related choice models. It is worth noting that the click-based MNL model can be viewed as a special case of the mixture (latent class) of MNL model with exponentially many customer segments. More specifically, in Appendix EC.1, we present a formal description of how any click-based MNL model can be converted to an equivalent latent class MNL model. For this purpose, we can view each consideration set as a distinct segment and express the choice probability $\pi(i, S)$ as a probabilistic mixture:

$$\pi(i, S) = \sum_{\substack{C \subseteq S: \\ i \in C}} \left(\prod_{j \in C} \lambda_j \cdot \prod_{j \in S \setminus C} (1 - \lambda_j) \right) \cdot \frac{w_i}{1 + w(C)}. \quad (2)$$

Unfortunately, this reduction provides little practical value, as it requires specifying a latent class MNL model over 2^n customer segments, meaning that existing algorithms for the assortment problem under the latent class MNL cannot be efficiently applied to these instances. Moreover, it is worth mentioning that even with polynomially many segments, the latent class MNL-based assortment optimization problem is known to be $\Omega(n^{1-\epsilon})$ -hard to approximate, as shown by Désir et al. (2022).

Interestingly, highly structured special cases of the click-based MNL model have surfaced in the recent literature. For example, the random consideration set model introduced by Manzini and Mariotti (2014) and further studied by Gallego and Li (2017) can be viewed as a limiting case of the click-based MNL where the second-stage choice decisions follow a single ranking over the items.² In fact, the model of Gallego and Li (2017) is also a special case of the Markov chain based choice model (Blanchet et al. 2016). In contrast, as we show in Section 2.1, an instance of the click-based MNL model cannot be represented as an equivalent polynomially sized instance of the Markov chain model, unless $P = NP$. The latter property implies a clear separation between the click-based MNL model in its general form and special cases examined in earlier literature.

To summarize, although our approach is closely related to several existing choice models, these connections cannot be exploited to efficiently solve the click-based MNL assortment problem. As we proceed to show in this paper, the probabilistic structure of the consideration sets in our model creates fundamental challenges that require new technical insights and algorithmic tools.

1.2. Main contributions

This paper studies the application of the click-based MNL model to assortment optimization. We tightly characterize the computational status of the resulting assortment optimization problem. We show that this problem is NP-hard even for very basic click-based MNL settings. On the positive side, we develop a polynomial-time approximation scheme (PTAS), showing that provably near-optimal assortments can be efficiently computed. This result builds on the development of algorithmic and probabilistic tools to approximate the outcomes of independent Bernoulli trials, representing how customers' clicks generate random consideration sets.

We provide an empirical validation of our modeling approach in terms of prediction accuracy through a collaboration with the online retailer Alibaba. We estimate the click-based MNL model

² Any instance of the model studied by Gallego and Li (2017) can be efficiently reduced to an instance of the click-based MNL assortment problem, while losing only an $O(\epsilon)$ -fraction of the optimal revenue, for every $\epsilon \geq 0$. Their model is based on the assumption that the customers' second-stage decisions follow a strict preference ranking $\sigma : [n] \rightarrow [n]$. By instantiating the click-based MNL model with preference weights $w_{\sigma(n)} = 1/\epsilon$ and $w_{\sigma(i)} = (1/\epsilon) \cdot w_{\sigma(i+1)}$ for every $i \in [n-1]$, it is not difficult to show that the expected revenue function of the resulting click-based MNL model approximates that of Gallego and Li (2017) up to an $O(\epsilon)$ -factor.

on large-scale data sets describing customers' purchase history on mobile platforms, where a customized selection of products is displayed to end users. We develop a simple estimation strategy that pieces together the use of clickstream data, machine learning methods, and maximum-likelihood estimation. We demonstrate that the click-based MNL model can offer improvements over the MNL model and linear-utility latent class MNL models in term of prediction accuracy, with a smaller computational cost for estimating the model parameters.

Hardness results and estimation of choice probabilities. In Section 2.1, we show that assortment optimization under the click-based MNL model is NP-hard, by relating this setting to the well-known set partition problem. Our reduction shows that the former problem is intractable even when there is only one item whose inclusion in the consideration set is random, whereas all remaining items are deterministically included. That said, as a preliminary indication that the click-based MNL model can still be rigorously analyzed, we show in Section 2.2 that the expected revenue $\mathcal{R}(S)$ of a given assortment $S \subseteq [n]$ can be efficiently estimated. Specifically, even though the screening phase of this two-stage choice model generates a distribution over exponentially-many consideration sets C_S , we devise a fully polynomial time approximation scheme (FPTAS) for computing its choice probabilities $\pi(i, S)$.

Approximation scheme. Our main algorithmic contribution comes in the form of a polynomial time approximation scheme (PTAS) for the assortment optimization problem. That is, for any accuracy level $\epsilon > 0$, our algorithm constructs an assortment whose expected revenue is within a factor of $1 - \epsilon$ of optimal, running in $O(n^{O(\frac{1}{\epsilon^4} \log \frac{1}{\epsilon})})$ time, as formally stated in Section 2.3. At a high-level, our algorithmic approach proceeds from a distinction between two regimes of parameters: *unlikely items* corresponding to low click probabilities, and *likely items* corresponding to high click probabilities. Our algorithm builds on the intuition that assortment optimization becomes easier to handle when each of these settings is studied independently of the other. For unlikely items, the probability that two items are simultaneously considered is small, and thus, the substitution effects are limited. From an algorithmic standpoint, this observation means that, in this regime, the assortment problem can be approximately “linearized” to obtain knapsack-like formulations. In contrast, for likely items, products have non-negligible probabilities of being clicked on. Here, the instance structure is closer in spirit to that of a standard MNL-assortment optimization problem, and therefore, we utilize a revenue-ordered-like policy. We devote Section 3 to a detailed exposition of our approximation scheme.

Numerical experiments. Although it is useful for our theoretical analyses, the running time complexity of our PTAS is prohibitive for large-scale numerical applications. Hence, in Section 4, we propose two practical algorithms: a simple greedy heuristic and a modified version of our PTAS that can also handle cardinality-constrained instances. We then conduct a set of numerical experiments

in which the efficacy of these approaches is tested on randomly generated instances consisting of 20 to 80 products. In this setting, for instances with 40 or fewer products, the optimal assortment can be recovered via a complete enumeration of all feasible assortments. For these smaller instances, our PTAS identifies assortments that are within fractions of a percent of optimal, on average, and that never exceed 1% in all instances tested. **Although the optimality gap of the greedy heuristic is small on average, it may exceed 5% in certain configurations of our generative setting. We observe that the greedy heuristic is very efficient when the consideration probabilities are small but its performance devolves when they are large — a setting closer to the standard MNL model. On larger instances with 60-80 products, we find that our modified PTAS may still achieve good performance tradeoffs between computational time and expected revenue.**

Case study on Alibaba's data. In Section 5, we conduct a case study that focuses on quantifying the value of clickstream data in choice modeling settings. We make use of real historical sales data from Tmall.com, one of Alibaba's largest online marketplaces that connects third party-sellers to customers. At a high level, customers landing on any particular seller's homepage are often presented with an option to select a discount coupon. Upon clicking on this coupon, the customer is transferred to a coupon sub-page that contains six products, each of which can be purchased at a discount using the acquired coupon. For the top ten sellers (traffic-wise) across Tmall.com, our data set consists of all information related to the clicks and purchases of every customer landing on a coupon sub-page over a two-week selling period. Beyond these records, our data set includes a collection of 25 feature values associated with each customer and offered product.

We benchmark the click-based MNL model against the standard MNL and latent class MNL models. The latter choice models are estimated via standard maximum likelihood estimation (MLE) methods. Similarly, for the click-based MNL model, the utility parameters of the second-stage MNL instance are estimated using standard MLE. On the other hand, the prediction of click probabilities can be cast as a binary classification problem. Hence, we leverage a gradient-boosted tree algorithm to estimate the click probabilities.

We find that the click-based MNL model significantly outperforms the standard MNL model in terms of out-of-sample log-likelihoods, with relative improvements of 1-13%. Even when benchmarked against the latent class MNL model, we find that click-based MNL is competitive and may yield small improvements in out-of-sample log-likelihoods. Given the connection between the latent class MNL and click-based MNL models (see Appendix EC.1), this suggests that click data provide a useful signal to model latent customer heterogeneity through random consideration sets. It is worth noting that the click-based MNL model also has an advantage from a computational standpoint. The estimation of click-based MNL models runs in under 20 minutes, whereas the estimation of latent class MNL models exceeds three hours.

We also study the extent to which the click-based MNL model leads to assortment recommendations that differ from those of the MNL model. Using semi-synthetic instances based on Alibaba's choice data, we show that the *cost of a model misspecification* can be significant. This notion quantifies the loss of revenue incurred by using an MNL-optimal assortment in place of an assortment that is optimal for the click-based MNL model, when the latter is the ground truth. To keep this paper as concise as possible, this simulation study is presented in the online companion (Appendix EC.7.5).

2. Theoretical Results

In what follows, we present our main theoretical results regarding the click-based MNL assortment problem. We first prove in Section 2.1 that this computational problem is NP-hard even in seemingly simple settings. In Section 2.2, we develop an FPTAS to compute the choice probabilities for any given assortment. In Section 2.3, we present our main algorithmic results for the click-based MNL assortment problem.

2.1. Hardness result

First, we prove that the assortment optimization problem, as formally defined in Section 1.1, is NP-hard. Our approach leverages ideas that are similar in spirit to the reduction proposed by Feldman and Topaloglu (2018) for an MNL-based choice model with nested consideration sets. To this end, we focus on the feasibility version of the former problem, where an additional revenue threshold K is specified. The goal of the *assortment feasibility problem* is to decide whether there exists an assortment whose expected revenue is at least K .

THEOREM 1. *The assortment feasibility problem is NP-complete.*

The proof is based on a reduction from set partition, which is one of Karp's 21 NP-complete problems (Karp 1972). This reduction is presented in Appendix EC.2.1. Interestingly, we map the set partition instances to click-based MNL settings in which all products are clicked on with probability 1, except for a single product, which is clicked on with probability $1/2$. Hence, this construction proves that the computational hardness comes from the basic probabilistic structure of the click-based MNL choice model, where the customer's random consideration set is formed through independent Bernoulli random variables.

2.2. FPTAS for computing choice probabilities

In addition to the computational hardness of its assortment optimization problem, the challenges surrounding the click-based MNL model start with the estimation of its customers' choice probabilities. Indeed, each assortment S induces a distribution over $2^{|S|}$ consideration sets, each of which occurs with positive probability. Therefore, computing the choice probability $\pi(i, S)$ of an item i in

any given assortment S is not straightforward, and in fact, we are not aware of any efficient method for computing $\pi(i, S)$ in an exact way. Nevertheless, as stated in the next claim, this quantity can be deterministically estimated within any degree of accuracy in polynomial time. In what follows, we let $w_{\min}$ and $w_{\max}$ denote the minimal and maximal MNL weights, respectively.

THEOREM 2. *For any $\epsilon \in (0, 1)$, there is a deterministic $O(\frac{n^2}{\epsilon} \cdot \log(n \frac{w_{\max}}{w_{\min}}))$ time algorithm that computes an estimate $\tilde{\pi}(i, S)$ for the choice probability $\pi(i, S)$ satisfying*

$$\pi(i, S) \leq \tilde{\pi}(i, S) \leq (1 + \epsilon) \cdot \pi(i, S) .$$

To establish Theorem 2, we formulate in Appendix EC.2.2 an approximate dynamic program that estimates the choice probabilities up to a factor of $1 + \epsilon$. In this formulation, items are sequentially processed in an arbitrary order with a state variable describing the cumulative preference weights of all items considered before reaching the current state. In the remainder of this paper, we make use of the dynamic programming method of Theorem 2 as a black-box to estimate the choice probabilities under the click-based MNL model. This positive result provides an initial indication that the click-based MNL assortment problem can be well approximated, despite the NP-hardness result of Section 2.1.

2.3. Approximation scheme

Here, we discuss our main algorithmic results for the click-based MNL assortment problem. For ease of presentation, we make a mild structural assumption, focusing on a slightly restricted class of instances. This assumption allows us to greatly simplify the design of our approximation scheme. Nonetheless, we conclude this section by stating our most general result, without any assumption whatsoever.

Bounded-ratio assumption. Let $\epsilon > 0$ be an extra input parameter to our approximation scheme, standing for its desired accuracy. Recall that $w_{\min} = \min\{w_i : i \in [n]\}$ and $w_{\max} = \max\{w_i : i \in [n]\}$ are the minimum and maximum preference weights over all items, respectively. The following assumption imposes an upper bound on the ratio $\frac{w_{\max}}{w_{\min}}$, taking the form of a polynomial function of n for every fixed $\epsilon > 0$.

ASSUMPTION 1. *All preference weights are polynomially related, i.e., $\frac{w_{\max}}{w_{\min}} = n^{O(\frac{1}{\epsilon})}$.*

The above assumption restricts our attention to a setting we refer to as the *bounded-ratio* regime, where the variability of the preference weights is limited. Intuitively, this regime ensures that certain statistics of optimal assortments can be recovered via polynomial-time enumeration. Assumption 1 is likely to be satisfied by instances encountered in practice, for a sufficiently small choice of $\epsilon > 0$. Nevertheless, as this property does not hold in general, we will develop further algorithmic ideas to handle arbitrary instances later on.

Main algorithmic results. Focusing on the bounded-ratio setting, we present our main algorithmic result. Compared with standard approximation algorithms for assortment optimization, one notable nuance is that our approach constructs a random assortment, rather than a deterministic one. In other words, the assortment decision is sampled from a distribution over assortments; this random experiment is independent from the customer's purchasing decision, as prescribed by the click-based MNL choice model. Moreover, we restrict attention to the class of *decomposable assortments*, which are generated by specifying a certain probability for each product to be included in the assortment. That is, given a sequence of independent Bernoulli random variables $B = (B_1, \dots, B_n)$, the realization of the decomposable assortment B is given by $S = \{i \in [n] : B_i = 1\}$. In what follows, we denote probabilities and expectations with regards to this distribution by $\Pr_{S \sim B}[\cdot]$ and $\mathbb{E}_{S \sim B}[\cdot]$, respectively. Furthermore, we denote by S^* an optimal (deterministic) assortment, which is fixed throughout our exposition. With these definitions at hand, we next state our main theorem, whose specifics are provided in Section 3.

THEOREM 3. *Suppose that Assumption 1 is satisfied. For any $\epsilon > 0$, we can construct in $O(n^{O(\frac{1}{\epsilon^4} \log \frac{1}{\epsilon})})$ time a decomposable assortment B , such that $\mathbb{E}_{S \sim B}[\mathcal{R}(S)] \geq (1 - \epsilon) \cdot \mathcal{R}(S^*)$.*

The theoretical significance of Theorem 3 resides in showing that the click-based MNL assortment problem can be efficiently approximated within any degree of accuracy under Assumption 1. We point out that decomposable assortment decisions, as stated above, can be derandomized using the method of conditional expectations (Alon and Spencer 2016, Chap. 16). For completeness, we explain in Appendix EC.2.3 how to efficiently convert any decomposable assortment B into a deterministic assortment S without any loss in optimality. The running time stated in Theorem 3 suggests that our approach might be computationally inefficient in practice. In Section 4, we will demonstrate that a variant of this algorithm, which preserves its core inner workings, can be implemented on large-scale cardinality-constrained instances.

Relaxing the bounded-ratio assumption. To conclude this section, we relax Assumption 1 and formally state our approximation result in its utmost generality. This result complements the NP-hardness result presented in Section 2.1.

THEOREM 4. *There exists a polynomial-time approximation scheme for the click-based MNL assortment problem. For any $\epsilon > 0$, this algorithm computes an assortment S such that $\mathcal{R}(S) \geq (1 - \epsilon) \cdot \mathcal{R}(S^*)$ in time $O((\frac{n}{\epsilon})^{O(\frac{1}{\epsilon^4} \log \frac{1}{\epsilon})})$.*

To handle instances in which the bounded-ratio assumption is not satisfied, the proof of Theorem 4 involves an additional layer of rather involved arguments, which are separately presented in Appendix EC.4. At a high level, we construct an approximation-preserving reduction from arbitrary instances of the click-based MNL assortment problem to constrained bounded-ratio instances.

For this purpose, we make use of dynamic programming ideas to decompose the original instance into a collection of bounded-ratio sub-instances, utilizing our approximation scheme in this context as a subroutine. The latter decomposition partitions the collection of items into classes of well-separated preference weights, meaning that we leave a sufficiently-large multiplicative gap between successive classes. Now, at least intuitively, the important observation is that “long-range” interactions between items selected across different classes are negligible, as far as their effects on the choice probabilities of these items. Thus, the assortment decisions can be decoupled across classes, leading to a compact state description within our dynamic program.

3. Algorithmic Approach

In what follows, we provide a detailed description of our approximation scheme for bounded-ratio instances, while deferring its performance analysis to Appendix EC.4. In Section 3.1, we start by introducing additional notation to designate specific classes of items. Next, in Section 3.2, we provide a high-level overview of our approximation scheme, with a breakdown of its main algorithmic steps in Sections 3.3-3.6. Along the way, we provide ample intuition regarding the specific properties that will be crucial in our theoretical analyses. Concurrently, we explain why each algorithmic step is computationally efficient and how it relates to the design of a near-optimal assortment.

3.1. Decomposition into classes

The starting point of our algorithm is to decompose the collection of items based on the magnitude of their consideration probabilities. We will distinguish between *likely* and *unlikely* items, employing specialized algorithmic ideas to handle each type of item. To formalize this distinction, we introduce some notation to single out specific classes of products. For every integer p , we denote by Λ_p the collection of items $i \in [n]$ with a consideration probability $\lambda_i \in [\epsilon \cdot (1 + \epsilon)^p, \epsilon \cdot (1 + \epsilon)^{p+1}]$. Similarly, for every integer $q \geq 1$, we denote by $\mathcal{W}_q$ the collection of items $i \in [n]$ with a preference weight $w_i \in (\frac{\epsilon}{n} \cdot (1 + \epsilon)^{q-1}, \frac{\epsilon}{n} \cdot (1 + \epsilon)^q]$. In addition, $\mathcal{W}_0 = \{i \in [n] : w_i \leq \frac{\epsilon}{n}\}$ is the collection of remaining items. Next, we define $Q_{\min}$ to be the minimal index q for which $\mathcal{W}_q \neq \emptyset$ and let $Q_{\max}$ be the maximal such index; we will use $\mathcal{Q} = [Q_{\min}, Q_{\max}]$ as an abbreviation for the range of weight class indices. Similarly, let $P_{\min}$ be the minimal index $p \in \mathbb{Z}$ for which $\Lambda_p \neq \emptyset$ and let $P_{\max}$ be the maximal such index.

With this notation at hand, we say that an item $i \in [n]$ is unlikely if $\lambda_i \in [0, \epsilon]$. Otherwise, $\lambda_i \in (\epsilon, 1]$, and we say that item i is likely. Consequently, for every $q \in \mathcal{Q}$, we will use $\mathcal{U}_q = \bigcup_{p \in [P_{\min}, 0]} (\Lambda_p \cap \mathcal{W}_q)$ to denote the set of unlikely items within class $\mathcal{W}_q$ and $\mathcal{N}_{\text{unlikely}} = \bigcup_{q \in \mathcal{Q}} \mathcal{U}_q$ to denote the collection of unlikely items. Similarly, we denote by $\mathcal{N}_{\text{likely}} = \bigcup_{p=1}^{P_{\max}} \Lambda_p$ the collection of likely items.

3.2. Outline of our approximation scheme

As described in Algorithm 1 below, our approach proceeds in four steps. In step 1, we modify the instance parameters by rounding the preference weights and consideration probabilities to powers of $1 + \epsilon$. In step 2, we generate a collection Ω of real-valued vectors that encode “aggregate statistics” of an optimal assortment. The precise meaning of these vectors will be revealed as part of our detailed explanations; for now, we treat them as input parameters for subsequent stages. In steps 3 and 4, we consider each vector in Ω and construct a corresponding decomposable assortment by employing two specialized subroutines, **MinKnapsack**($\cdot$) and **Greedy**($\cdot$), on the classes of unlikely and likely items, respectively. We eventually pick the decomposable assortment with the largest expected revenue by leveraging the FPTAS of Theorem 2 as an evaluation oracle for the choice probabilities involved. Although the latter approximation scheme was developed for deterministic assortments, it is also applicable to decomposable assortments, through the derandomization method mentioned in Section 2.3. In what follows, we will separately describe each algorithmic step in greater detail.

Algorithm 1 Approximation scheme under Assumption 1

Step 1: Parameter rounding (Section 3.3)

For every $p \in [P_{\min}, P_{\max}]$, $q \in \mathcal{Q}$:

$$w^{(q)} \leftarrow \frac{\epsilon}{n} \cdot (1 + \epsilon)^q.$$

For every $i \in \Lambda_p \cap \mathcal{W}_q$:

$$\lambda_i \leftarrow \epsilon \cdot (1 + \epsilon)^p, \quad w_i \leftarrow w^{(q)}, \quad \text{and} \quad r_i \leftarrow \frac{w_i}{w^{(q)}} \cdot r_i.$$

Step 2: Enumerating over assortment statistics (Section 3.4)

Generate the collection Ω of assortment statistics $\{(\hat{c}_q, \hat{\alpha}_q)\}_{q \in \mathcal{Q}}, \{(\hat{c}_{p,q})\}_{(p,q) \in [P_{\max}] \times \mathcal{Q}}$.

Initialize $\mathcal{R}_{\max} \leftarrow 0$ and $B_{\max} \leftarrow \emptyset$.

For every $\{(\hat{c}_q, \hat{\alpha}_q)\}_{q \in \mathcal{Q}}, \{(\hat{c}_{p,q})\}_{(p,q) \in [P_{\max}] \times \mathcal{Q}} \in \Omega$:

Step 3: Min-knapsack algorithm (Section 3.5)

Compute $B_{\text{unlikely}} = \text{MinKnapsack}(\{(\hat{c}_q, \hat{\alpha}_q)\}_{q \in \mathcal{Q}})$.

Step 4: Greedy algorithm (Section 3.6)

Compute $B_{\text{likely}} = \text{Greedy}(\{(\hat{c}_{p,q})\}_{(p,q) \in [P_{\max}] \times \mathcal{Q}}, B_{\text{unlikely}})$.

Compute an estimate $\hat{r}$ of the expected revenue of $B = B_{\text{unlikely}} \cup B_{\text{likely}}$.

If $\hat{r} > \mathcal{R}_{\max}$:

$$\mathcal{R}_{\max} \leftarrow \hat{r}, \quad B_{\max} \leftarrow B$$

Output: Return the decomposable assortment $B_{\max}$ and its estimated expected revenue $\mathcal{R}_{\max}$.

3.3. Step 1: Parameter rounding

We begin by rounding the consideration probabilities, preference weights, and selling prices. For each item $i \in \mathcal{W}_q \cap \Lambda_p$, its rounded weight $\tilde{w}_i$ is defined by $\tilde{w}_i = \frac{\epsilon}{n} \cdot (1 + \epsilon)^q$ and its rounded consideration probability $\tilde{\lambda}_i$ is defined by $\tilde{\lambda}_i = \epsilon \cdot (1 + \epsilon)^p$. Finally, the rounded selling price of item i is set to $\tilde{r}_i = \frac{w_i}{\tilde{w}_i} \cdot r_i$. Here, we use specific notation (i.e., w_i and $\tilde{w}_i$) to distinguish between the original and the rounded parameters. However, in the pseudo-code of Algorithm 1, we overloaded the original notation $\{w_i\}_{i \in [n]}$, $\{\lambda_i\}_{i \in [n]}$, and $\{r_i\}_{i \in [n]}$ to indicate that the expected revenue is eventually determined using the modified instance parameters.

From a computational standpoint, these rounded-form parameters will be leveraged in subsequent steps to conduct efficient enumerations. For now, we observe that the rounding errors involved in step 1 leave the optimal expected revenue almost unaltered. To establish this claim, we conduct a *sensitivity analysis*, showing that these modifications of the instance parameters affect the optimal expected revenue only by a factor $1 - O(\epsilon)$. As a side note, a factor- $(1 - O(n\epsilon))$ approximation can easily be derived by observing that the expected revenue function is a degree- n polynomial in the consideration probabilities $\{\lambda_i\}_{i \in [n]}$; however, such bounds are insufficient for our purposes.

The desired property requires a finer analysis based on the probabilistic structure of the click-based MNL model. For this purpose, let $\tilde{\mathcal{R}}(\cdot)$ denote the expected revenue function with respect to the rounded parameters $\{\tilde{\lambda}_i\}_{i \in [n]}$, $\{\tilde{w}_i\}_{i \in [n]}$, and $\{\tilde{r}_i\}_{i \in [n]}$. For every assortment $S \subseteq [n]$, we consider the decomposable assortment B^S , where each item $i \in S$ is picked with probability $1 - \epsilon$ whereas items $i \in [n] \setminus S$ are not picked at all. The next claim, whose proof is provided in Appendix EC.3.1, establishes our sensitivity bounds for $\tilde{\mathcal{R}}(S)$ with regards to $\mathcal{R}(S)$ and $\mathbb{E}_{S' \sim B^S}[\mathcal{R}(S')]$.

LEMMA 1. *For every $S \subseteq [n]$, we have $(1 - 2\epsilon) \cdot \mathcal{R}(S) \leq \tilde{\mathcal{R}}(S) \leq (1 + 3\epsilon) \cdot \mathbb{E}_{S' \sim B^S}[\mathcal{R}(S')]$.*

This claim implies that the click-based MNL assortment problem is well-behaved with respect to our alterations of the instance parameters. Indeed, given an α -approximation $\tilde{S}$ of the modified instance, the corresponding decomposable assortment $B^{\tilde{S}}$ satisfies

$$\mathbb{E}_{S' \sim B^{\tilde{S}}}[\mathcal{R}(S')] \geq (1 - 3\epsilon) \cdot \tilde{\mathcal{R}}(\tilde{S}) \geq (1 - 3\epsilon) \alpha \cdot \tilde{\mathcal{R}}(S^*) \geq (1 - 5\epsilon) \alpha \cdot \mathcal{R}(S^*),$$

where the first and last inequalities follow from Lemma 1. This analysis shows that operating with the rounded parameters $\{\tilde{\lambda}_i\}_{i \in [n]}$, $\{\tilde{w}_i\}_{i \in [n]}$, and $\{\tilde{r}_i\}_{i \in [n]}$ in place of the original ones incurs only an $O(\epsilon)$ -factor loss for the design of approximation algorithms.

3.4. Step 2: Enumeration

Let S^* be any optimal assortment with respect to the modified instance generated in step 1; this assortment will serve as a fixed benchmark throughout our exposition and analysis. Next, our algorithm “guesses” a real-valued vector containing aggregate statistics of S^* . This vector aims to

describe how S^* is structured with respect to the different classes of products. More specifically, for every $q \in \mathcal{Q}$, we define $c_q^* = \sum_{i \in S^* \cap \mathcal{U}_q} r_i \cdot \pi(i, S^*)$ as the *revenue contribution* of the class of items $\mathcal{U}_q$ in the optimal assortment S^* . Similarly, for every $(q, p) \in \mathcal{Q} \times [P_{\min}, P_{\max}]$, we define $c_{p,q}^* = \sum_{i \in S^* \cap \mathcal{W}_q \cap \Lambda_p} r_i \cdot \pi(i, S^*)$ as the revenue contribution of the class of items $\mathcal{W}_q \cap \Lambda_p$. In addition, for every $S \subseteq [n]$ and $q \in \mathcal{Q}$, let $\alpha(S, q) = \mathbb{E}[\frac{w^{(q)}}{1+w^{(q)}+w(C_S)}]$ be the *coefficient* of the class of items $\mathcal{U}_q$ in the assortment S , where $w^{(q)} = \frac{\epsilon}{n} \cdot (1 + \epsilon)^q$ is the rounded preference weight within class $\mathcal{W}_q$.

Clearly, the quantities $\{(c_q^*, \alpha(S^*, q))\}_{q \in \mathcal{Q}}$ and $\{c_{p,q}^*\}_{(p,q) \in [P_{\max}] \times \mathcal{Q}}$ are initially unknown since they depend on the optimal assortment S^* . Nevertheless, the specialized algorithms **MinKnapsack**($\cdot$) and **Greedy**($\cdot$) will use input parameters that “closely approximate” these quantities. To be specific, **MinKnapsack**($\cdot$) is executed with input parameters $\{\hat{c}_q, \hat{\alpha}_q\}_{q \in \mathcal{Q}}$ as a proxy for $\{(c_q^*, \alpha(S^*, q))\}_{q \in \mathcal{Q}}$, and **Greedy**($\cdot$) is executed with $\{\hat{c}_{p,q}\}_{(p,q) \in [P_{\max}] \times \mathcal{Q}}$ as a proxy for $\{c_{p,q}^*\}_{(p,q) \in [P_{\max}] \times \mathcal{Q}}$.

At first glance, it is unclear how to efficiently guess these input parameters. Indeed, from a computational standpoint, there are exponentially many possible values for $\{(c_q^*, \alpha(S^*, q))\}_{q \in \mathcal{Q}}$ and $\{c_{p,q}^*\}_{(p,q) \in [P_{\max}] \times \mathcal{Q}}$. However, the important observation is that the number of input parameters is relatively small, i.e., $|\mathcal{Q}| = Q_{\max} - Q_{\min} + 1 = O(\frac{1}{\epsilon^2} \log n)$ by Assumption 1, and $P_{\max} = O(\frac{1}{\epsilon} \cdot \log \frac{1}{\epsilon})$ given that the consideration probabilities of likely items are within the range $(\epsilon, 1]$. Thus, basic counting arguments show that the input parameters can be approximated by enumerating over polynomially many vectors, as formally stated in the next claim, whose proof is deferred to Appendix EC.3.2. In what follows, we use $\mathcal{L} = |\mathcal{Q}| \cdot (P_{\max} + 1)$ for ease of notation.

LEMMA 2. *There is a collection Ω of input parameters, such that $|\Omega| = O(n^{O(\frac{1}{\epsilon^4} \log \frac{1}{\epsilon})})$ and such that there exists $(\{\hat{c}_q, \hat{\alpha}_q\}_{q \in \mathcal{Q}}, \{\hat{c}_{p,q}\}_{(p,q) \in [P_{\max}] \times \mathcal{Q}}) \in \Omega$ satisfying:*

1. $\hat{c}_{p,q} \leq c_{p,q}^* < \hat{c}_{p,q} + \frac{2\epsilon}{\mathcal{L}} \cdot \mathcal{R}(S^*)$ for every $(p, q) \in [P_{\max}] \times \mathcal{Q}$.
2. $\hat{c}_q \leq c_q^* < \hat{c}_q + \frac{2\epsilon}{\mathcal{L}} \cdot \mathcal{R}(S^*)$ for every $q \in \mathcal{Q}$.
3. $(1 - \epsilon)^2 \cdot \hat{\alpha}_q \leq \alpha(S^*, q) \leq (1 - \epsilon) \cdot \hat{\alpha}_q$ for every $q \in \mathcal{Q}$.

Moreover, the set Ω can be constructed in time $O(n^{O(\frac{1}{\epsilon^4} \log \frac{1}{\epsilon})})$.

The above claim states that by enumerating over Ω , we will consider in particular input parameters $\{\hat{c}_q, \hat{\alpha}_q\}_{q \in \mathcal{Q}}$ and $\{\hat{c}_{p,q}\}_{(p,q) \in [P_{\max}] \times \mathcal{Q}}$ that closely approximate $\{(c_q^*, \alpha(S^*, q))\}_{q \in \mathcal{Q}}$ and $\{c_{p,q}^*\}_{(p,q) \in [P_{\max}] \times \mathcal{Q}}$. Indeed, Property 1 and 2 ensure that the revenue contribution parameters are accurate, while Property 3 ensures that the coefficient parameters are accurate; the latter have a slight negative bias that is required in our subsequent analysis.

3.5. Step 3: Min-knapsack algorithm for unlikely items

In this section, we devise the specialized algorithm **MinKnapsack**($\cdot$) for unlikely items. To motivate this algorithm, we informally describe a linear approximation of the click-based MNL assortment problem, which lends itself to the formulation of min-knapsack problems.

Linear approximation idea. Generally speaking, the tradeoff in assortment decisions is between increasing market share and mitigating the cannibalization of revenue from low-price items. Focusing on a specific class of unlikely items $\mathcal{U}_q$, we argue that this tradeoff can be approximately resolved by formulating a certain instance of the min-knapsack problem, which makes use of the aggregate statistics guessed in step 2.

We first capture the “cannibalization” component in this tradeoff. Based on representation (1) of the choice probabilities, for every assortment $S \subseteq [n]$ and item $i \in S \cap \mathcal{U}_q$, we have:

$$\pi(i, S) = \lambda_i \cdot \mathbb{E} \left[\frac{w_i}{1 + w_i + w(C_S^{-i})} \right].$$

Intuitively, the amount of cannibalization exerted by the items $S \setminus \{i\}$ on item i 's choice probability scales with the random variable $w(C_S^{-i})$. To be specific, the quantity $\mathbb{E}[\frac{w_i}{1 + w_i + w(C_S^{-i})}]$ is decreasing convex with respect to $w(C_S^{-i})$. The items of $S \cap \mathcal{U}_q$ have a total contribution to $w(C_S^{-i})$ of exactly $w^{(q)} \cdot |C_{S \cap \mathcal{U}_q}^{-i}|$. Here, the random variable $|C_{S \cap \mathcal{U}_q}^{-i}|$ represents the number of items considered amongst $S \cap \mathcal{U}_q$ other than item i , which is a sum of independent Bernoulli outcomes.

The main technical insight for unlikely items is that the distribution of $|C_{S \cap \mathcal{U}_q}^{-i}|$ can be approximated using a single *summary statistic*, which is its expectation, $\mathbb{E}[|C_{S \cap \mathcal{U}_q}^{-i}|]$. This insight proceeds from the next lemma, whose proof is provided in Appendix EC.3.3, showing that sums of independent Bernoulli outcomes are well-approximated by Poisson surrogates. In what follows, for $\lambda \in [0, 1]$, we denote by $P(\lambda)$ a Poisson random variable with parameter λ . Two random variables Z_1 and Z_2 are said to be in the decreasing convex order $Z_1 \succeq_{\text{dcx}} Z_2$ if, for every decreasing convex function ϕ , we have $\mathbb{E}[\phi(Z_1)] \geq \mathbb{E}[\phi(Z_2)]$, provided these expectations exist (Shaked and Shanthikumar 2007, Chap. 4).

LEMMA 3. *Suppose that $\epsilon \in [0, \sqrt{2} - 1]$. Let $Y = \sum_{i=1}^k X_i$, where $\{X_i\}_{i \in [k]}$ are independent Bernoulli random variables with success probabilities $\{\lambda_i\}_{i \in [k]}$ in $[0, \epsilon]$. Letting $\lambda = \sum_{i=1}^k \lambda_i$, we have $P(\lambda) \succeq_{\text{dcx}} Y \succeq_{\text{dcx}} P((1 + \epsilon) \cdot \lambda)$.*

This claim shares similarities with the Poissonization method (Le Cam 1960). The proof is derived from first principles using stochastic orderings. Informally, by leveraging Lemma 3, we will argue in Appendix EC.4 that the cannibalization effects due to the assortment decisions $S \cap \mathcal{U}_q$ can be “mitigated” by ensuring that $\mathbb{E}[|C_{S \cap \mathcal{U}_q}|]$ is sufficiently small. Moreover, $\mathbb{E}[|C_{S \cap \mathcal{U}_q}|] = \sum_{j \in S \cap \mathcal{U}_q} \lambda_j$ is clearly linear with respect to the assortment decisions $S \cap \mathcal{U}_q$.

Next, we turn our attention to the “revenue” component in the tradeoff mentioned earlier. How do we quantify the revenue contribution of an individual unlikely item? By equation (1), in order to compute the choice probability $\pi(i, S)$ for an item $i \in S \cap \mathcal{U}_q$, we need to estimate the quantity $\mathbb{E}[\frac{w_i}{1 + w_i + w(C_S^{-i})}]$, which clearly depends on the assortment S and the precise values of $q \in \mathcal{Q}$ and

$p \in [P_{\min}, 0]$ that govern the values of w_i and the distribution of $w(C_S^{-i})$, respectively. Interestingly, in the next claim, we reveal that the dependency on $p \in [P_{\min}, 0]$ plays a minor role for unlikely items. To this end, recall that $\alpha(S, q) = \mathbb{E}[\frac{w^{(q)}}{1+w^{(q)}+w(C_S)}]$ is the coefficient of the class of items $\mathcal{U}_q$ in the assortment S , a quantity that only depends on S and q .

LEMMA 4. *For all $S \subseteq [n]$ and $i \in S \cap \mathcal{U}_q$, we have $(1 - \epsilon) \cdot \pi(i, S) \leq \lambda_i \cdot \alpha(S, q) \leq \pi(i, S)$.*

The proof is provided in Appendix EC.3.4. Consequently, the contribution of items $i \in S \cap \mathcal{U}_q$ to the expected revenue $\mathbb{E}[\mathcal{R}(S)]$ is well-approximated by the expression $\sum_{i \in S \cap \mathcal{U}_q} r_i \lambda_i \cdot \alpha(S, q)$. The important observation is that, should we use the input parameter $\hat{\alpha}_q$ as a constant proxy for $\alpha(S, q)$, the resulting expression $\sum_{i \in S \cap \mathcal{U}_q} r_i \lambda_i \cdot \hat{\alpha}_q$ is linear with respect to S .

All in all, the above observations suggest that the assortment decisions within $\mathcal{U}_q$ can be approximated by a min-knapsack problem with the following linear objective and linear constraint: Minimize $\mathbb{E}[|C_{S \cap \mathcal{U}_q}|]$ by selecting $S \cap \mathcal{U}_q$ subject to $\sum_{i \in S \cap \mathcal{U}_q} r_i \lambda_i \cdot \hat{\alpha}_q \geq \hat{c}_q$, where $\hat{\alpha}_q > 0$ is the chosen coefficient for the class of items $\mathcal{U}_q$ and $\hat{c}_q \geq 0$ is the desired revenue contribution from this class of items. Here, the minimization of $\mathbb{E}[|C_{S \cap \mathcal{U}_q}|]$ allows us to mitigate the cannibalization effects, and the constraint $\sum_{i \in S \cap \mathcal{U}_q} r_i \lambda_i \cdot \hat{\alpha}_q \geq \hat{c}_q$ ensures that a sufficient revenue contribution is generated.

Min-knapsack algorithm. To summarize the above discussion, $\text{MinKnapsack}(\cdot)$ proceeds by solving a collection of min-knapsack problems. Here, for every $q \in \mathcal{Q}$, we assume that a coefficient $\hat{\alpha}_q$ and a revenue contribution $\hat{c}_q$ for the class of items $\mathcal{U}_q$ are given according to the enumeration procedure of Lemma 2. Given these inputs, the algorithm $\text{MinKnapsack}(\cdot)$ constructs a decomposable assortment B_{unlike} over the collection of unlikely items, i.e., $B_{\text{unlike}} = \text{MinKnapsack}(\{(\hat{c}_q, \hat{\alpha}_q)\}_{q \in \mathcal{Q}})$.

First, for each $q \in \mathcal{Q}$, we solve the following min-knapsack instance: Find a subset of items $U \subseteq \mathcal{U}_q$ that minimizes $\mathbb{E}[|C_U|] = \sum_{i \in U} \lambda_i$ subject to the linear constraint $\sum_{i \in U} \gamma_i \geq (1 - \epsilon) \cdot \frac{n}{\epsilon}$, where $\gamma_i = \lfloor \frac{n \lambda_i r_i \hat{\alpha}_q}{\epsilon \hat{c}_q} \rfloor$. To gain some insight into the latter constraint, note that by eliminating floors, it can be rearranged as $\hat{\alpha}_q \cdot \sum_{i \in U} r_i \lambda_i \geq (1 - \epsilon) \cdot \hat{c}_q$. Hence, the constraint precisely ensures that the desired revenue contribution of $\hat{c}_q$ is approximately generated by the items $U \subseteq \mathcal{U}_q$.

Now, let $U_q \subseteq \mathcal{U}_q$ be the resulting optimal subset of items; if the min-knapsack instance is infeasible, we simply return $U_q = \emptyset$. Note that for the correct guess of input parameters $\{(\hat{c}_q, \hat{\alpha}_q)\}_{q \in \mathcal{Q}} \in \Omega$, our analysis will reveal that the corresponding min-knapsack instance must be feasible, and thus, the infeasibility case of $U_q = \emptyset$ can be logically ignored. Given U_q , we define the decomposable assortment B^q , where each item $i \in U_q$ is independently picked with probability $1 - \epsilon$; namely, $B^q = (B_i^q)_{i \in U_q}$ is a collection of IID Bernoulli random variables with probability of success of $1 - \epsilon$. Finally, our algorithm returns the decomposable assortment B_{unlike} , defined by concatenating the assortments B^q over $q \in \mathcal{Q}$. **Intuitively, this randomization procedure is meant to reduce the cannibalization effects from products in U_q . Dropping each item with probability $1 - \epsilon$ mimics a**

reduction of the consideration probabilities by the same factor, which enables using the Poisson approximation inequalities of Lemma 3.

From a computational perspective, the min-knapsack instance constructed above can be solved to optimality in $O(\frac{n^2}{\epsilon})$ time via standard dynamic programming ideas (see, e.g., Vazirani (2013, Chap 8.1)), since the γ_i -parameters are integral and upper-bounded by $\frac{n}{\epsilon}$.

3.6. Step 4: Greedy algorithm for likely items

The final piece of our approach concerns likely items. Here, the optimal assortment decisions over $\mathcal{N}_{\text{likely}}$ can be approximated using a greedy procedure motivated by the following structural observation.

Revenue-ordered property. We establish a generalized revenue-ordered-like property for the click-based MNL assortment problem. To this end, we say that an assortment S is *revenue-ordered by class* if, for every class $\mathcal{W}_q \cap \Lambda_p$, the subset $S \cap \mathcal{W}_q \cap \Lambda_p$ consists of the $|S \cap \mathcal{W}_q \cap \Lambda_p|$ highest-revenue items in $\mathcal{W}_q \cap \Lambda_p$, breaking ties arbitrarily.

LEMMA 5. *There exists an optimal assortment S^* which is revenue-ordered by class.*

The proof is presented in Appendix EC.3.5. The important observation is that this property is very useful from an algorithmic perspective. By Lemma 5, within each class, higher price items should always be offered before lower price items. Consequently, our algorithm greedily selects items within each class of items $\mathcal{W}_q \cap \Lambda_p$ by order of decreasing prices until reaching the desired threshold $\hat{c}_{p,q}$; recall that this quantity approximates the revenue contribution within $\mathcal{W}_q \cap \Lambda_p$.

Greedy algorithm. Suppose that a revenue contribution $\hat{c}_{p,q}$ is given for each class $\mathcal{W}_q \cap \Lambda_p$. We also assume that the decomposable assortment B_{unlikely} over the unlikely items has already been determined via the $\text{MinKnapsack}(\cdot)$ algorithm of Section 3.5. Given these inputs, $\text{Greedy}(\cdot)$ constructs the decomposable assortment $B_{\text{likely}} = \text{Greedy}(\{\hat{c}_{p,q}\}_{(p,q) \in [P_{\max}] \times \mathcal{Q}}, B_{\text{unlikely}})$ over the set of likely items.

Starting from the decomposable assortment B_{unlikely} , by order of decreasing revenues, we greedily add a single likely item from any class $\Lambda_p \cap \mathcal{W}_q$ that satisfies the following two conditions: (i) At least one item from $\Lambda_p \cap \mathcal{W}_q$ is yet to be chosen, and (ii) The items already chosen from $\Lambda_p \cap \mathcal{W}_q$ contribute less than $(1 - \epsilon) \cdot \hat{c}_{p,q}$ to the expected revenue. If any of these conditions, (i) or (ii), fails, the algorithm terminates. To formalize this idea, the $\text{Greedy}(\cdot)$ algorithm proceeds as follows:

1. We start with $S_0 = \emptyset$.
2. Given S_t , our algorithm halts if the following termination criterion is met: For every $p \in [P_{\max}]$ and $q \in [Q_{\min}, Q_{\max}]$, either $S_t \cap \Lambda_p \cap \mathcal{W}_q = \Lambda_p \cap \mathcal{W}_q$ or

$$\sum_{i \in S_t \cap \Lambda_p \cap \mathcal{W}_q} r_i \cdot \mathbb{E}_{S \sim B_{\text{unlikely}}} [\tilde{\pi}(i, S \cup S_t)] \geq (1 - \epsilon) \cdot \hat{c}_{p,q} . \quad (3)$$

Otherwise, let $\Lambda_p \cap \mathcal{W}_q$ be a class for which this criterion does not hold. Letting i_t be the maximum price item in $(\Lambda_p \cap \mathcal{W}_q) \setminus S_t$, we define the next assortment as $S_{t+1} = S_t \cup \{i_t\}$.

3. Let S_T be the assortment obtained in step 2, at the last iteration T . We return the decomposable assortment $B_{\text{likely}} = (B_i)_{i \in S_T}$ where $\Pr[B_i = 1] = 1$ for all $i \in S_T$.

From a computational perspective, the number of iterations in step 2 is at most n , and thus, **Greedy**($\cdot$) makes $O(n^2)$ calls to the FPTAS of Theorem 2 to estimate the choice probabilities, which results in an overall running time of $O(\frac{1}{\epsilon^2} n^4 \log n)$ by Assumption 1.

3.7. Conclusion and extension

The specification of **Greedy**($\cdot$) in Section 3.6 completes the exposition of our PTAS. As previously mentioned, the analysis of these algorithmic ideas is deferred to Appendix EC.4, where we establish the near-optimal performance guarantee of Algorithm 1 based on the intuition provided in steps 1-4. In this analysis, Assumption 1 is only important to make sure that the enumeration over Ω can be implemented in polynomial time. In general, when this assumption is violated, we can no longer construct a collection of input parameters Ω that achieves the properties of Lemma 2, while ensuring that $|\Omega| = O(n^{O(\frac{1}{\epsilon^4} \log \frac{1}{\epsilon})})$. To handle this more general context, we will extend Algorithm 1 using dynamic programming ideas, which are separately presented in Appendix EC.5.

Under Assumption 1, our approximation scheme can be extended to the cardinality-constrained setting, where an upper bound is imposed on the size of the selected assortment. We omit the detailed exposition of the resulting algorithm, which closely follows Algorithm 1.³ Modeling an equality constraint on the assortment cardinality rather than an upper bound can be helpful in practical applications. This variant of the assortment optimization problem is less frequent in previous literature, although it arises in applications such as recommendation algorithms for online platforms, where a fixed number of options (typically a small number of product alternatives) must be displayed to the end user. For the MNL model, such fixed-cardinality problems can be solved to optimality in polynomial time using known methods, e.g., using the LP formulation with totally unimodular constraints (Sumida et al. 2021) or parametric search algorithm (Rusmevichientong et al. 2010). In the next section, we develop a fixed-parameter PTAS in this setting, when the required assortment cardinality is treated as a constant.

³ The main observation is that, for some guess of the assortment statistics in Ω , **Greedy**($\cdot$) halts with a subset of the products contained in an optimal assortment. For **MinKnapsack**($\cdot$), we need to allocate the cardinality constraint across the unlikely item classes $q \in \mathcal{Q}$ using further approximate enumeration and then account for the resulting cardinality constraint in the min-knapsack algorithm. Ultimately, our decomposable assortment may violate the cardinality upper bound constraint by a factor of at most $1 + \epsilon$. Therefore, as post-processing, we greedily eliminate a fraction ϵ of the products to meet the cardinality constraint, while preserving a fraction $1 - \epsilon$ of the expected revenue.

4. Computational Experiments

In this section, we implement and employ our PTAS on a variety of randomly generated instances with up to 80 products. Additionally, motivated by the revenue-ordered-like property of Lemma 5, we propose a greedy heuristic for the click-based MNL assortment problem, which serves both as a benchmark for our PTAS and as a candidate solution approach when very fast running times are required. We find that the greedy heuristic has a very satisfactory performance on most instances, although it is sub-optimal in the worst case. **Our PTAS significantly outperforms this heuristic on certain instances, and it returns near-optimal assortments in almost all test cases, with optimality gaps often smaller than our $O(\epsilon)$ -dependent performance guarantee.** Additionally, we present a running time analysis for our implemented PTAS as we scale up the number of available products.

4.1. Instance generator

We randomly generate cardinality-constrained instances of the click-based MNL assortment problem with $n \in \{20, 40, 60, 80\}$ items. Specifically, we enforce that each recommended assortment must include exactly six products. The motivation for this additional constraint is two-fold. First, in many e-commerce settings, product recommendation pages are composed of displays that consist of a fixed number of products. This is the case, for example, in the Alibaba discount coupon setting considered in Section 5, where Alibaba is tasked with presenting personalized six product displays of discounted products to arriving customers. Second, by enforcing this exact cardinality constraint, we limit both the total number of feasible assortments to $\binom{n}{6}$ and the number of potential consideration sets to 2^6 for any fixed assortment. As such, when n is reasonably small (i.e., $n \leq 40$), we can completely enumerate all feasible assortments to recover the optimal one, which enables us to report an exact optimality gap as our measure of each recommended assortment's profitability. For larger instances, with $n \in \{60, 80\}$, we benchmark our PTAS against the greedy heuristic approach.

The prices, weights and consideration probabilities of each product are generated as follows:

- *Prices:* For each item $i \in [n]$, we generate r_i from a log-normal distribution with location 0 and scale $\sigma \in \{0.1, 0.5\}$. Let $r_{\min}$ and $r_{\max}$ be the respective smallest and largest prices generated for a particular instance.
- *Preference weights:* The preference weight of item i is set to $w_i = e^{\alpha_w - \beta_w r_i}$, where $\alpha_w = r_{\min} \cdot \beta_w$ and we vary $\beta_w \in \{0.5, 1, 2\}$.
- *Consideration probabilities:* We first compute unnormalized consideration probabilities $\lambda'_i = e^{\alpha_c - \beta_c r_i}$ for items $i \in [n]$, and then set the true consideration probability of item i to be $\lambda_i = \lambda_{\text{cap}} \cdot \frac{\lambda'_i}{\max_{j \in [n]} \lambda'_j}$. Here, we set $\alpha_c = \beta_c \cdot r_{\min}$, and vary $\beta_c \in \{0.5, 1, 2\}$, $\lambda_{\text{cap}} \in \{0.1, 0.5, 1\}$.

We choose α_w and α_c as described above to ensure that the preference weights and click probabilities do not vary too widely. Furthermore, we note that the parameter λ_{cap} represents the maximum possible consideration probability that can arise in any particular problem instance.

This set-up allows us to characterize each test case through a particular configuration of the five parameters $(n, \sigma, \beta_w, \beta_c, \lambda_{\text{cap}})$. Since we vary $n \in \{20, 40, 60, 80\}$, $\sigma \in \{0.1, 0.5\}$, $\beta_w, \beta_c \in \{0.5, 1, 2\}$, and $\lambda_{\text{cap}} \in \{0.1, 0.5, 1\}$, there are a total of 216 unique test cases. For each test case, we generate 20 unique streams of prices, which ultimately yields 4,320 distinct assortment problem instances to consider. For each of these instances, we apply two algorithmic approaches, that are detailed next.

4.2. Tested algorithms

In what follows, two algorithms are developed in order to compute assortment recommendations for the click-based MNL instances generated in Section 4.1. The first is a version of our PTAS that we adapt and specialize to the cardinality-constrained instances at hand. In addition, we devise a greedy heuristic approach that draws inspiration from Lemma 5, although it is not accompanied by any non-trivial performance guarantee.

The two algorithms both begin by rounding the preference weights, revenues, and consideration probabilities, as described in Section 3.3. Consequently, the items are partitioned into the classes $\Lambda_p \cap \mathcal{W}_q$, where $p \in [P_{\min}, P_{\max}]$ and $q \in [Q_{\min}, Q_{\max}]$. In what follows, since we are choosing assortments consisting of six products, the choice probabilities with respect to any assortment can be exactly computed by enumerating over all 2^6 consideration sets.

Product Recommendation PTAS. Our first approach is a slightly modified variant of our PTAS that is adapted to the cardinality constrained setting. In what follows, we provide the key implementation details that are incorporated to improve the practical running time of our PTAS, while also preserving its core algorithmic features lead to its efficacy in generating profitable assortments.

- *Rare items:* First, we re-define the unlikely items to be those with consideration probabilities $\lambda_i \in [\epsilon/n, \epsilon]$. The items with consideration probability $\lambda_i < \epsilon/n$ are now termed “rare”. The definition of likely items remains unchanged.
- *Modified step 3:* For the unlikely items, we first guess the quantity $|S^* \cap \mathcal{U}_q|$, which represents the number of unlikely items from each item class in the optimal assortment. Then, we carry out step 3 of our approximation scheme, where $\text{MinKnapsack}(\{(\hat{c}_q, \hat{\alpha}_q)\}_{q \in \mathcal{Q}})$ follows Section 3.5, with the addition of exact cardinality constraints, ensuring that $|S^* \cap \mathcal{U}_q|$ items are added for every $q \in [Q_{\min}, Q_{\max}]$.
- *Modified step 4:* We select B_{likely} by leveraging the restriction that exactly 6 products should be offered. Specifically, in light of the revenue-ordered-by-class property, we guess B_{likely} by simply enumerating over all ways to divide the $C = 6$ products among the item classes $(p, q) \in [P_{\max}] \times \mathcal{Q}$.

This procedure is equivalent to carrying out step 4 of our original approach with a sufficiently small ϵ .

- *New step 5 (Leveraging rare items)*: We construct our choice of rare items B_{rare} by myopically selecting the rare items of largest ρ_i -quantities until the cardinality constraint is met, where $\rho_i = r_i \cdot \pi(i, B_{\text{unlikely}} \cup B_{\text{likely}} \cup \{i\})$. In Appendix EC.6, we show that this simple greedy approach degrades the expected revenue of the final assortment by only an $O(\epsilon)$ -factor. As such, not only does introducing this third class of items lighten the burden of step 3 by reducing the total number of unlikely items considered, but we can easily determine near-optimal assortments of rare items via the just-mentioned greedy approach.

Finally, we note that for each item $i \in B_{\text{likely}} \cup B_{\text{unlikely}} \cup B_{\text{rare}}$, we let B_i denote a Bernoulli random variable with success probability 1, i.e. due to the exact cardinality constraint, we no longer sparsify any of the returned assortments.

In Section 4.3, we demonstrate that this implementation incurs running times of under a minute on average, even for the instances in which $n = 80$ items are considered, and for an accuracy tolerance of $\epsilon = 0.05$. These results show that our implementation scales to problems of typical size, as seen in the recent numerical studies of Bertsimas and Mišić (2019), Berbeglia et al. (2022), and Désir et al. (2021). Our Product Recommendation PTAS was implemented in Java 1.8 on an Intel Core i5 with 3.8GHz CPU and 64GB of RAM.

Greedy heuristic. This approach is motivated by the revenue-ordered-by-class property highlighted in Section 3.5. Specifically, we have shown that there exists an optimal assortment S^* that picks the $|S^* \cap \Lambda_p \cap \mathcal{W}_q|$ largest revenue items in $\Lambda_p \cap \mathcal{W}_q$, for every $p \in [P_{\min}, P_{\max}]$ and $q \in [Q_{\min}, Q_{\max}]$. Here, we develop a greedy heuristic that constructs assortments satisfying this property, by iteratively adding the product that generates the largest marginal increase in the expected revenue. More formally, we initially start with an empty assortment $S_0 = \emptyset$, and in each step, we add exactly one product to our assortment until the required capacity of 6 products is reached. Using S_t to denote our current assortment, we construct the collection of candidate assortments $\mathcal{S}_{t+1} = \{S_t \cup \{i_{t,p,q}\} : p \in [P_{\min}, P_{\max}], q \in [Q_{\min}, Q_{\max}]\}$, where $i_{t,p,q}$ is the largest revenue item out of $(\Lambda_p \cap \mathcal{W}_q) \setminus S_t$. Consequently, we define $S_{t+1} = \arg \max_{S \in \mathcal{S}_{t+1}} \mathcal{R}(S)$. Ultimately, our heuristic returns the assortment S_6 hence defined. It is worth noting that, as explained in Section 5, since Alibaba is required to display precisely 6 products, items are added up until reaching this capacity, even if the expected revenue decreases. The greedy heuristic was implemented in Python 3.6 on an Intel Core i5 with 3.8GHz CPU and 64GB of RAM.

4.3. Results

In this section, we present a detailed assessment of the performance of our PTAS across the testbed of instances generated in Section 4.1. From this point on, we use PTAS as a shorthand

for our updated approximation scheme and GR for the greedy heuristic. Furthermore, for each algorithm $\mathcal{A} \in \{\text{PTAS}, \text{GR}\}$, let $S^{\mathcal{A}}$ be the assortment returned by algorithm $\mathcal{A}$, and let S^* be the optimal assortment. For smaller instances, with $n \in \{20, 40\}$, we measure the revenue performance of algorithm $\mathcal{A}$ through the optimality gap, which is computed as $100 \cdot \frac{\mathcal{R}(S^*) - \mathcal{R}(S^{\mathcal{A}})}{\mathcal{R}(S^*)}$. For larger instances, with $n \in \{60, 80\}$, we assess the profitability of the assortments returns by PTAS through their relative revenue improvement as compared to the assortments returned by GR, which is precisely measured as $100 \cdot \frac{\mathcal{R}(S^{\text{PTAS}}) - \mathcal{R}(S^{\text{GR}})}{\mathcal{R}(S^{\text{PTAS}})}$.

In Tables 1 and 2, we present four summary statistics of the distributions of optimal gaps for PTAS and GR. For these small- n instances, we employed PTAS with an accuracy level of $\epsilon = 0.05$, where for GR, an accuracy level of $\epsilon = 0.01$ was used.⁴ In each row of these two tables, we have aggregated the results across the mean price parameter $\sigma \in \{0.1, 0.5\}$, since this parameter turned out to have little effect on the high-level performance trends of either approach. Furthermore, in addition to aggregating across σ , the various statistics are also computed across the 20 trials generated for each test case, meaning that each statistic is determined with respect to optimality gaps computed for 40 distinct problem instances.

With only a superficial scan of these two tables, it is clear that the revenue performance of PTAS is near-optimal; its worst-case optimality gap never exceeds 1%, and its average performance is always within hundredths of a percent of optimal. Interestingly, for test cases with $\lambda_{\text{cap}} = 0.1$, we observe that GR only lags slightly behind PTAS in terms of its ability to generate near-optimal assortments. However, as λ_{cap} is increased to 0.5 and then to 1, we observe a sharp drop in the performance of GR across all four summary statistics. This salient trend is likely a direct consequence of the algorithmic insight we make in Appendix EC.6 regarding rare products (recall that, informally, these are products with very small λ_i). Namely, we show that simple greedy algorithms, closely resembling GR, yield ϵ -optimal assortments for instances that exclusively consist of rare products. At a high level, this observation follows by noting that, when each product is associated with a small consideration probability, our random consideration set consists of a single product with high probability, and therefore, the MNL-based substitution effects become negligible. However, for instances that do not exclusively consist of small- λ_i products (i.e., test cases with $\lambda_{\text{cap}} \in \{0.5, 1\}$), consideration sets with more than one product are likely to realize and re-ignite the MNL-based substitution dynamics. To understand why this phenomenon is detrimental to a greedy approach, note that in the extreme case where all $\lambda_1 = \dots = \lambda_n = 1$, we recover the standard MNL model, for which it is easy to construct instances where GR performs arbitrarily bad.

⁴ We do not present any plots or statistics regarding the running time of GR, since it is clearly a highly efficient approach.

λ_{cap}	β_c	β_w	Avg. Opt. Gap		75-th% Opt. Gap		90-th% Opt. Gap		Max Opt. Gap	
			PTAS	GR	PTAS	GR	PTAS	GR	PTAS	GR
0.1	0.5	0.5	0.00	0.04	0.00	0.02	0.00	0.07	0.07	0.79
0.1	0.5	1.0	0.04	0.11	0.07	0.17	0.13	0.31	0.17	0.51
0.1	0.5	2.0	0.00	0.01	0.00	0.02	0.01	0.03	0.03	0.24
0.1	1.0	0.5	0.04	0.08	0.08	0.13	0.12	0.19	0.22	0.43
0.1	1.0	1.0	0.04	0.02	0.02	0.02	0.13	0.06	0.41	0.26
0.1	1.0	2.0	0.00	0.07	0.00	0.01	0.00	0.03	0.01	1.76
0.1	2.0	0.5	0.07	0.03	0.14	0.02	0.21	0.05	0.47	0.54
0.1	2.0	1.0	0.06	0.14	0.11	0.02	0.24	0.06	0.35	2.18
0.1	2.0	2.0	0.01	0.01	0.00	0.02	0.00	0.02	0.28	0.04
0.5	0.5	0.5	0.00	0.24	0.00	0.20	0.00	1.05	0.11	1.64
0.5	0.5	1.0	0.10	0.89	0.11	1.32	0.33	1.56	0.97	2.13
0.5	0.5	2.0	0.01	1.57	0.01	1.97	0.03	2.16	0.17	4.83
0.5	1.0	0.5	0.00	2.38	0.00	3.32	0.00	3.74	0.03	4.35
0.5	1.0	1.0	0.02	2.47	0.01	2.99	0.07	3.29	0.13	4.58
0.5	1.0	2.0	0.00	0.96	0.00	1.67	0.01	2.69	0.04	3.96
0.5	2.0	0.5	0.01	1.38	0.00	2.24	0.01	3.77	0.08	4.80
0.5	2.0	1.0	0.01	0.66	0.00	1.16	0.04	2.03	0.15	3.55
0.5	2.0	2.0	0.00	0.25	0.00	0.06	0.00	0.89	0.01	3.74
1.0	0.5	0.5	0.00	0.27	0.00	0.01	0.00	1.32	0.01	2.65
1.0	0.5	1.0	0.11	2.53	0.21	4.13	0.34	5.09	0.47	6.85
1.0	0.5	2.0	0.04	4.89	0.03	5.93	0.14	7.09	0.52	9.09
1.0	1.0	0.5	0.00	4.19	0.00	4.93	0.00	5.75	0.00	7.28
1.0	1.0	1.0	0.08	4.71	0.05	5.97	0.28	6.69	0.86	7.23
1.0	1.0	2.0	0.01	4.98	0.00	5.63	0.01	7.67	0.14	9.31
1.0	2.0	0.5	0.00	5.62	0.00	6.34	0.00	9.19	0.08	11.73
1.0	2.0	1.0	0.01	5.83	0.01	7.67	0.02	10.09	0.06	13.09
1.0	2.0	2.0	0.00	3.30	0.00	4.43	0.02	7.60	0.05	10.19

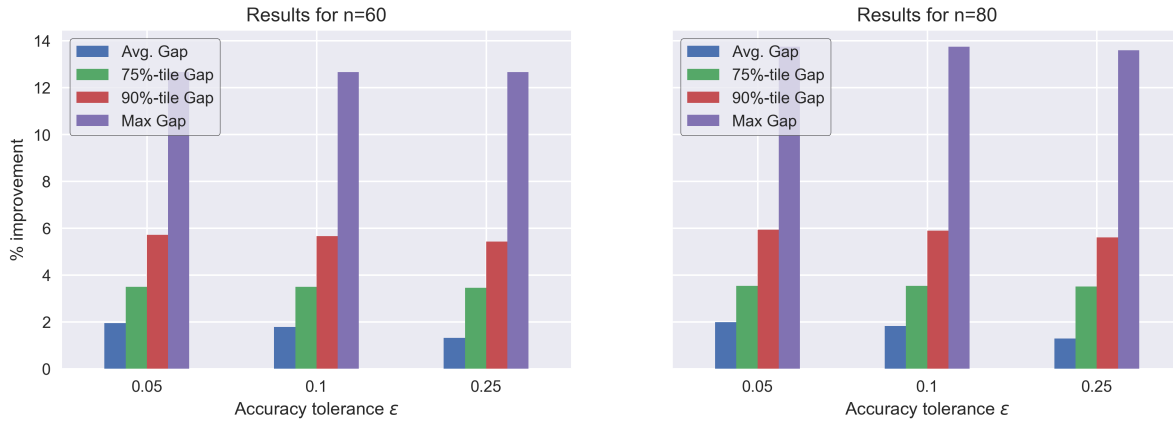
Table 1 Optimality gaps of GR and PTAS for instances with $n = 20$

The performance of PTAS relative to GR for $n \in \{60, 80\}$ items, aggregated across all test instances are presented in Figure 1. Here, we implement PTAS with accuracy levels of $\epsilon \in \{0.05, 0.1, 0.25\}$, while still employing GR with $\epsilon = 0.01$. Figure 2 provides a picture of the running times for PTAS as function of n and ϵ . We note that for all instances with $n \in \{20, 40\}$, we only carried out PTAS with $\epsilon = 0.05$, whose running time summary statistics are depicted by the solid line within the plots of Figure 2. Overall, there are two main take-home insights that can be extracted from these two figures. First, as shown in Figure 1, the performance of PTAS once again far surpasses GR, and also deteriorates very little as ϵ is increased from 0.05 to 0.25. Second, from Figure 2, we observe that the running time of PTAS never exceeds 3 minutes, even for $\epsilon = 0.05$. Moreover, when we choose $\epsilon = 0.25$, its running time never exceeds two seconds, and on average, runs in 0.03 seconds for $n = 80$.

5. Case Study with Alibaba Sales Data

In this section, we present a case study in which click-based MNL models and latent class MNL (LC-MNL) models are fit to historical sales and click data from Alibaba, one of the largest online retailers in China (Lim 2016). Our primary goal is to understand and quantify the benefits of using

λ_{cap}	β_c	β_w	Avg. Opt. Gap		75-th% Opt. Gap		90-th% Opt. Gap		Max Opt. Gap	
			PTAS	GR	PTAS	GR	PTAS	GR	PTAS	GR
0.1	0.5	0.5	0.02	0.04	0.03	0.07	0.06	0.10	0.11	0.18
0.1	0.5	1.0	0.06	0.18	0.11	0.30	0.15	0.36	0.32	0.45
0.1	0.5	2.0	0.01	0.04	0.00	0.05	0.04	0.12	0.08	0.33
0.1	1.0	0.5	0.08	0.13	0.12	0.18	0.16	0.28	0.24	0.35
0.1	1.0	1.0	0.08	0.05	0.13	0.08	0.20	0.11	0.43	0.32
0.1	1.0	2.0	0.01	0.05	0.00	0.02	0.03	0.21	0.05	0.48
0.1	2.0	0.5	0.06	0.06	0.12	0.04	0.19	0.12	0.25	0.86
0.1	2.0	1.0	0.09	0.07	0.18	0.03	0.30	0.25	0.66	0.80
0.1	2.0	2.0	0.01	0.06	0.00	0.02	0.01	0.03	0.27	0.88
0.5	0.5	0.5	0.00	0.47	0.00	0.86	0.00	1.28	0.07	1.80
0.5	0.5	1.0	0.10	1.53	0.15	1.79	0.26	2.06	0.56	2.21
0.5	0.5	2.0	0.01	2.15	0.01	2.45	0.02	2.67	0.13	2.82
0.5	1.0	0.5	0.01	3.09	0.00	4.11	0.03	4.58	0.12	5.02
0.5	1.0	1.0	0.02	2.79	0.03	3.45	0.06	3.71	0.11	4.27
0.5	1.0	2.0	0.01	1.37	0.01	1.86	0.02	2.79	0.06	4.81
0.5	2.0	0.5	0.01	1.89	0.02	3.01	0.04	4.31	0.09	5.53
0.5	2.0	1.0	0.01	1.52	0.01	3.09	0.05	4.36	0.11	5.25
0.5	2.0	2.0	0.00	0.83	0.00	1.46	0.00	2.56	0.02	4.11
1.0	0.5	0.5	0.00	1.84	0.00	3.63	0.00	4.87	0.00	5.59
1.0	0.5	1.0	0.17	3.48	0.24	4.79	0.51	5.42	0.82	7.33
1.0	0.5	2.0	0.01	6.12	0.02	7.01	0.06	7.41	0.07	7.70
1.0	1.0	0.5	0.02	4.58	0.00	5.50	0.00	6.22	0.91	7.02
1.0	1.0	1.0	0.03	5.52	0.01	6.10	0.08	7.04	0.21	8.14
1.0	1.0	2.0	0.01	5.51	0.00	5.89	0.04	8.03	0.06	9.94
1.0	2.0	0.5	0.00	6.58	0.00	7.60	0.00	9.27	0.04	11.17
1.0	2.0	1.0	0.01	6.15	0.00	8.63	0.02	9.96	0.18	10.99
1.0	2.0	2.0	0.00	4.45	0.00	7.33	0.01	9.02	0.09	10.37

Table 2 Optimality gaps of GR and PTAS for instances with $n = 40$ **Figure 1** Performance of PTAS relative to GR for the large- n instances

click behavior as a proxy for the set of products considered by each customer. First, we demonstrate that the click-based MNL is competitive against the latent class MNL model in terms of predictive accuracy. Moreover, the running time required to fit the click-based MNL model is significantly smaller than the time required to fit latent class MNL models. Our results indicate that click behavior provides a strong signal regarding the set of products considered by each customer, and

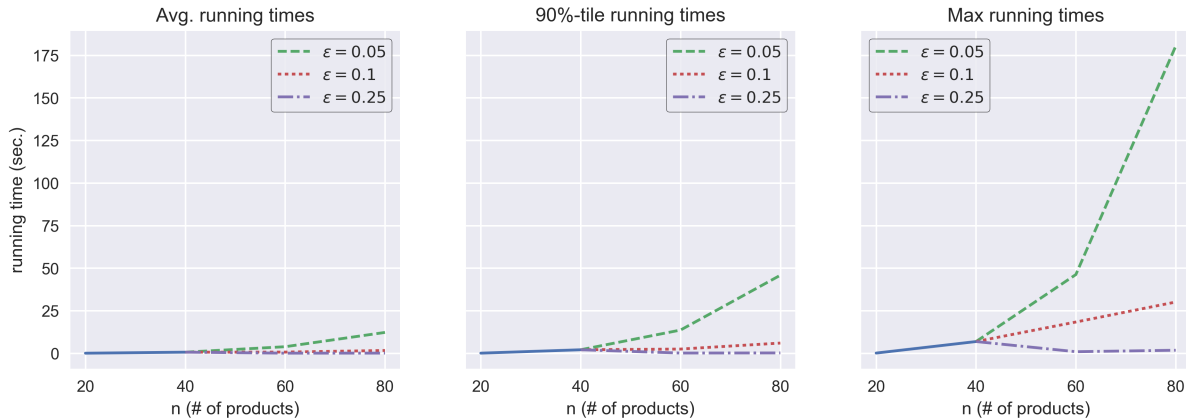


Figure 2 Running times of PTAS

hence a choice model that incorporates this screening behavior can capture customer purchasing patterns more accurately than one that ignores these signals.

Outline. In Section 5.1, we begin by presenting the exact retailing setting at Alibaba, as considered in our case study, which is followed by an inclusive summary of the available historical sales data in Section 5.2. In Section 5.3, the fitting accuracies of our two models of interest are benchmarked against each other as well as against a standard MNL model. It is worth noting that we also conduct semi-synthetic experiments to study the cost of model misspecification, where the assortment recommendation of the MNL and click-based MNL models are compared in terms of revenue. To keep this paper concise, the latter results are presented in Appendix EC.7.5.

5.1. General setting

The sales data we examine were collected at Tmall.com, one of Alibaba’s largest online marketplaces that connects third-party sellers to customers. More specifically, Tmall.com is a large-scale third-party business-to-consumer platform for branded goods, such as Nike and Adidas. Throughout this section, the term “seller” refers to any business that offers its products on that marketplace.

Customers landing on the front page of Tmall.com can then navigate to seller-specific pages, which only offer products sold by that particular seller. Upon visiting a particular seller’s front page, the customer can acquire a seller-specific coupon by clicking on a coupon icon at the top of that page. Clicking on this icon brings the customer to a “coupon sub-page” that contains a personalized display of six products, each of which can now be purchased at a discount price. Alibaba selects the six products to display with the goal of maximizing the aggregate revenue generated by coupon-claiming customers. As formalized in Appendix EC.7.5, this decision is analogous to an assortment optimization problem. Figure 3 demonstrates how a customer progresses from a seller’s front page to the coupon sub-page and finally to the six displayed products. For each customer who arrives to one of these coupon sub-pages, our data set includes the set of offered products as well as the subset of products clicked on and ultimately purchased by that customer.



Figure 3 The process of landing on the seller-specific coupon sub-page

5.2. Available sales and clickstream data

In what follows, we describe the sales data shared by Alibaba. We begin by discussing the make-up of the available historical sales data used to fit our customer choice models.

Seller statistics. Alibaba provided us with two weeks of sales data acquired from the top-10 sellers on Tmall.com in terms of overall traffic. Due to confidentiality agreements, the precise identities of these sellers cannot be revealed. Table 3 provides key sales statistics for each of these ten sellers, including the category of products sold by the particular seller in addition to information on the number of products clicked and purchased over the two-week selling horizon. The rightmost column of this table specifies the conversion rate, which is the fraction of customers who arrived to a coupon sub-page with six displayed products and who made a purchase. Typically, in the context of e-commerce, this statistic is between two and five percent, which is precisely what we observe. It should be noted that, since multiple products were purchased in approximately 0.01% of customer visits, the conversion rate does not exactly evaluate to the number of purchases (column five) divided by the number of customer arrivals (column six).

Available sales data. For each seller, we let $\mathcal{N} = \{1, \dots, n\}$ be the set of all potential products that could be displayed on coupon sub-pages (column three in Table 3). Furthermore, we assume the sales data is composed of historical sales data from τ customers (column six). To help unclutter notation, we do not indicate the dependency of $\mathcal{N}$ and τ on the specific seller, even though these quantities clearly differ from one seller to another, as indicated in Table 3. For each arriving customer t , we denote by $S_t \subseteq \mathcal{N}$ the set of six displayed products, where each product $i \in S_t$ is stored

Seller	Product Category	# products	# clicks	# purchases	# customers	conversion %
1	Electronics	169	8,338	2,045	41,765	4.88
2	Men's Apparel	1,047	11,508	1,956	213,678	0.88
3	Diapers	132	10,296	2,979	90,467	3.01
4	Bed Linens	115	6,975	1,767	39,494	4.43
5	Perfume	103	32,535	8,478	131,822	6.16
6	Women's Apparel	501	7,267	2,127	63,466	3.23
7	Furniture	49	4,949	1,937	33,579	5.75
8	Cooking Appliances	82	4,220	1,448	40,108	3.59
9	Women's Apparel	118	17,792	2,163	139,853	1.49
10	Cooking Appliances	38	3,376	2,180	37,925	5.75

Table 3 Key seller statistics

as a vector of representative feature values X_{it} . More specifically, we use the 25 features with the highest importance scores according to the machine learning approaches utilized by Alibaba's engineers for demand estimation purposes. Among these top 25 features are product-specific features such as price, the image quality of the associated picture displayed to each customer, and various measures of historical click-through rates and customer sentiment towards the product as reflected by past reviews. In addition, we use customer-specific features such as the given customer's spending and the total number of products added to the shopping cart both in the last week and in the last month. Beyond these rather straightforward product/customer features, we also have access to joint features that are specific to each customer and product pair. For example, one such joint feature is a collaborative filtering score, that takes into account past purchase and click behavior of the given customer and other customers who are deemed to have similar purchasing preferences to compute a single score signifying the extent to which the particular product will appeal towards the particular customer. Once again, due to confidentiality agreements, we cannot disclose the complete list of all 25 features.

The sales data for each customer t can be described by a 4-tuple $(S_t, X_t, \text{Click}_t, z_t)$, where $X_t = \{X_{it} : i \in S_t\}$ specifies the feature values of each offered product, $\text{Click}_t = \{c_{it} \in \{0, 1\} : i \in S_t\}$ is the subset of displayed products that were clicked on, and z_t indicates the purchased product. When customer t did not make a purchase, we set $z_t = 0$; in the rare event that $k \geq 2$ different products were purchased during a single visit, we create k copies of the 4-tuple $(S_t, X_t, \text{Click}_t, z_t)$, one for each of the unique values for z_t . With regards to Click_t , we set $c_{it} = 1$ if product $i \in S_t$ is clicked on by customer t , and $c_{it} = 0$ otherwise. Within the scope of our click-based MNL model, for each customer t , we assume that $C_{S_t} = \{i \in S_t : c_{it} = 1\}$. In other words, we use the click events to indicate whether each product in the displayed assortment is part of the given customer's consideration set. For each seller, we represent the full set of historical sales as $\text{PurchaseHistory} = \{(S_t, X_t, C_t, z_t) : t = 1, \dots, \tau\}$.

MLE estimation. We fit the click-based MNL model to the sales data using maximum log-likelihood estimation and a parametrization based on the features X_{it} that describe the interaction of each customer t with an offered product $i \in S_t$. For each seller, we randomly select 75% of the data to fit the choice models (i.e., train set) and hold out the remaining 25% of the data to test the accuracy of these models (i.e., test set). Because we observe both the clicks and the purchases, we can fully decompose the estimation of click rates and that of MNL preference weights. Specifically, we employ gradient-boosted classification trees (Catboost) to estimate the click rates and a linear-utility specification to estimate the MNL model. Details of this procedure are provided in Appendix EC.7.1. We report summary statistics for the fitted click-based MNL models in Appendix EC.7.2. Similarly, we estimate latent class MNL models on the same training set. We follow a standard estimation procedure combining MLE with a train/validate approach to select the number of segments to avoid overfitting and conduct robustness checks using alternative estimation procedures. Our approach is summarized in Appendix EC.7.3

5.3. Fitting accuracy results

The results of our main set of experiments are presented in Table 4. We report average predictive performance based on 20 trials, where each single trial consists of the following three steps: randomly splitting the data into 75/25 train and test sets, solving the corresponding MLE problem on train data, and computing the out-of-sample log-likelihood on test data. The first column of Table 4 identifies the seller number. The second, third, and fifth columns give the average log-likelihood on the testing set of the fitted click-based MNL, MNL, and LC-MNL models. The fourth and sixth columns specify the relative percentage improvements in average log-likelihood of the click-based MNL compared to the standard MNL model and the LC-MNL models, i.e., $\frac{LL_{LC-MNL}}{LL_{CMNL}} - 1$.

Seller #	Click-based MNL	MNL model		LC-MNL model	
	LL	LL	% Diff	LL	% Diff
1	-1791.18	-2027.79	13.21%	-1788.85	-0.13%
2	-2369.30	-2484.44	4.86%	-2413.84	1.88%
3	-2820.93	-2908.66	3.11%	-2831.64	0.38%
4	-1595.40	-1637.83	2.66%	-1634.96	2.48%
5	-6633.32	-6760.67	1.92%	-6691.02	0.87%
6	-2070.23	-2211.00	6.80%	-2081.82	0.56%
7	-1667.22	-1721.40	3.25%	-1668.33	0.067%
8	-1295.56	-1331.05	2.74%	-1292.83	-0.21%
9	-2468.47	-2492.90	0.99%	-2491.67	0.94%
10	-1698.77	-1757.20	3.44%	-1717.45	1.10%

Table 4 Predictive performance of the fitted models on each seller data, including both the average out-of-sample log-likelihoods (LL) and relative percentage improvement of the click-based MNL model (% Diff).

The most salient result is the significant gain in predictive accuracy when comparing the click-based MNL model with the standard MNL model. In fact, for eight out of the ten sellers, this percentage improvement is at least 2.5%, and exceeds 4.8% for Sellers 1, 2, and 6. Second, and somewhat surprisingly, we observe that the prediction accuracy of the click-based MNL model is competitive against that of the latent class MNL model, and even slightly outperforms it for the majority of sellers. These results suggest that clickstream data contain valuable information to refine the predictions of customer choices and that the click-based MNL model seems to incorporate this information effectively and efficiently. Although the class of LC-MNL models is more general in theory than the click-based MNL model, it requires specifying a very large number of customer segments in the mixture. Moreover, the LC-MNL model does not leverage clickstream data in the estimation process.

We provide summary statistics describing fitted click-based MNL models in Appendix EC.7.2. Additionally, we report a more detailed comparison of the predictive performance of the MNL, click-based MNL, and LC-MNL models on each seller's dataset in Appendix EC.7.4. We find that the differences in average out-of-sample log-likelihoods between the click-based MNL and MNL models are indeed statistically significant for most sellers, but differences between our model and LC-MNLs are not. Moreover, we observe that the predictive edge over the MNL model requires using a machine-learning method to predict click rates, whereas simple predictors such as logistic regression might not yield any particular advantage over the standard MNL model. This observation suggests that reducing bias in the first-stage estimation of consideration set distributions is important to achieve satisfactory performance.

6. Concluding Remarks

We conclude this paper by proposing two practical enhancements to the click-based MNL model, correlated clicks and contextual effects, that may potentially increase its ability to capture realistic browsing and purchasing behaviors in e-commerce applications. Such extensions create new challenges both in formulating compact parametric choice models and in devising efficient algorithms for estimation and optimization. These directions might be interesting for future research.

Correlated clicks. A fundamental assumption of the click-based MNL model is that each offered product is considered independently of all others. Although this simplifying assumption is typical of most clickstream models in the literature (Richardson et al. 2007, Chapelle and Zhang 2009, Zhu et al. 2010, Gao et al. 2022), it is worth investigating ways in which correlations can be introduced into our model. In Appendix EC.7.2, we describe the correlations between click events in the Alibaba clickstream data. We find that customers' clicks tend to be slightly positively correlated across the offered products, even after controlling for certain observed factors. Although

our click-based MNL model achieves a robust predictive performance on this data set, it may be valuable to relax the assumption of click independence in future research to enhance the predictive power and/or broaden the applicability of this model. A more flexible correlation structure may be relevant when considering larger assortments, compared to Alibaba's product recommendation setting. We mention in passing that our model can be generalized by incorporating the following positive correlation structure: customers do not click on any product with probability p_{no} , and with the residual probability $(1 - p_{no})$, their behavior is described by a click-based MNL model with independent clicks. Our algorithmic results extend to this more general setting.

Contextual assortment-dependent effects. A second fundamental assumption underlying the click-based MNL model is that the consideration probabilities and MNL-based preference weights are assortment-independent, i.e., they are not a function of the offered assortment. Again, this assumption is typical in the choice-based revenue management literature. That said, there have been a handful of attempts to relax this assumption, leading up to so-called contextual choice models. For example, Lo and Topaloglu (2019) incorporate pairwise interaction effects into the standard MNL model to capture synergies between offered products. Adding modeling elements of this flavor to the click-based MNL model represents an intriguing direction for future work.

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Online Companion

EC.1. The Connection between the Click-Based MNL and the Latent Class MNL Models

The goal of this section is to formalize the connection between the click-based MNL and latent class MNL choice models. Specifically, we first show how to represent any click-based MNL model in terms of an equivalent latent class MNL model. We then argue that since the resulting latent class MNL model requires specifying exponentially-many customer segments, existing approaches for assortment optimization in this context are computationally impractical, except possibly for very small scale instances.

To begin, recall that under the latent class MNL model, the customer population is partitioned into K customer segments, where each segment $k \in [K]$ is associated with an arrival probability θ_k and a vector of MNL-based preference weights $w_1^k, \dots, w_n^k$. The arrival probability of each segment can be interpreted as the fraction of customers belonging to this segment. In this context, the choice probability $\pi(i, S)$ of item i out of the assortment S is given by:

$$\pi(i, S) = \sum_{k \in [K]} \theta_k \cdot \frac{w_i^k}{1 + w_k(S)},$$

where $w_k(S) = \sum_{i \in S} w_i^k$.

We can convert any click-based MNL model with consideration probabilities $\{\lambda_i\}_{i \in [n]}$ and MNL-based preference weights $\{w_i\}_{i \in [n]}$ to a choice-equivalent latent class MNL model as follows. First, we introduce a customer segment for each subset of items $C \subseteq [n]$. The stochastic nature of the consideration set formation phase in the click-based MNL model is captured by defining the arrival probability of segment C as

$$\theta_C = \prod_{i \in C} \lambda_i \cdot \prod_{i \notin C} (1 - \lambda_i).$$

The preference weights for customers of segment C are defined as $w_i^C = w_i$ if $i \in C$, and $w_i^C = 0$ otherwise. Under this specification, the choice probability of item i out of the assortment S is

$$\begin{aligned} \pi(i, S) &= \sum_{C \subseteq [n]: i \in C} \theta_C \cdot \frac{w_i^C}{1 + w_C(S)} \\ &= \sum_{C \subseteq S: i \in C} \left(\prod_{j \in C} \lambda_j \cdot \prod_{j \notin S \setminus C} (1 - \lambda_j) \right) \cdot \frac{w_i}{1 + w(C)} \\ &= \lambda_i \cdot \sum_{C \subseteq S \setminus \{i\}} \Pr[C_S^{-i} = C] \cdot \frac{w_i}{1 + w_i + w(C)} \\ &= \lambda_i \cdot \mathbb{E} \left[\frac{w_i}{1 + w_i + w(C_S^{-i})} \right]. \end{aligned}$$

The important observation is that the last expression is exactly the choice probability dictated by the click-based MNL model, as specified in Section 1.1. Hence, the click-based MNL model can be viewed as a parameterized latent class MNL model in the sense that its $2n$ parameters fully specify a latent class MNL model over 2^n customer segments.

Given that there are a handful of previously developed approaches to tackle the assortment problem under a latent class MNL model, it is natural to wonder if any of these algorithms can be combined with the above-mentioned reduction to attain any non-trivial guarantees for the click-based MNL model. Unfortunately, there is little merit in this idea, since the click-based MNL model's representation as a latent class MNL model requires creating 2^n customer classes. In addition, it is worth pointing out that even with polynomially-many segments, the latent class MNL-based assortment optimization problem is known to be $\Omega(n^{1-\epsilon})$ -hard to approximate, as shown by Désir et al. (2022).

EC.2. Proofs from Section 2

EC.2.1. Proof of Theorem 1

To prove the desired result, we present a reduction from set partition, which is one of Karp's 21 NP-complete problems (Karp 1972). Here, we are given a collection of positive integers $w_1, \dots, w_n$. The partition problem asks whether these numbers can be divided into two subsets, each summing to precisely $L = \frac{1}{2} \sum_{i=1}^n w_i$.

Given a partition instance of this form, we define an instance of the assortment feasibility problem as follows:

- There are $n + 1$ products.
- For every $i \in [n]$, the revenue of product i is given by $r_i = 2$. The revenue of product $n + 1$ is $r_{n+1} = 7$.
- For every $i \in [n]$, the preference weight of product i is w_i , while that of product $n + 1$ is $4L$. In addition, the no-purchase option is associated with a preference weight of L .
- Each of the products $1, \dots, n$ has a consideration probability of 1, while product $n + 1$ is considered with probability $1/2$.
- Finally, the expected revenue threshold is $K = 3$.

In the remainder of this proof, we show that there exists a set of products to offer with an expected revenue of at least K if and only if there exists a vector $x \in \{0, 1\}^n$ for which $\sum_{i=1}^n w_i x_i = L$.

We first observe that, since product $n + 1$ has the largest revenue among all products, it is not difficult to show that offering this product cannot decrease the expected revenue of any set of products. Therefore, the residual question for the assortment feasibility problem is that of deciding whether one can pick a subset of the products $1, \dots, n$, along with the preselected product $n + 1$,

to obtain an expected revenue of at least $K = 3$. For this purpose, we use the binary vector $x = (x_1, \dots, x_n)$ to indicate our product choices, meaning that $x_i = 1$ if product i is offered and $x_i = 0$ otherwise. Letting $W(x) = \sum_{i=1}^n w_i x_i$ and recalling that product $n+1$ is preselected, the expected revenue of the assortment decisions represented by x can be written as

$$\frac{1}{2} \cdot \frac{2W(x)}{L + W(x)} + \frac{1}{2} \cdot \frac{2W(x) + 28L}{L + W(x) + 4L} = \frac{W(x)}{L + W(x)} + \frac{W(x) + 14L}{5L + W(x)}.$$

Therefore, there exists a set of products that provides an expected revenue of at least $K = 3$ if and only if there exists a vector $x \in \{0, 1\}^n$ for which

$$\frac{W(x)}{L + W(x)} + \frac{W(x) + 14L}{5L + W(x)} \geq 3.$$

The inequality above can equivalently be rewritten as $(W(x) - L)^2 \leq 0$. Therefore, an expected revenue of at least $K = 3$ can be obtained if and only if there exists a vector $x \in \{0, 1\}^n$ for which $W(x) = L$, which is precisely what the partition problem is interested in.

EC.2.2. Proof of Theorem 2

The probabilistic question. In order to establish this result, we consider a slightly more general setting. Specifically, let $X_1, \dots, X_n$ be a collection of n independent Bernoulli random variables; we use λ_i to denote the success probability of X_i . In addition, let $w_1, \dots, w_n$ be non-negative weights, and consider the random variable $W = \sum_{i \in [n]} w_i X_i$. Given $\beta \geq w_{\min}$, we wish to approximately estimate the expectation $\mathbb{E}[\frac{1}{\beta + W}]$. It is easy to see that addressing the latter question would immediately provide the required estimate for $\pi(i, S)$. Indeed, due to representation (1) of the choice probabilities, where we have shown that $\pi(i, S) = \lambda_i w_i \cdot \mathbb{E}[\frac{1}{1 + w_i + w(C_S^{-i})}]$, the connection is made by setting $\beta = 1 + w_i$ and $W = \sum_{j \in S \setminus \{i\}} w_j X_j$.

The continuous dynamic program. Now let us first introduce the following notation:

- For an integer $0 \leq t \leq n$, let $W_t = \sum_{i \in [t]} w_i X_i$.
- For $\alpha \in [\beta, \beta + \sum_{i=t+1}^n w_i]$, let $F(t, \alpha) = \mathbb{E}[\frac{1}{\alpha + W_t}]$.

Clearly, our objective is to estimate $F(n, \beta) = \mathbb{E}[\frac{1}{\beta + W_n}] = \mathbb{E}[\frac{1}{\beta + W}]$. With this notation, we argue that the value function F can be expressed in a recursive way:

- Initialization: For $t = 0$ and $\alpha \in [\beta, \beta + \sum_{i=1}^n w_i]$, we have $F(0, \alpha) = \frac{1}{\alpha}$.
- General step: For $t = 1, \dots, n$ and $\alpha \in [\beta, \beta + \sum_{i=t+1}^n w_i]$, we have

$$\begin{aligned} F(t, \alpha) &= \mathbb{E} \left[\frac{1}{\alpha + W_t} \right] \\ &= \mathbb{E} \left[\frac{1}{\alpha + W_{t-1} + w_t X_t} \right] \\ &= \Pr[X_t = 1] \cdot \mathbb{E} \left[\frac{1}{\alpha + W_{t-1} + w_t X_t} \middle| X_t = 1 \right] \end{aligned}$$

$$\begin{aligned}
& + \Pr[X_t = 0] \cdot \mathbb{E} \left[\frac{1}{\alpha + W_{t-1} + w_t X_t} \middle| X_t = 0 \right] \\
& = \lambda_t \cdot \mathbb{E} \left[\frac{1}{\alpha + w_t + W_{t-1}} \right] + (1 - \lambda_t) \cdot \mathbb{E} \left[\frac{1}{\alpha + W_{t-1}} \right] \\
& = \lambda_t \cdot F(t-1, \alpha + w_t) + (1 - \lambda_t) \cdot F(t-1, \alpha) .
\end{aligned} \tag{EC.1}$$

It is worth noting that the parameter α is continuous, meaning that the characterization above is still not an algorithmic result.

Efficient discretization. In order to discretize the continuous dynamic program F , we define an approximate version $\tilde{F}$. Here, the parameter α is restricted to the set $\mathcal{E}$, consisting of all numbers in $[\beta, \beta + \sum_{i=1}^n w_i]$ that can be written as $\beta \cdot (1 + \delta)^k$ for some integer $k \geq 0$, where $\delta = \frac{\epsilon}{2(n+1)}$. It is easy to verify that $|\mathcal{E}| = O(\frac{1}{\delta} \cdot \log(n \frac{w_{\max}}{w_{\min}})) = O(\frac{n}{\epsilon} \cdot \log(n \frac{w_{\max}}{w_{\min}}))$, since $\beta \geq w_{\min}$.

To define our modified recursive equations, let $\lfloor \cdot \rfloor_{\mathcal{E}}$ be an operator that rounds its argument down to the nearest number in $\mathcal{E}$. We mention in passing that each occurrence of this operator in the upcoming analysis is indeed well-defined, due to applying it only to arguments in $[\beta, \beta + \sum_{i=1}^n w_i]$. With this notation, the approximate value function $\tilde{F}$ is defined as follows:

- Initialization: For $t = 0$ and $\alpha \in \mathcal{E}$, we set $\tilde{F}(0, \alpha) = \frac{1}{\alpha}$.
- General step: For $t = 1, \dots, n$ and $\alpha \in \mathcal{E} \cap [\beta, \beta + \sum_{i=t+1}^n w_i]$, we set

$$\tilde{F}(t, \alpha) = \lambda_t \cdot \tilde{F}(t-1, \lfloor \alpha + w_t \rfloor_{\mathcal{E}}) + (1 - \lambda_t) \cdot \tilde{F}(t-1, \alpha) .$$

In the next claim, we show that $\tilde{F}(n, \lfloor \beta \rfloor_{\mathcal{E}})$ over-estimates $F(n, \beta)$ by a factor of at most $1 + \epsilon$.

LEMMA EC.1. $F(n, \beta) \leq \tilde{F}(n, \lfloor \beta \rfloor_{\mathcal{E}}) \leq (1 + \epsilon) \cdot F(n, \beta)$.

Proof To establish the desired claim, we first show that for every $0 \leq t \leq n$ and $\alpha \in [\beta, \beta + \sum_{i=t+1}^n w_i]$,

$$F(t, \alpha) \leq \tilde{F}(t, \lfloor \alpha \rfloor_{\mathcal{E}}) \leq (1 + \delta)^{t+1} \cdot F(t, \alpha) , \tag{EC.2}$$

by induction on t . The base case is $t = 0$, where $\tilde{F}(0, \lfloor \alpha \rfloor_{\mathcal{E}}) = \frac{1}{\lfloor \alpha \rfloor_{\mathcal{E}}}$ and $F(0, \alpha) = \frac{1}{\alpha}$, clearly indicating that $F(0, \alpha) \leq \tilde{F}(0, \lfloor \alpha \rfloor_{\mathcal{E}}) \leq (1 + \delta) \cdot F(0, \alpha)$.

Now, for the general case of $t \geq 1$, note that

$$\begin{aligned}
\tilde{F}(t, \lfloor \alpha \rfloor_{\mathcal{E}}) & = \lambda_t \cdot \tilde{F}(t-1, \lfloor \lfloor \alpha \rfloor_{\mathcal{E}} + w_t \rfloor_{\mathcal{E}}) + (1 - \lambda_t) \cdot \tilde{F}(t-1, \lfloor \alpha \rfloor_{\mathcal{E}}) \\
& \geq \lambda_t \cdot F(t-1, \lfloor \alpha \rfloor_{\mathcal{E}} + w_t) + (1 - \lambda_t) \cdot F(t-1, \alpha) \\
& = \lambda_t \cdot \mathbb{E} \left[\frac{1}{\lfloor \alpha \rfloor_{\mathcal{E}} + w_t + W_{t-1}} \right] + (1 - \lambda_t) \cdot F(t-1, \alpha) \\
& \geq \lambda_t \cdot \mathbb{E} \left[\frac{1}{\alpha + w_t + W_{t-1}} \right] + (1 - \lambda_t) \cdot F(t-1, \alpha) \\
& = \lambda_t \cdot F(t-1, \alpha + w_t) + (1 - \lambda_t) \cdot F(t-1, \alpha)
\end{aligned}$$

$$= F(t, \alpha) ,$$

where the first inequality follows from the induction hypothesis, the second and third equalities hold by definition of $F(t-1, \cdot)$, and the last equality is precisely the recursive equation (EC.1). Using similar arguments in the opposite direction, we have

$$\begin{aligned} \tilde{F}(t, \lfloor \alpha \rfloor_{\mathcal{E}}) &= \lambda_t \cdot \tilde{F}(t-1, \lfloor \lfloor \alpha \rfloor_{\mathcal{E}} + w_t \rfloor_{\mathcal{E}}) + (1 - \lambda_t) \cdot \tilde{F}(t-1, \lfloor \alpha \rfloor_{\mathcal{E}}) \\ &\leq (1 + \delta)^t \cdot (\lambda_t \cdot F(t-1, \lfloor \alpha \rfloor_{\mathcal{E}} + w_t) + (1 - \lambda_t) \cdot F(t-1, \alpha)) \\ &= (1 + \delta)^t \cdot \left(\lambda_t \cdot \mathbb{E} \left[\frac{1}{\lfloor \alpha \rfloor_{\mathcal{E}} + w_t + W_{t-1}} \right] + (1 - \lambda_t) \cdot F(t-1, \alpha) \right) \\ &\leq (1 + \delta)^t \cdot \left(\lambda_t \cdot (1 + \delta) \cdot \mathbb{E} \left[\frac{1}{\alpha + w_t + W_{t-1}} \right] + (1 - \lambda_t) \cdot F(t-1, \alpha) \right) \\ &\leq (1 + \delta)^{t+1} \cdot (\lambda_t \cdot F(t-1, \alpha + w_t) + (1 - \lambda_t) \cdot F(t-1, \alpha)) \\ &= (1 + \delta)^{t+1} \cdot F(t, \alpha) . \end{aligned}$$

Based on inequality (EC.2), it follows that $F(n, \beta) \leq \tilde{F}(n, \lfloor \beta \rfloor_{\mathcal{E}})$ regardless of the value of δ . In addition, since we have previously chosen $\delta = \frac{\epsilon}{2(n+1)}$, an upper bound on our over-estimation error is obtained by observing that

$$\frac{\tilde{F}(n, \lfloor \beta \rfloor_{\mathcal{E}})}{F(n, \beta)} \leq (1 + \delta)^{n+1} \leq e^{(n+1) \cdot \delta} = e^{\epsilon/2} \leq 1 + \epsilon ,$$

where the last inequality holds since $\epsilon \in (0, 1)$. $\square$

To conclude our proof, it remains to upper bound the number of states needed in order to compute $\tilde{F}(n, \lfloor \beta \rfloor_{\mathcal{E}})$. The size of the state space is $O(n \cdot |\mathcal{E}|)$, and the approximate value function for each such state can be evaluated in $O(1)$ time. Consequently, the overall running time is $O(n \cdot |\mathcal{E}|) = O(\frac{n^2}{\epsilon} \cdot \log(n \frac{w_{\max}}{w_{\min}}))$, precisely as stated in Theorem 2. $\square$

EC.2.3. Derandomization

Given a decomposable assortment B , we show how to efficiently compute a deterministic assortment $S_d \subseteq [n]$ with an expected revenue of $\mathcal{R}(S_d) \geq \mathbb{E}_{S \sim B}[\mathcal{R}(S)]$. To this end, we apply the method of conditional expectation (Alon and Spencer 2016, Chap. 16) with respect to the sequence of random variables $(B_1, \dots, B_n)$. The idea is to iteratively replace each random variable by one of its possible realizations, without decreasing the conditional expected revenue in each iteration, given the set of deterministic values chosen thus far.

In order to apply this derandomization method, one should just be able to compute the expected revenue generated by a decomposable assortment. Indeed, for any binary sequence $(b_1, \dots, b_k) \in \{0, 1\}^k$ with $k \in [0, n]$, we wish to compute the expected revenue $\mathbb{E}_{S \sim B^k}[\mathcal{R}(S)]$, where $B^k = (b_1, \dots, b_k, B_{k+1}, \dots, B_n)$. In turn, we are left with the task of computing the choice probability

of each item $i \in [n]$ under the decomposable assortment B^k , denoted by $\mathbb{E}_{S \sim B^k}[\pi(i, S)]$. The key observation is that our FPTAS for computing the choice probabilities with respect to deterministic assortments, described in Theorem 2, can be readily leveraged to estimate $\mathbb{E}_{S \sim B^k}[\pi(i, S)]$. Indeed, in Claim EC.1 below, we show that computing the choice probabilities of a decomposable assortment is equivalent to computing the choice probabilities of a deterministic assortment with modified consideration probabilities. Specifically, for every item $i \in [n]$, we define $\hat{\lambda}_i = \lambda_i \cdot \Pr[B_i^k = 1]$. Henceforth, we denote by $\hat{\pi}(i, S)$ the choice probability of item i in the assortment S with respect to the modified consideration probabilities $(\hat{\lambda}_1, \dots, \hat{\lambda}_n)$.

CLAIM EC.1. $\mathbb{E}_{S \sim B^k}[\pi(i, S)] = \hat{\pi}(i, [n])$.

Proof. The equivalence between the choice probabilities proceeds from the observation that:

$$\begin{aligned} \Pr_{S \sim B^k}[i \in C_S] &= \lambda_i \cdot \Pr_{S \sim B^k}[i \in S] \\ &= \lambda_i \cdot \Pr[B_i^k = 1] \\ &= \hat{\lambda}_i \\ &= \Pr[i \in \hat{C}_{[n]}] , \end{aligned}$$

where $\hat{C}_{[n]}$ is the consideration set induced by the assortment $[n]$ under the modified consideration probabilities $\hat{\lambda}_i$. By remarking that $\mathbb{I}[1 \in C_S], \dots, \mathbb{I}[n \in C_S]$ and $\mathbb{I}[1 \in \hat{C}_{[n]}], \dots, \mathbb{I}[n \in \hat{C}_{[n]}]$ are sequences of independent Bernoulli random variables, we infer from the above equality that $w(C_S^{-i})$ and $w(\hat{C}_{[n]}^{-i})$ follow the same distribution when $S \sim B^k$, for every $i \in [n]$. Hence, based on representation (1) of the choice probabilities, we obtain that

$$\begin{aligned} \mathbb{E}_{S \sim B^k}[\pi(i, S)] &= \Pr_{S \sim B^k}[i \in C_S] \cdot \mathbb{E}_{S \sim B^k} \left[\frac{w_i}{1 + w_i + w(C_S^{-i})} \right] \\ &= \Pr[i \in \hat{C}_{[n]}] \cdot \mathbb{E} \left[\frac{w_i}{1 + w_i + w(\hat{C}_{[n]}^{-i})} \right] \\ &= \hat{\pi}(i, [n]) . \end{aligned}$$

□

EC.3. Proofs from Section 3

EC.3.1. Proof of Lemma 1

The proof of this result proceeds in two steps, where we separately analyze the effects of our rounding procedures. First, we examine the effects of utilizing the rounded weights $\{\tilde{w}_i\}_{i \in [n]}$ and selling prices $\{\tilde{r}_i\}_{i \in [n]}$, keeping the original consideration probabilities $\{\lambda_i\}_{i \in [n]}$ unchanged. We use $\mathcal{R}_{\tilde{w}, \tilde{r}}(\cdot)$ to denote the resulting expected revenue function. The following claim is established in Appendix EC.3.1.1.

CLAIM EC.2. *For every $S \subseteq [n]$, we have $(1 - \epsilon) \cdot \mathcal{R}(S) \leq \mathcal{R}_{\tilde{w}, \tilde{r}}(S) \leq (1 + 2\epsilon) \cdot \mathcal{R}(S)$.*

Next, suppose that in addition to considering the rounded weights $\{\tilde{w}_i\}_{i \in [n]}$ and selling prices $\{\tilde{r}_i\}_{i \in [n]}$, we also consider the rounded consideration probabilities $\{\tilde{\lambda}_i\}_{i \in [n]}$ in place of $\{\lambda_i\}_{i \in [n]}$. Recall that $\tilde{\mathcal{R}}(\cdot)$ denotes the resulting expected revenue function. Our analysis proceeds from the next two claims, whose respective proofs appear in Appendices EC.3.1.2 and EC.3.1.3.

CLAIM EC.3. *$\tilde{\mathcal{R}}(S) \geq (1 - \epsilon) \cdot \mathcal{R}_{\tilde{w}, \tilde{r}}(S)$, for every $S \subseteq [n]$.*

CLAIM EC.4. *$\mathbb{E}_{S' \sim BS}[\mathcal{R}_{\tilde{w}, \tilde{r}}(S')] \geq (1 - \epsilon) \cdot \tilde{\mathcal{R}}(S)$, for every $S \subseteq [n]$.*

To conclude the proof of Lemma 1, note that

$$\mathbb{E}_{S' \sim BS}[\mathcal{R}(S')] \geq (1 - 2\epsilon) \cdot \mathbb{E}_{S' \sim BS}[\mathcal{R}_{\tilde{w}, \tilde{r}}(S')] \geq (1 - 3\epsilon) \cdot \tilde{\mathcal{R}}(S) ,$$

where the former inequality is due to Claim EC.2 and the latter follows from Claim EC.4. Reciprocally, we have

$$\tilde{\mathcal{R}}(S) \geq (1 - \epsilon) \cdot \mathcal{R}_{\tilde{w}, \tilde{r}}(S) \geq (1 - 2\epsilon) \cdot \mathcal{R}(S) ,$$

where the inequalities above are derived by applying Claims EC.3 and EC.2 in succession.

EC.3.1.1. Proof of Claim EC.2. Let $\pi_w(i, S)$ be the choice probability of item i out of the assortment S , with respect to the original weight parameters w . The analogous choice probability $\pi_{\tilde{w}}(i, S)$ with respect to $\tilde{w}$ is defined similarly, noting that both quantities are independent of the price parameters. In order to establish the desired claim, it suffices to prove that, for every item $i \in S$,

$$(1 - \epsilon) \cdot r_i \cdot \pi_w(i, S) \leq \tilde{r}_i \cdot \pi_{\tilde{w}}(i, S) \leq (1 + 2\epsilon) \cdot r_i \cdot \pi_w(i, S) .$$

To obtain the first inequality, rather than exploiting representation (1) of the choice probabilities, we express the latter by conditioning on how the consideration set C_S is realized. Based on this idea, we have

$$\begin{aligned} \tilde{r}_i \cdot \pi_{\tilde{w}}(i, S) &= \tilde{r}_i \cdot \sum_{T \subseteq S: i \in T} \Pr[C_S = T] \cdot \frac{\tilde{w}_i}{1 + \tilde{w}(T)} \\ &\geq r_i \cdot \sum_{T \subseteq S: i \in T} \Pr[C_S = T] \cdot \frac{w_i}{1 + \epsilon + (1 + \epsilon) \cdot w(T)} \\ &\geq (1 - \epsilon) \cdot r_i \cdot \sum_{T \subseteq S: i \in T} \Pr[C_S = T] \cdot \frac{w_i}{1 + w(T)} \\ &= (1 - \epsilon) \cdot r_i \cdot \pi_w(i, S) , \end{aligned}$$

where the first inequality holds since $\tilde{r}_i \tilde{w}_i = r_i w_i$ and since

$$\tilde{w}(T) = \sum_{i \in T \cap \mathcal{W}_0} \tilde{w}_i + \sum_{q \geq 1} \sum_{i \in T \cap \mathcal{W}_q} \tilde{w}_i \leq \frac{|T \cap \mathcal{W}_0|}{n} \cdot \epsilon + (1 + \epsilon) \cdot \sum_{q \geq 1} \sum_{i \in T \cap \mathcal{W}_q} w_i \leq \epsilon + (1 + \epsilon) \cdot w(T) .$$

Using similar arguments in the opposite direction, we have

$$\begin{aligned}
\tilde{r}_i \cdot \pi_{\tilde{w}}(i, S) &= \tilde{r}_i \cdot \sum_{T \subseteq S: i \in T} \Pr[C_S = T] \cdot \frac{\tilde{w}_i}{1 + \tilde{w}(T)} \\
&\leq r_i \cdot \sum_{T \subseteq S: i \in T} \Pr[C_S = T] \cdot \frac{w_i}{1 + w(T) - \epsilon} \\
&\leq (1 + 2\epsilon) \cdot r_i \cdot \sum_{T \subseteq S: i \in T} \Pr[C_S = T] \cdot \frac{w_i}{1 + w(T)} \\
&= (1 + 2\epsilon) \cdot r_i \cdot \pi_w(i, S) ,
\end{aligned}$$

Here, the first inequality holds since

$$\tilde{w}(T) \geq \sum_{q \geq 1} \sum_{i \in T \cap \mathcal{W}_q} \tilde{w}_i \geq \sum_{q \geq 1} \sum_{i \in T \cap \mathcal{W}_q} w_i + \left(\sum_{i \in T \cap \mathcal{W}_0} w_i - \epsilon \right) = w(T) - \epsilon .$$

EC.3.1.2. Proof of Claim EC.3. Let $\tilde{\pi}(i, S)$ be the choice probability of item i out of the assortment S with respect to the rounded parameters. To prove the desired claim, it suffices to show that $\tilde{\pi}(i, S) \geq (1 - \epsilon) \cdot \pi_{\tilde{w}}(i, S)$. For this purpose, we utilize representation (1) of the choice probabilities to obtain

$$\tilde{\pi}(i, S) = \tilde{\lambda}_i \cdot \mathbb{E} \left[\frac{\tilde{w}_i}{1 + \tilde{w}_i + \tilde{w}(\tilde{C}_S^{-i})} \right] ,$$

where $\tilde{C}_S^{-i}$ represents the random consideration set induced by $S \setminus \{i\}$ with respect to the rounded consideration probabilities $\{\tilde{\lambda}_i\}_{i \in [n]}$. Consequently, we have

$$\begin{aligned}
\tilde{\pi}(i, S) &= \tilde{\lambda}_i \cdot \mathbb{E} \left[\frac{\tilde{w}_i}{1 + \tilde{w}_i + \tilde{w}(\tilde{C}_S^{-i})} \right] \\
&\geq (1 - \epsilon) \cdot \lambda_i \cdot \mathbb{E} \left[\frac{\tilde{w}_i}{1 + \tilde{w}_i + \tilde{w}(\tilde{C}_S^{-i})} \right] \\
&\geq (1 - \epsilon) \cdot \lambda_i \cdot \mathbb{E} \left[\frac{\tilde{w}_i}{1 + \tilde{w}_i + \tilde{w}(C_S^{-i})} \right] \\
&= (1 - \epsilon) \cdot \pi_{\tilde{w}}(i, S) .
\end{aligned}$$

Here, the first inequality follows from the construction of the rounded consideration probabilities. The second inequality proceeds from the stochastic orderings $\tilde{w}(\tilde{C}_S^{-i}) \sim \sum_{j \in S \setminus \{i\}} \tilde{w}_j \cdot B(\tilde{\lambda}_j) \preceq \sum_{j \in S \setminus \{i\}} \tilde{w}_j \cdot B(\lambda_j) \sim \tilde{w}(C_S^{-i})$, where the inequality holds since the usual stochastic order is closed under convolution and since $\tilde{\lambda}_j \leq \lambda_j$ for every $j \in S \setminus \{i\}$.

EC.3.1.3. Proof of Claim EC.4. To prove the desired claim, it is sufficient to show that $\mathbb{E}_{S' \sim BS}[\pi_{\tilde{w}}(i, S')] \geq (1 - \epsilon) \cdot \tilde{\pi}(i, S)$. To this end, we note that

$$\mathbb{E}_{S' \sim BS}[\pi_{\tilde{w}}(i, S')] = \mathbb{E}_{S' \sim BS} \left[(1 - \epsilon) \cdot \lambda_i \cdot \mathbb{E} \left[\frac{\tilde{w}_i}{1 + \tilde{w}_i + \tilde{w}(C_{S'}^{-i})} \right] \right]$$

$$\begin{aligned}
&\geq \lambda_i \cdot \mathbb{E} \left[\frac{\tilde{w}_i}{1 + \tilde{w}_i + \tilde{w}(\tilde{C}_S^{-i})} \right] \\
&\geq (1 - \epsilon) \cdot \tilde{\lambda}_i \cdot \mathbb{E} \left[\frac{\tilde{w}_i}{1 + \tilde{w}_i + \tilde{w}(\tilde{C}_S^{-i})} \right] \\
&= (1 - \epsilon) \cdot \tilde{\pi}(i, S) ,
\end{aligned}$$

where the equalities above follow from representation (1) of the choice probabilities. The first inequality proceeds from the stochastic ordering $\tilde{w}(C_{S'}^{-i}) \sim \sum_{j \in S \setminus \{i\}} \tilde{w}_j \cdot B((1 - \epsilon) \cdot \lambda_j) \preceq \sum_{j \in S \setminus \{i\}} \tilde{w}_j \cdot B(\tilde{\lambda}_j) \sim \tilde{w}(\tilde{C}_S^{-i})$, where S' is the random realization of the decomposable assortment B^S . Here, the stochastic inequality holds since the usual stochastic order is closed under convolution and since $\tilde{\lambda}_j \geq (1 - \epsilon) \cdot \lambda_j$ for every $j \in S \setminus \{i\}$.

EC.3.2. Proof of Lemma 2

The proof proceeds by defining the parameters $(\{\hat{c}_q, \hat{\alpha}_q\}_{q \in \mathcal{Q}}, \{\hat{c}_{p,q}\}_{(p,q) \in [P_{\max}] \times \mathcal{Q}})$ in a way that satisfies the properties stated in Lemma 2. Subsequently, we explain how these parameters can be determined through efficient enumeration.

Defining the input parameters. Let $\hat{\mathcal{R}}^*$ be an under-estimate of the optimal expected revenue $\mathcal{R}(S^*)$ such that $\hat{\mathcal{R}}^* \leq \mathcal{R}(S^*) \leq 2 \cdot \hat{\mathcal{R}}^*$. By noting that $\mathcal{R}(\{i^*\}) \leq \mathcal{R}(S^*) \leq n \cdot \mathcal{R}(\{i^*\})$, where i^* is the item that maximizes $\mathcal{R}(\{i\})$ over all items $i \in [n]$, we pick $\hat{\mathcal{R}}^*$ within the set $\{2^s \cdot \mathcal{R}(\{i^*\}) : 0 \leq s \leq \lceil \log_2 n \rceil\}$.

- *Revenue contribution vectors:* We approximately estimate the revenue contributions given by $\{c_{p,q}^*\}_{(p,q) \in [P_{\max}] \times \mathcal{Q}}$ and $\{c_q^*\}_{q \in \mathcal{Q}}$, through the respective quantities $\{\hat{c}_{p,q}\}_{(p,q) \in [P_{\max}] \times \mathcal{Q}}$ and $\{\hat{c}_q\}_{q \in \mathcal{Q}}$. These vectors are jointly defined as follows. Letting $\delta = \frac{\epsilon}{\mathcal{L}} \cdot \hat{\mathcal{R}}^*$, where $\mathcal{L} = (P_{\max} + 1) \cdot |\mathcal{Q}|$, for every $p \in [P_{\max}]$ and $q \in \mathcal{Q}$ the quantity $\hat{c}_{p,q}$ is of the form $\hat{c}_{p,q} = n_{p,q} \cdot \delta$, where $n_{p,q}$ is the unique integer for which

$$\hat{c}_{p,q} = n_{p,q} \cdot \delta \leq c_{p,q}^* < (n_{p,q} + 1) \cdot \delta = \hat{c}_{p,q} + \delta . \quad (\text{EC.3})$$

Similarly, for every $q \in \mathcal{Q}$, the approximate quantity $\hat{c}_q$ is of the form $\hat{c}_q = n_q \cdot \delta$, where n_q is the unique integer for which

$$\hat{c}_q = n_q \cdot \delta \leq c_q^* < (n_q + 1) \cdot \delta = \hat{c}_q + \delta . \quad (\text{EC.4})$$

- *Coefficient vector:* We approximately estimate the coefficients $\alpha(S^*, q) = \mathbb{E}[\frac{w^{(q)}}{1 + w^{(q)} + w(C_{S^*})}]$ of the unlikely items $\mathcal{U}_q$ in the optimal assortment S^* for every $q \in \mathcal{Q}$, through the quantities $\{\hat{\alpha}_q\}_{q \in \mathcal{Q}}$. Letting $\beta = \frac{1}{1 + (n+1) \cdot w_{\max}}$, the estimated quantity $\hat{\alpha}_q$ is of the form $\hat{\alpha}_q = w^{(q)} \beta \cdot (1 + \epsilon)^{r_q}$, where r_q is the unique integer for which

$$(1 - \epsilon)^2 \cdot \hat{\alpha}_q \leq \alpha(S^*, q) \leq (1 - \epsilon) \cdot \hat{\alpha}_q . \quad (\text{EC.5})$$

Inequalities (EC.3)-(EC.5) immediately imply that the input parameters $(\{\hat{c}_q, \hat{\alpha}_q\}_{q \in \mathcal{Q}})$ and $(\{\hat{c}_{p,q}\}_{(p,q) \in [P_{\max}] \times \mathcal{Q}})$ satisfy Properties 1-3, as stated in Lemma 2.

Efficient enumeration. Now, we argue these estimates can be determined by enumerating over polynomially many distinct vectors. To this end, we bound the cardinality of the set Ω formed by all possible input vectors of the form specified in (EC.3)-(EC.5). We first observe that $\hat{\mathcal{R}}^*$ is chosen within a set of $O(\log n)$ distinct values. In order to obtain an upper bound on the number of distinct contribution vectors, we observe that

$$\sum_{p=1}^{P_{\max}} \sum_{q=Q_{\min}}^{Q_{\max}} n_{p,q} + \sum_{q=Q_{\min}}^{Q_{\max}} n_q \leq \frac{2\mathcal{L}}{\epsilon},$$

where the inequality follows from (EC.3) and (EC.4), noting that $\sum_{p=1}^{P_{\max}} \sum_{q=Q_{\min}}^{Q_{\max}} c_{p,q}^* + \sum_{q=Q_{\min}}^{Q_{\max}} c_q^* = \mathcal{R}(S^*) \leq 2 \cdot \hat{\mathcal{R}}^*$. Consequently, the vector formed by $\{n_{p,q}\}_{(p,q) \in [P_{\max}] \times \mathcal{Q}}$ and $\{n_q\}_{q \in \mathcal{Q}}$ can be viewed as a partition of some integer $\mu \in [0, \frac{2\mathcal{L}}{\epsilon}]$ into precisely $\mathcal{L} = |\mathcal{Q}| \cdot (P_{\max} + 1)$ integers. Therefore, the contributions vector can be recovered by enumerating over $O(2^{O(\frac{\mathcal{L}}{\epsilon})}) = O(n^{O(\frac{1}{\epsilon^4} \log \frac{1}{\epsilon})})$ distinct partitions, since $\mathcal{L} = O(\frac{1}{\epsilon^3} \log \frac{1}{\epsilon} \log n)$.

In order to derive an upper bound on the number of distinct coefficient vectors, we remark that the quantity $\frac{\alpha(S^*, q)}{w(q)} = \mathbb{E}[\frac{1}{1+w(q)+w(C_{S^*})}]$ is monotone non-increasing in $q \in \mathcal{Q}$, and it is contained in the range $[\beta, \frac{1}{1+w_{\min}}]$, where $\beta = \frac{1}{1+(n+1)w_{\max}}$. Thus, by inequality (EC.5), the corresponding sequence of exponents $\{r_q\}_{q=Q_{\min}}^{Q_{\max}}$ within our estimates $\hat{\alpha}_q = w^{(q)} \beta \cdot (1 + \epsilon)^{r_q}$ is monotone non-increasing, and contained in the interval $[\frac{1}{\epsilon} \cdot (\log(n+1) + |\mathcal{Q}|)]$. Each such sequence is uniquely determined by a partition of an integer $\mu \in [\frac{1}{\epsilon} \cdot (\log(n+1) + |\mathcal{Q}|)]$ into precisely $|\mathcal{Q}|$ integers, meaning that there are only $O(2^{O(\frac{1}{\epsilon} \log n + \frac{1}{\epsilon^3} \log n)}) = O(n^{O(\frac{1}{\epsilon^3})})$ distinct candidates since $|\mathcal{Q}| = O(\frac{1}{\epsilon} \log \frac{w_{\max}}{w_{\min}}) = O(\frac{1}{\epsilon^2} \log n)$ by Assumption 1.

All in all, we have shown that parameters $(\{\hat{c}_q, \hat{\alpha}_q\}_{q \in \mathcal{Q}})$ and $(\{\hat{c}_{p,q}\}_{(p,q) \in [P_{\max}] \times \mathcal{Q}})$ can be recovered by enumerating over a collection Ω of cardinality $|\Omega| = O(n^{O(\frac{1}{\epsilon^4} \log \frac{1}{\epsilon})})$.

EC.3.3. Proof of Lemma 3

For $\mu \in [0, 1]$, we use $B(\mu)$ to denote a Bernoulli random variable with probability of success μ . We remind the reader that two random variables Z_1 and Z_2 are in the (standard) convex order $Z_1 \succeq_{\text{cx}} Z_2$ if, for every convex function ϕ , we have $\mathbb{E}[\phi(Z_1)] \geq \mathbb{E}[\phi(Z_2)]$, provided the expectations exist. It is worth noting that the convex order $Z_1 \succeq_{\text{cx}} Z_2$ implies in particular that the decreasing convex relationship $Z_1 \succeq_{\text{dcx}} Z_2$ holds.

Our proof proceeds from two basic claims relating $B(\mu)$ and $P(\mu)$ through stochastic orders. Claim EC.5 argues that these two random variables are convexly ordered.

CLAIM EC.5. $P(\mu) \succeq_{\text{cx}} B(\mu)$, for every $\mu \in [0, 1]$

This inequality is well-known; see, for example, Lemma 3 in the paper by Boutsikas and Vagge-latou (2002) who invoke a general criterion due to Karlin and Novikoff (1963). Since the convex

order is closed under convolution (e.g., see Thm. 3.A.12(d) in Shaked and Shanthikumar (2007)), Claim EC.5 implies that

$$P(\lambda) \sim \sum_{i=1}^k P(\lambda_i) \succeq_{\text{cx}} \sum_{i=1}^k B(\lambda_i) \sim \sum_{i=1}^k X_i \sim Y .$$

Since the usual convex order implies the decreasing convex order, we conclude that $P(\lambda) \succeq_{\text{dcx}} Y$.

It remains to show that $Y \succeq_{\text{dcx}} P((1+\epsilon) \cdot \lambda)$. To this end, when the Poisson arrival rate is made slightly larger than λ , we establish in the next claim a reverse inequality to that of Claim EC.5, according to the usual stochastic order.

CLAIM EC.6. *For every $\epsilon \in [0, \sqrt{2} - 1]$ and $\mu \in [0, \epsilon]$, we have $P((1+\epsilon) \cdot \mu) \succeq_{\text{st}} B(\mu)$.*

Proof To prove the claim, it suffices to show that $\Pr[P((1+\epsilon) \cdot \mu) = 0] \leq \Pr[B(\mu) = 0]$. For this purpose, note that

$$\begin{aligned} \Pr[P((1+\epsilon) \cdot \mu) = 0] &= e^{-(1+\epsilon) \cdot \mu} \\ &\leq 1 - (1+\epsilon) \cdot \mu + \frac{1}{2}(1+\epsilon)^2 \cdot \mu^2 \\ &\leq 1 - \mu \\ &= \Pr[B(\mu) = 0] , \end{aligned}$$

where the second inequality holds since $\epsilon \in [0, \sqrt{2} - 1]$ and $\mu \in [0, \epsilon]$. $\square$

By noting that the usual stochastic order implies a decreasing convex relationship in reverse order, we have

$$Y \sim \sum_{i=1}^k X_i \sim \sum_{i=1}^k B(\lambda_i) \succeq_{\text{dcx}} \sum_{i=1}^k P((1+\epsilon) \cdot \lambda_i) \sim P((1+\epsilon) \cdot \lambda) ,$$

where the stochastic inequality above follows from Claim EC.6 and the fact that the usual stochastic order is closed under convolution.

EC.3.4. Proof of Lemma 4

Focusing on an unlikely item $i \in S \cap \mathcal{U}_q$, and letting $p \in [P_{\min}, 0]$ be the unique index for which $i \in S \cap \mathcal{W}_q \cap \Lambda_p$, we first observe that

$$\begin{aligned} \pi(i, S) &= \lambda_i \cdot \mathbb{E} \left[\frac{w_i}{1 + w_i + w(C_S^{-i})} \right] \\ &\geq \lambda_i \cdot \mathbb{E} \left[\frac{w^{(q)}}{1 + w^{(q)} + w(C_S)} \right] \\ &= \lambda_i \cdot \alpha(S, q) , \end{aligned}$$

where the inequality above holds since $w_i = w^{(q)}$ and $w(C_S) \succeq_{\text{st}} w(C_S^{-i})$. In the opposite direction, we have

$$\begin{aligned} \lambda_i \cdot \alpha(S, q) &= \lambda_i \cdot \mathbb{E} \left[\frac{w^{(q)}}{1 + w^{(q)} + w(C_S)} \right] \\ &\geq \lambda_i \cdot \Pr[i \notin C_S] \cdot \mathbb{E} \left[\frac{w^{(q)}}{1 + w^{(q)} + w(C_S^{-i})} \middle| i \notin C_S \right] \\ &\geq \lambda_i \cdot (1 - \epsilon) \cdot \mathbb{E} \left[\frac{w_i}{1 + w_i + w(C_S^{-i})} \right] \\ &= (1 - \epsilon) \cdot \pi(i, S) , \end{aligned}$$

where the second inequality proceeds by noting that the random variables $\mathbb{I}[i \notin C_S]$ and $w(C_S^{-i})$ are independent, and that $\Pr[i \notin C_S] = 1 - \lambda_i \geq 1 - \epsilon$ since item i is unlikely.

EC.3.5. Proof of Lemma 5

We assume without loss of generality that items are indexed such that $r_1 \geq \dots \geq r_n$, and let $S^* \subseteq [n]$ be an assortment for which the quantity $\sum_{i \in S^*} i$ is minimized over all optimal assortments. We argue that S^* is necessarily revenue-ordered by class. To arrive at a contradiction, suppose that there exists a class $\mathcal{W}_q \cap \Lambda_p$ and a pair of items $i_1, i_2 \in \mathcal{W}_q \cap \Lambda_p$ such that $i_1 < i_2$, with $i_2 \in S^*$ and $i_1 \notin S^*$. In this case, we construct the altered assortment $\tilde{S} = (S^* \setminus \{i_2\}) \cup \{i_1\}$ where i_1 is swapped into S^* in place of i_2 . The important observation is that, since these items have precisely the same consideration probabilities and preference weights, which govern the random choice outcomes, we have $\pi(i_1, \tilde{S}) = \pi(i_2, S^*)$ as well as $\pi(j, \tilde{S}) = \pi(j, S^*)$ for every other item $j \in S^* \setminus \{i_2\}$. Thus, we obtain $\mathcal{R}(\tilde{S}) - \mathcal{R}(S^*) = r_{i_1} \cdot \pi(i_1, \tilde{S}) - r_{i_2} \cdot \pi(i_2, S^*) \geq 0$, since $r_{i_1} \geq r_{i_2}$ and $\pi(i_1, \tilde{S}) = \pi(i_2, S^*)$, meaning that the assortment $\tilde{S}$ is also optimal. However, as $i_1 < i_2$, we have $\sum_{i \in \tilde{S}} i < \sum_{i \in S^*} i$, contradicting the choice of S^* .

EC.4. Proof of Theorem 3: Analysis

In what follows, we formally establish near-optimal performance guarantees for the approximation scheme presented in Section 3. From a running time standpoint, Algorithm 1 can be implemented in $O(n^{O(\frac{1}{\epsilon^4} \log \frac{1}{\epsilon})})$ time as stated in of Theorem 3. Indeed, the rounding operations of step 1 can be completed in $O(n)$ time. For each instantiation of the input parameters picked in Ω , steps 3 and 4 are very efficient with an overall running time of $O(\frac{1}{\epsilon^2} \cdot n^4 \log n)$ as explained in Sections 3.5 and 3.6. Therefore, the desired running time immediately follows from the upper bound on $|\Omega| = O(n^{O(\frac{1}{\epsilon^4} \log \frac{1}{\epsilon})})$, which was established in Lemma 2.

Going forward, we will prove that the decomposable assortment returned by Algorithm 1 is revenue-wise $(1 - O(\epsilon))$ -close to the optimum. To simplify the analysis, we assume that the instance parameters are already specified in their rounded form, as enforced by step 1. As demonstrated by our sensitivity analysis in Section 3.3, this simplifying assumption comes with only an

$O(\epsilon)$ -approximation error. First, we will establish in Appendix EC.4.1 various structural properties that mirror the technical insights of Section 3. Next, in Appendices EC.4.2 and EC.4.3, we lower bound the revenue contributions of the unlikely and likely items selected by the specialized algorithms **MinKnapsack**($\cdot$) and **Greedy**($\cdot$), respectively. Finally, by combining these results in Appendix EC.4.4, we complete the proof of Theorem 3.

Preliminaries and notation. In what follows, we fix a collection of input parameters $(\{\hat{c}_q, \hat{\alpha}_q\}_{q \in \mathcal{Q}}, \{\hat{c}_{p,q}\}_{(p,q) \in [P_{\max}] \times \mathcal{Q}}) \in \Omega$ that satisfy Properties 1-3 of Lemma 2. We denote by $\hat{B} = \hat{B}_{\text{unlikely}} \cup \hat{B}_{\text{likely}}$ the resulting decomposable assortment, generated in steps 3 and 4 of our approximation scheme, i.e., $\hat{B}_{\text{unlikely}} = \text{MinKnapsack}(\{(\hat{c}_q, \hat{\alpha}_q)\}_{q \in \mathcal{Q}})$ and $\hat{B}_{\text{likely}} = \text{Greedy}(\{\hat{c}_{p,q}\}_{(p,q) \in [P_{\max}] \times \mathcal{Q}}, \hat{B}_{\text{unlikely}})$. We let $\hat{U}_q$ be the collection of items generated by solving the min-knapsack instance over the class of items $\mathcal{U}_q$, and let $\hat{B}_q$ be the resulting decomposable assortment specified by **MinKnapsack**($\cdot$) within this class. Similarly, we let $\hat{S}_t \subseteq \mathcal{N}_{\text{likely}}$ be the subset of likely items constructed in the iteration t of **Greedy**($\cdot$).

EC.4.1. Main structural properties

In what follows, we establish key structural properties of the decomposable assortments $\hat{B}_{\text{unlikely}}$ and $\hat{B}_{\text{likely}}$, which validate the design principles that guide these specialized algorithms.

Properties of the min-knapsack procedure (step 3). Recall that, in our description of **MinKnapsack**($\cdot$) in Section 3.5, the objective of the min-knapsack instances was to construct a subset of items S so as to minimize the quantity $\mathbb{E}[|C_{S \cap \mathcal{U}_q}|]$ within each class of items $\mathcal{U}_q$. This formulation was (informally) related to a mitigation of the “cannibalization effects” due to the inclusion of items in the assortment. The next claim confirms that, for every $q \in \mathcal{Q}$, the choice probabilities are no more “cannibalized” by our assortment decisions $\hat{B}_q$ within $\mathcal{U}_q$ than by the optimal assortment decisions $S^* \cap \mathcal{U}_q$.

CLAIM EC.7. *Let W be a positive random variable, independent of the consideration sets $\{C_{S \cap \mathcal{U}_q}\}_{q \in \mathcal{Q}}$ and $\{C_{S^* \cap \mathcal{U}_q}\}_{q \in \mathcal{Q}}$, where S denotes a random assortment sampled according to the decomposable assortment $\hat{B}_{\text{unlikely}}$. Then,*

$$\mathbb{E}_{S \sim \hat{B}_{\text{unlikely}}} \left[\frac{1}{W + \sum_{q \in \mathcal{Q}} w^{(q)} \cdot |C_{S \cap \mathcal{U}_q}|} \right] \geq \mathbb{E} \left[\frac{1}{W + \sum_{q \in \mathcal{Q}} w^{(q)} \cdot |C_{S^* \cap \mathcal{U}_q}|} \right].$$

The proof of Claim EC.7 is presented in Appendix EC.4.5, crucially relying on the Poissonization idea of Lemma 3.

Properties of the greedy procedure (step 4). In what follows, we establish an invariant of **Greedy**($\cdot$), arguing that at any point in time, the constructed assortment forms a subset of the likely items picked by the optimal assortment S^* .

CLAIM EC.8. $\hat{S}_t \subseteq S^* \cap \mathcal{N}_{\text{likely}}$, for every $t \in [0, T]$.

The proof is presented in Appendix EC.4.6. At a high-level, we show that the termination criterion placed on **Greedy**($\cdot$) prevents our algorithm from over-selecting items in any given class $\mathcal{W}_q \cap \Lambda_p$.

EC.4.2. Bounding the contributions of unlikely items

Armed with the properties of Appendix EC.4.1, we are ready analyze the contributions of the unlikely items picked by $\hat{B}_{\text{unlikely}}$ to the expected revenue $\mathbb{E}_{S \sim \hat{B}}[\mathcal{R}(S)]$.

CLAIM EC.9. $\mathbb{E}_{S \sim \hat{B}}[\sum_{i \in S \cap \mathcal{N}_{\text{unlikely}}} r_i \cdot \pi(i, S)] \geq (1 - 4\epsilon) \cdot \sum_{q=Q_{\min}}^{Q_{\max}} \hat{c}_q$.

Proof. Observe that, for every $q \in \mathcal{Q}$, the expected choice probability $\mathbb{E}_{S \sim \hat{B}}[\pi(i, S)]$ of every unlikely item $i \in \hat{U}_q$ can be lower-bounded as follows:

$$\begin{aligned} \mathbb{E}_{S \sim \hat{B}}[\pi(i, S)] &= \Pr_{S \sim \hat{B}}[i \in C_S] \cdot \mathbb{E}_{S \sim \hat{B}} \left[\frac{w_i}{1 + w_i + w(C_S^{-i})} \right] \\ &\geq (1 - \epsilon) \cdot \lambda_i \cdot \mathbb{E}_{S \sim \hat{B}} \left[\frac{w_i}{1 + w_i + w(C_S)} \right] \end{aligned} \quad (\text{EC.6})$$

$$\begin{aligned} &= (1 - \epsilon) \cdot \lambda_i \cdot \mathbb{E}_{S \sim \hat{B}} \left[\frac{w_i}{1 + w_i + \sum_{q'=Q_{\min}}^{Q_{\max}} w^{q'} \cdot |C_{S \cap \mathcal{U}_{q'}}| + w(C_{S \cap \mathcal{N}_{\text{likely}}})} \right] \\ &\geq (1 - \epsilon) \cdot \lambda_i w_i \cdot \mathbb{E}_{S \sim \hat{B}} \left[\frac{1}{1 + w_i + \sum_{q'=Q_{\min}}^{Q_{\max}} w^{q'} \cdot |C_{S \cap \mathcal{U}_{q'}}| + w(C_{S^* \cap \mathcal{N}_{\text{likely}}})} \right] \end{aligned} \quad (\text{EC.7})$$

$$\begin{aligned} &\geq (1 - \epsilon) \cdot \lambda_i w_i \cdot \mathbb{E} \left[\frac{1}{1 + w_i + \sum_{q' \in \mathcal{Q}} w^{q'} \cdot |C_{S^* \cap \mathcal{U}_{q'}}| + w(C_{S^* \cap \mathcal{N}_{\text{likely}}})} \right] \quad (\text{EC.8}) \\ &= (1 - \epsilon) \cdot \lambda_i \cdot \mathbb{E} \left[\frac{w^{(q)}}{1 + w^{(q)} + w(C_{S^*})} \right] \\ &= (1 - \epsilon) \cdot \lambda_i \cdot \alpha(S^*, q) \\ &\geq (1 - 3\epsilon) \cdot \lambda_i \cdot \hat{\alpha}_q. \end{aligned} \quad (\text{EC.9})$$

Here, inequality (EC.6) is obtained by noting that $\Pr_{S \sim \hat{B}}[i \in C_S] = (1 - \epsilon) \cdot \lambda_i$ and that $w(C_S^{-i}) \preceq_{\text{st}} w(C_S)$ for any subset of items S . Inequality (EC.7) holds since $w(C_{S^* \cap \mathcal{N}_{\text{likely}}}) \succeq_{\text{st}} w(C_{S_T})$ as implied by Claim EC.8, and by noting that $S \cap \mathcal{N}_{\text{likely}}$ is equal to S_T with probability 1, since S is sampled from the decomposable assortment $\hat{B}$ whose likely items are offered with probability 1. Inequality (EC.8) follows from Claim EC.7, instantiated with $W = 1 + w_i + w(C_{S^* \cap \mathcal{N}_{\text{likely}}})$. Finally, inequality (EC.9) proceeds from Property 3 of Lemma 2, stating in particular that $\alpha(S^*, q) \geq (1 - \epsilon)^2 \cdot \hat{\alpha}_q$.

Consequently, for the expected revenue contribution due to likely items, we obtain:

$$\mathbb{E}_{S \sim \hat{B}} \left[\sum_{i \in S \cap \mathcal{N}_{\text{unlikely}}} r_i \cdot \pi(i, S) \right] \geq (1 - 3\epsilon) \cdot \sum_{q=Q_{\min}}^{Q_{\max}} \sum_{i \in U_q} r_i \lambda_i \cdot \hat{\alpha}_q$$

$$\begin{aligned}
&\geq (1 - 3\epsilon) \cdot \sum_{q=Q_{\min}}^{Q_{\max}} \left(\frac{\epsilon}{n} \cdot \hat{c}_q \cdot \sum_{i \in \hat{U}_q} \gamma_i \right) \\
&\geq (1 - 4\epsilon) \cdot \sum_{q=Q_{\min}}^{Q_{\max}} \hat{c}_q,
\end{aligned}$$

where the second inequality proceeds from eliminating floors in our definition of γ_i , where $\gamma_i = \lfloor \frac{n\lambda_i r_i \hat{\alpha}_q}{\epsilon \hat{c}_q} \rfloor$, and the last inequality holds since $\sum_{i \in S^* \cap \mathcal{U}_q} \gamma_i \geq (1 - \epsilon) \cdot \frac{n}{\epsilon}$, as shown by inequality (EC.16) in the proof of Claim EC.7. $\square$

EC.4.3. Bounding the contributions of likely items

We now turn our attention to the contributions of the likely items picked by $\hat{B}_{\text{likely}}$ to the expected revenue $\mathbb{E}_{S \sim \hat{B}}[\mathcal{R}(S)]$. Specifically, we establish the following claim.

CLAIM EC.10. $\mathbb{E}_{S \sim \hat{B}}[\sum_{i \in S \cap \mathcal{N}_{\text{likely}}} r_i \cdot \pi(i, S)] \geq (1 - \epsilon) \cdot \sum_{p=1}^{P_{\max}} \sum_{q=Q_{\min}}^{Q_{\max}} \hat{c}_{p,q}.$

Proof. We begin by noting that $S \cap \mathcal{N}_{\text{likely}}$ is equal to $\hat{S}_T$ with probability 1, since the decomposable assortment $\hat{B}_{\text{likely}}$, generated in the last iteration of **Greedy**($\cdot$), includes each item of S_T with probability 1. Thus, in order to prove the desired claim, it suffices to show that

$$\sum_{i \in \hat{S}_T \cap \Lambda_p \cap \mathcal{W}_q} r_i \cdot \mathbb{E}_{S \sim \hat{B}_{\text{unlikely}}} [\tilde{\pi}(i, S \cup \hat{S}_T)] \geq (1 - \epsilon) \cdot \hat{c}_{p,q},$$

for every $p \in [P_{\max}]$ and $q \in \mathcal{Q}$. For this purpose, by Claim EC.8, we have in particular $(\hat{S}_T \cap \Lambda_p \cap \mathcal{W}_q) \subseteq (S^* \cap \Lambda_p \cap \mathcal{W}_q)$. Therefore, given the termination criterion of **Greedy**($\cdot$), the inequalities $|\hat{S}_T \cap \Lambda_p \cap \mathcal{W}_q| < |S^* \cap \Lambda_p \cap \mathcal{W}_q|$ and $\sum_{i \in \hat{S}_T \cap \Lambda_p \cap \mathcal{W}_q} r_i \cdot \mathbb{E}_{S \sim \hat{B}_{\text{unlikely}}} [\tilde{\pi}(i, S \cup \hat{S}_T)] < (1 - \epsilon) \cdot \hat{c}_{p,q}$ cannot simultaneously hold at the last iteration T . If the latter inequality does not hold, we are clearly done, noting that this inequality is exactly the opposite of what we want to show. We therefore consider the case where the former inequality does not hold, i.e., $|\hat{S}_T \cap \Lambda_p \cap \mathcal{W}_q| \geq |S^* \cap \Lambda_p \cap \mathcal{W}_q|$. Clearly, in this case, we have $(\hat{S}_T \cap \Lambda_p \cap \mathcal{W}_q) = (S^* \cap \Lambda_p \cap \mathcal{W}_q)$. Consequently, letting $S_{\text{likely}}^* = S^* \cap \mathcal{N}_{\text{likely}}$, it follows that

$$\begin{aligned}
\sum_{i \in \hat{S}_T \cap \Lambda_p \cap \mathcal{W}_q} r_i \cdot \mathbb{E}_{S \sim \hat{B}_{\text{unlikely}}} [\tilde{\pi}(i, S \cup \hat{S}_T)] &= \sum_{i \in S^* \cap \Lambda_p \cap \mathcal{W}_q} r_i \cdot \mathbb{E}_{S \sim \hat{B}_{\text{unlikely}}} [\tilde{\pi}(i, S \cup \hat{S}_T)] \\
&\geq \sum_{i \in S^* \cap \Lambda_p \cap \mathcal{W}_q} r_i \cdot \mathbb{E}_{S \sim \hat{B}_{\text{unlikely}}} [\pi(i, S \cup \hat{S}_T)] \\
&\geq \sum_{i \in S^* \cap \Lambda_p \cap \mathcal{W}_q} r_i \cdot \mathbb{E}_{S \sim \hat{B}_{\text{unlikely}}} [\pi(i, S \cup S_{\text{likely}}^*)] \quad (\text{EC.10})
\end{aligned}$$

where the first inequality holds since the FPTAS of Theorem 2 ensures that $\mathbb{E}_{S \sim \hat{B}_{\text{unlikely}}} [\tilde{\pi}(i, S \cup \hat{S}_T)] \geq \mathbb{E}_{S \sim \hat{B}_{\text{unlikely}}} [\pi(i, S \cup \hat{S}_T)]$, and the second inequality holds since $\hat{S}_T \subseteq S_{\text{likely}}^*$ by Claim EC.8. In the

remainder of this proof, we show that the expression on the right-hand-side of inequality (EC.10) can be lower-bounded by $\hat{c}_{p,q}$. To this end, we observe that

$$\begin{aligned}
& \sum_{i \in S^* \cap \Lambda_p \cap \mathcal{W}_q} r_i \cdot \mathbb{E}_{S \sim \hat{B}_{\text{unlikely}}} [\pi(i, S \cup S^*_{\text{likely}})] \\
&= \sum_{i \in S^* \cap \Lambda_p \cap \mathcal{W}_q} r_i \lambda_i w_i \cdot \mathbb{E}_{S \sim \hat{B}_{\text{unlikely}}} \left[\frac{1}{1 + w_i + w(C_S) + w(C_{S^* \cap \mathcal{N}_{\text{likely}}}^{-i})} \right] \\
&\geq \sum_{i \in S^* \cap \Lambda_p \cap \mathcal{W}_q} r_i \lambda_i w_i \cdot \mathbb{E} \left[\frac{1}{1 + w_i + w(C_{S^* \cap \mathcal{N}_{\text{unlikely}}}) + w(C_{S^* \cap \mathcal{N}_{\text{likely}}}^{-i})} \right] \\
&= \sum_{i \in S^* \cap \Lambda_p \cap \mathcal{W}_q} r_i \cdot \pi(i, S^*) \\
&= c_{p,q}^* \\
&\geq \hat{c}_{p,q} .
\end{aligned}$$

Here, the first inequality proceeds from Claim EC.7, instantiated with $W = 1 + w_i + w(C_{S^* \cap \mathcal{N}_{\text{likely}}}^{-i})$. The last inequality follows from Property 1 of Lemma 2. $\square$

EC.4.4. Proving the performance guarantee of Theorem 3

To conclude the proof of Theorem 3, we show that the decomposable assortment B generates an expected revenue within factor $1 - O(\epsilon)$ of the optimal expected revenue $\mathcal{R}(S^*)$. Specifically, by combining the respective bounds obtained for unlikely and likely items in Claims EC.9 and EC.10, we have

$$\begin{aligned}
& \mathbb{E}_{S \sim B} [\mathcal{R}(S)] \\
&\geq (1 - \epsilon) \cdot \mathbb{E}_{S \sim \hat{B}} [\mathcal{R}(S)] \\
&= (1 - \epsilon) \cdot \mathbb{E}_{S \sim \hat{B}} \left[\sum_{i \in S \cap \mathcal{N}_{\text{likely}}} r_i \cdot \pi(i, S) + \sum_{i \in S \cap \mathcal{N}_{\text{unlikely}}} r_i \cdot \pi(i, S) \right] \\
&\geq (1 - 5\epsilon) \cdot \left(\sum_{p=1}^{P_{\max}} \sum_{q=Q_{\min}}^{Q_{\max}} \hat{c}_{p,q} + \sum_{q=Q_{\min}}^{Q_{\max}} \hat{c}_q \right) \\
&\geq (1 - 5\epsilon) \cdot \left(\sum_{p=1}^{P_{\max}} \sum_{q=Q_{\min}}^{Q_{\max}} \left(c_{p,q}^* - \frac{2\epsilon}{\mathcal{L}} \cdot \mathcal{R}(S^*) \right) + \sum_{q=Q_{\min}}^{Q_{\max}} \left(c_q^* - \frac{2\epsilon}{\mathcal{L}} \cdot \mathcal{R}(S^*) \right) \right) \\
&\geq (1 - 7\epsilon) \cdot \mathcal{R}(S^*) . \tag{EC.11}
\end{aligned}$$

Here, the first inequality follows by noting that the decomposable assortment B dominates $\hat{B}$ in terms of revenue by steps 3 and 4 of Algorithm 1. The second inequality follows from Claims EC.9 and EC.10. The next inequality is due to Properties 1 and 2 of Lemma 2. The last inequality follows by recalling that $\mathcal{L} = (P_{\max} + 1) \cdot |\mathcal{Q}|$.

EC.4.5. Proof of Claim EC.7

Let $\mathcal{Q} = \{q_1, \dots, q_{|\mathcal{Q}|}\}$ be an arbitrary indexing of the elements of $\mathcal{Q}$. The proof proceeds by an induction over $t \in [0, |\mathcal{Q}|]$ showing that:

$$\begin{aligned} & \mathbb{E}_{S \sim \hat{B}_{\text{unlike}}} \left[\frac{1}{W + \sum_{q \in \mathcal{Q}} w^{(q)} \cdot |C_{S \cap \mathcal{U}_q}|} \right] \\ & \geq \mathbb{E}_{S \sim \hat{B}_{\text{unlike}}} \left[\frac{1}{W + \sum_{k=1}^t w^{q_k} \cdot |C_{S^* \cap \mathcal{U}_{q_k}}| + \sum_{k=t+1}^{|\mathcal{Q}|} w^{q_k} \cdot |C_{S \cap \mathcal{U}_{q_k}}|} \right]. \end{aligned}$$

The base case of the induction $t = 0$ is immediate, since this inequality is satisfied with an equality.

Next, we show that the induction hypothesis propagates to every $1 \leq t \leq |\mathcal{Q}|$. To this end, letting

$V_{-t} = W + \sum_{k=1}^{t-1} w^{q_k} \cdot |C_{S^* \cap \mathcal{U}_{q_k}}| + \sum_{k=t+1}^{|\mathcal{Q}|} w^{q_k} \cdot |C_{S \cap \mathcal{U}_{q_k}}|$, we have:

$$\begin{aligned} & \mathbb{E}_{S \sim \hat{B}_{\text{likely}}} \left[\frac{1}{W + \sum_{k=1}^t w^{q_k} \cdot |C_{S^* \cap \mathcal{U}_{q_k}}| + \sum_{k=t+1}^{|\mathcal{Q}|} w^{q_k} \cdot |C_{S \cap \mathcal{U}_{q_k}}|} \right] \\ & = \mathbb{E} \left[\frac{1}{w^{q_t} \cdot |C_{S^* \cap \mathcal{U}_{q_t}}| + V_{-t}} \right] \\ & = \mathbb{E}_{V_{-t}} \left[\mathbb{E} \left[\frac{1}{w^{q_t} \cdot |C_{S^* \cap \mathcal{U}_{q_t}}| + V_{-t}} \middle| V_{-t} \right] \right] \\ & \leq \mathbb{E}_{V_{-t}} \left[\mathbb{E} \left[\frac{1}{w^{q_t} \cdot P(\sum_{i \in S^* \cap \mathcal{U}_{q_t}} \lambda_i) + V_{-t}} \middle| V_{-t} \right] \right] \tag{EC.12} \end{aligned}$$

$$\leq \mathbb{E}_{V_{-t}} \left[\mathbb{E} \left[\frac{1}{w^{q_t} \cdot P(\sum_{i \in \hat{\mathcal{U}}_{q_t}} \lambda_i) + V_{-t}} \middle| V_{-t} \right] \right] \tag{EC.13}$$

$$\leq \mathbb{E}_{V_{-t}} \left[\mathbb{E}_{S \sim \hat{B}_{\text{likely}}} \left[\frac{1}{w^{q_t} \cdot |C_{S \cap \mathcal{U}_{q_t}}| + V_{-t}} \middle| V_{-t} \right] \right] \tag{EC.14}$$

$$\begin{aligned} & = \mathbb{E}_{S \sim \hat{B}_{\text{likely}}} \left[\frac{1}{w^{q_t} \cdot |C_{S \cap \mathcal{U}_{q_t}}| + V_{-t}} \right] \\ & = \mathbb{E}_{S \sim \hat{B}_{\text{likely}}} \left[\frac{1}{W + \sum_{k=1}^{t-1} w^{q_k} \cdot |C_{S^* \cap \mathcal{U}_{q_k}}| + \sum_{k=t}^{|\mathcal{Q}|} w^{q_k} \cdot |C_{S \cap \mathcal{U}_{q_k}}|} \right] \\ & \leq \mathbb{E}_{S \sim \hat{B}_{\text{likely}}} \left[\frac{1}{W + \sum_{q \in \mathcal{Q}} w^{(q)} \cdot |C_{S \cap \mathcal{U}_q}|} \right], \tag{EC.15} \end{aligned}$$

where inequality (EC.15) immediately follows from our induction hypothesis.

Inequality (EC.12) proceeds from Lemma 3 and the independence between the random variables V_{-t} and $|C_{S^* \cap \mathcal{U}_{q_t}}|$. To apply Lemma 3, observe that $|C_{S^* \cap \mathcal{U}_{q_t}}| = \sum_{i \in S^* \cap \mathcal{U}_{q_t}} X_i$ where $\{X_i\}_{i \in S^* \cap \mathcal{U}_{q_t}}$ is a collection of independent Bernoulli random variables with probabilities of success $\{\lambda_i\}_{i \in S^* \cap \mathcal{U}_{q_t}}$. Hence, the first inequality of Lemma 3 implies that $P(\sum_{i \in S^* \cap \mathcal{U}_{q_t}} \lambda_i) \succeq_{\text{dex}} |C_{S^* \cap \mathcal{U}_{q_t}}|$. This relationship yields (EC.12) since the function $x \mapsto \frac{1}{w^{q_t} \cdot x + v}$ is decreasing convex over $\mathbb{R}^+$ for any $v > 0$.

Inequality (EC.13) is justified by showing that $\sum_{i \in S^* \cap \mathcal{U}_{q_t}} \lambda_i \geq \sum_{i \in \hat{U}_{q_t}} \lambda_i$. To this end, it suffices to show that the subset $S^* \cap \mathcal{U}_{q_t}$ is feasible with respect to the min-knapsack instance for which $\hat{U}_q$ is optimal (see Section 3.5). Since Property 3 in Lemma 2 is satisfied, we have

$$\sum_{i \in S^* \cap \mathcal{U}_q} r_i \lambda_i \hat{\alpha}_q \geq \frac{1}{1-\epsilon} \cdot \sum_{i \in S^* \cap \mathcal{U}_q} r_i \lambda_i \alpha(S^*, q) \geq \sum_{i \in S^* \cap \mathcal{U}_q} r_i \pi(i, S^*) = c_q^* \geq \hat{c}_q,$$

where the second inequality proceeds from Lemma 4, and the third inequality follows from Property 2 of Lemma 2. Now, since $\gamma_i = \lfloor \frac{n \lambda_i r_i \hat{\alpha}_q}{\epsilon \hat{c}_q} \rfloor$, we get

$$\sum_{i \in S^* \cap \mathcal{U}_q} \gamma_i \geq \sum_{i \in S^* \cap \mathcal{U}_q} \left(\frac{n \lambda_i r_i \hat{\alpha}_q}{\epsilon \hat{c}_q} - 1 \right) \geq \frac{n}{\epsilon} - |S^* \cap \mathcal{U}_q| \geq (1-\epsilon) \cdot \frac{n}{\epsilon}. \quad (\text{EC.16})$$

Inequality (EC.16) precisely states that $S^* \cap \mathcal{U}_{q_t}$ is feasible with respect to the min-knapsack instance associated with the class of items $\mathcal{U}_{q_t}$, thereby completing the proof of (EC.13).

Inequality (EC.14) proceeds from Lemma 3 and the independence between the random variables V_{-t} and $|C_{S \cap \mathcal{U}_{q_t}}|$, where S designates a random assortment sampled according to the decomposable assortment $\hat{B}_{\text{unlike}}$. Indeed, observe that $|C_{S \cap \mathcal{U}_{q_t}}| = \sum_{i \in \hat{U}_{q_t}} \tilde{X}_i$ where $\{\tilde{X}_i\}_{i \in \hat{U}_{q_t}}$ is a collection of independent Bernoulli random variables with probabilities of success $\tilde{\lambda}_i = (1-\epsilon) \cdot \lambda_i$ for every $i \in \hat{U}_{q_t}$. (Recall from Section 3.5 that $\hat{U}_q$ is the set of items that are picked with positive probability by the decomposable assortment $\hat{B}_{\text{unlike}}$ within $\mathcal{U}_q$.) The second inequality of Lemma 3 implies that $|C_{S \cap \mathcal{U}_{q_t}}| \succeq_{\text{dcx}} P((1+\epsilon) \cdot \sum_{i \in \hat{U}_{q_t}} (1-\epsilon) \cdot \lambda_i) \succeq_{\text{dcx}} P(\sum_{i \in \hat{U}_{q_t}} \lambda_i)$.

EC.4.6. Proof of Claim EC.8

This result is proven by induction over $t \in [0, T]$. The base case $t = 0$ is clearly satisfied since $\hat{S}_0 = \emptyset$ by definition. For the general case $t \geq 1$, we have $\hat{S}_{t-1} \subseteq S^* \cap \mathcal{N}_{\text{likely}}$ by the induction hypothesis. Recall that the set of items $\hat{S}_t$ was constructed by adding item i_{t-1} to $\hat{S}_{t-1}$ in iteration $t-1$; we let $p \in [P_{\max}]$ and $q \in \mathcal{Q}$ be the unique indices for which $i_{t-1} \in \Lambda_p \cap \mathcal{W}_q$. Now, suppose that $i_{t-1} \notin S^*$. By the revenue-ordered-by-class property stated in Lemma 5, the subset $S^* \cap \Lambda_p \cap \mathcal{W}_q$ is formed by the $k_{p,q}$ highest revenue items in $\Lambda_p \cap \mathcal{W}_q$, where $k_{p,q} = |S^* \cap \Lambda_p \cap \mathcal{W}_q|$. We first observe that $|\hat{S}_{t-1} \cap \Lambda_p \cap \mathcal{W}_q| = k_{p,q}$. Otherwise, if $|\hat{S}_{t-1} \cap \Lambda_p \cap \mathcal{W}_q| < k_{p,q}$, our greedy selection rule ensures that item i_{t-1} is necessarily among the $k_{p,q}$ highest revenue items in $\Lambda_p \cap \mathcal{W}_q$, implying that $i_{t-1} \in S^* \cap \Lambda_p \cap \mathcal{W}_q$, in contradiction to having $i_{t-1} \notin S^*$. Therefore, combining this observation with the induction hypothesis, we infer that $(\hat{S}_{t-1} \cap \Lambda_p \cap \mathcal{W}_q) = (S^* \cap \Lambda_p \cap \mathcal{W}_q)$. Now, just before picking item i_{t-1} at iteration $t-1$ of the greedy algorithm, we had

$$\begin{aligned} & \sum_{i \in \hat{S}_{t-1} \cap \Lambda_p \cap \mathcal{W}_q} r_i \cdot \mathbb{E}_{S \sim \hat{B}_{\text{unlike}}} \left[\pi(i, S \cup \hat{S}_{t-1}) \right] \\ & \leq \sum_{i \in \hat{S}_{t-1} \cap \Lambda_p \cap \mathcal{W}_q} r_i \cdot \mathbb{E}_{S \sim \hat{B}_{\text{unlike}}} \left[\tilde{\pi}(i, S \cup \hat{S}_{t-1}) \right] \end{aligned}$$

$$\begin{aligned}
&< (1 - \epsilon) \cdot \hat{c}_{p,q} \\
&< \hat{c}_{p,q} ,
\end{aligned} \tag{EC.17}$$

where the first inequality holds since the FPTAS of Theorem 2 ensures that $\mathbb{E}_{S \sim \hat{B}_{\text{unlike}}}[\pi(i, S \cup \hat{S}_{t-1})] \leq \mathbb{E}_{S \sim \hat{B}_{\text{unlike}}}[\tilde{\pi}(i, S \cup \hat{S}_{t-1})]$. The second inequality follows from the termination criterion of our greedy procedure, since inequality (3) is necessarily violated.

On the other hand,

$$\begin{aligned}
&\sum_{i \in \hat{S}_{t-1} \cap \Lambda_p \cap \mathcal{W}_q} r_i \cdot \mathbb{E}_{S \sim \hat{B}_{\text{unlike}}} [\pi(i, S \cup \hat{S}_{t-1})] \\
&= \sum_{i \in S^* \cap \Lambda_p \cap \mathcal{W}_q} r_i \cdot \mathbb{E}_{S \sim \hat{B}_{\text{unlike}}} [\pi(i, S \cup \hat{S}_{t-1})] \\
&= \sum_{i \in S^* \cap \Lambda_p \cap \mathcal{W}_q} r_i \lambda_i w_i \cdot \mathbb{E}_{S \sim \hat{B}_{\text{unlike}}} \left[\frac{1}{1 + w_i + w(C_S) + w(C_{\hat{S}_{t-1}}^{-i})} \right] \\
&\geq \sum_{i \in S^* \cap \Lambda_p \cap \mathcal{W}_q} r_i \lambda_i w_i \cdot \mathbb{E}_{S \sim \hat{B}_{\text{unlike}}} \left[\frac{1}{1 + w_i + w(C_S) + w(C_{S^* \cap \mathcal{N}_{\text{likely}}}^{-i})} \right] \\
&\geq \sum_{i \in S^* \cap \Lambda_p \cap \mathcal{W}_q} r_i \lambda_i w_i \cdot \mathbb{E} \left[\frac{1}{1 + w_i + w(C_{S^* \cap \mathcal{N}_{\text{unlike}}}) + w(C_{S^* \cap \mathcal{N}_{\text{likely}}}^{-i})} \right] \\
&= \sum_{i \in S^* \cap \Lambda_p \cap \mathcal{W}_q} r_i \cdot \pi(i, S^*) \\
&= c_{p,q}^* .
\end{aligned}$$

Here, the first equality holds since $(\hat{S}_{t-1} \cap \Lambda_p \cap \mathcal{W}_q) = (S^* \cap \Lambda_p \cap \mathcal{W}_q)$, as shown above. The first inequality holds since $\hat{S}_{t-1} \subseteq S^* \cap \mathcal{N}_{\text{likely}}$ by the induction hypothesis, implying that $w(C_{S^* \cap \mathcal{N}_{\text{likely}}}^{-i}) \succeq_{\text{st}} w(C_{\hat{S}_{t-1}}^{-i})$. In addition, we note that C_S is independent of $C_{\hat{S}_{t-1}}^{-i}$ and $C_{S^* \cap \mathcal{N}_{\text{likely}}}^{-i}$, where S is the random realization of the decomposable assortment $\hat{B}_{\text{unlike}}$. The second inequality proceeds from Claim EC.7, instantiated with $W = 1 + w_i + w(C_{S^* \cap \mathcal{N}_{\text{likely}}}^{-i})$. Combining the latter inequality with (EC.17), we obtain $c_{p,q}^* < \hat{c}_{p,q}$, which contradicts Property 1 of Lemma 2. It follows that $i_{t-1} \in S^* \cap \Lambda_p \cap \mathcal{W}_q$, and therefore $\hat{S}_t \subseteq S^* \cap \mathcal{N}_{\text{likely}}$, as desired.

EC.5. Proof of Theorem 4

In this section, we focus our attention on designing an approximation scheme for the click-based MNL assortment problem in its utmost generality. Specifically, our purpose is to bypass Assumption 1, stating that $\frac{w_{\max}}{w_{\min}} = n^{O(\frac{1}{\epsilon})}$, which means that the ratio of extremal preference weights is polynomially bounded. To handle arbitrary instances, in which the extremal preference weights are not necessarily polynomially related, we will show how to approximately decompose any given instance into a collection of bounded-ratio ones, glued together by means of dynamic programming.

Within this framework, our approach directly leverages Algorithm 1 as a subroutine. To simplify the exposition, we assume that the instance parameters are already specified in their rounded form, similarly to Appendix EC.4. Specifically, we assume upfront that each item $i \in \mathcal{W}_q \cap \Lambda_p$ has a preference weight of $w_i = w^{(q)} = \frac{\epsilon}{n} \cdot (1 + \epsilon)^q$ and a consideration probability of $\lambda_i = \epsilon \cdot (1 + \epsilon)^p$.

Outline. First, in Appendix EC.5.1, we establish a constrained version of Theorem 3, showing that our approximation scheme can easily handle an additional knapsack-like constraint on the selected decomposable assortment. Next, in Appendix EC.5.2, we present a decomposition method that breaks any given instance into subproblems that satisfy the bounded-ratio assumption. In Appendices EC.5.3 and EC.5.4, we formulate a dynamic program that sequentially examines these subproblems; at each state, this dynamic program employs our approximation scheme for the constrained bounded-ratio setting as a subroutine.

EC.5.1. Approximation scheme for the emptiness-constrained problem

In what follows, we argue that the approximation scheme of Theorem 3 can be generalized to handle an additional constraint on the decomposable assortment, termed *the emptiness constraint*. This constraint imposes a lower bound on the probability that the induced consideration set is empty. Specifically, given $\Phi \in [0, 1]$, the emptiness constraint requires that the decomposable assortment B satisfies $\Pr_{S \sim B}[C_S = \emptyset] \geq \Phi$.

In the next theorem, we state our main algorithmic result in the emptiness-constrained setting, while still considering instances that satisfy Assumption 1. Below, S^* is an optimal (deterministic) assortment for the click-based MNL assortment problem subject to the emptiness constraint with parameter Φ .

THEOREM EC.1. *For any accuracy level $\epsilon > 0$, we can compute a decomposable assortment B such that $\mathbb{E}_{S \sim B}[\mathcal{R}(S)] \geq (1 - \epsilon) \cdot \mathcal{R}(S^*)$ and $\Pr_{S \sim B}[C_S = \emptyset] \geq \Phi$. The running time of our algorithm is $O(n^{O(\frac{1}{\epsilon^4} \log \frac{1}{\epsilon})})$.*

We devote the remainder of this section to proving Theorem EC.1. Since our new algorithm is nearly identical to Algorithm 1, we only focus on the differences between these approaches. Our analysis directly builds on the one presented in Appendix EC.4 for the unconstrained setting.

Modified approximation scheme. We begin by describing how our approximation scheme is modified to account for the emptiness constraint. The algorithm is nearly identical to that of the unconstrained setting with the only exception being, as we enumerate over the collection of input parameters Ω , we only consider decomposable assortments that meet the emptiness criterion. Specifically, we employ our specialized algorithms with all input parameters $(\{\hat{c}_q, \hat{\alpha}_q\}_{q \in \mathcal{Q}}, \{\hat{c}_{p,q}\}_{(p,q) \in [P_{\max}] \times \mathcal{Q}}) \in \Omega$ to construct $B_{\text{unlike}} = \text{MinKnapsack}(\{\{\hat{c}_q, \hat{\alpha}_q\}_{q \in \mathcal{Q}}\})$ and $B_{\text{likely}} = \text{Greedy}(\{\{\hat{c}_{p,q}\}_{(p,q) \in [P_{\max}] \times \mathcal{Q}}\}, B_{\text{unlike}})$. The algorithm eventually returns the resulting decomposable assortment $B = B_{\text{unlike}} \cup B_{\text{likely}}$ of

maximum expected revenue out of all those that satisfy the constraint $\Pr_{S \sim B}[C_S = \emptyset] \geq \Phi$. That is, any decomposable assortment B with $\Pr_{S \sim B}[C_S = \emptyset] < \Phi$ is now discarded. To summarize, our modified algorithm proceeds as follows:

1. Generate the collection of input parameters Ω .
2. For every $(\{\hat{c}_q, \hat{\alpha}_q\}_{q \in \mathcal{Q}}, \{\hat{c}_{p,q}\}_{(p,q) \in [P_{\max}] \times \mathcal{Q}}) \in \Omega$:
 - (a) Compute $B_{\text{unlike}} = \text{MinKnapsack}(\{\{\hat{c}_q, \hat{\alpha}_q\}\}_{q \in \mathcal{Q}})$.
 - (b) Compute $B_{\text{likely}} = \text{Greedy}(\{\hat{c}_{p,q}\}_{(p,q) \in [P_{\max}] \times \mathcal{Q}}, B_{\text{unlike}})$.
3. Return the decomposable assortment $B = B_{\text{unlike}} \cup B_{\text{likely}}$ of maximum expected revenue out of those generated in step 2 that additionally satisfy the requirement $\Pr_{S \sim B}[C_S = \emptyset] \geq \Phi$.

As a side note, we have relabelled the steps of the modified approximation scheme in comparison to Algorithm 1. In particular, the initial rounding operations, which constitute step 1 of Algorithm 1, are no longer listed, since we have assumed upfront that the instance parameters are already specified in the desired rounded form.

Analysis. Recall that, for purposes of analysis, we fix a collection of input parameters $(\{\hat{c}_q, \hat{\alpha}_q\}_{q \in \mathcal{Q}}, \{\hat{c}_{p,q}\}_{(p,q) \in [P_{\max}] \times \mathcal{Q}}) \in \Omega$ that satisfy Properties 1-3 of Lemma 2. We denote by $\hat{B} = \hat{B}_{\text{unlike}} \cup \hat{B}_{\text{likely}}$ the resulting decomposable assortment, generated in step 2 of our modified approximation scheme, i.e., $\hat{B}_{\text{unlike}} = \text{MinKnapsack}(\{\{\hat{c}_q, \hat{\alpha}_q\}\}_{q \in \mathcal{Q}})$ and $\hat{B}_{\text{likely}} = \text{Greedy}(\{\hat{c}_{p,q}\}_{(p,q) \in [P_{\max}] \times \mathcal{Q}}, \hat{B}_{\text{unlike}})$. We let $\hat{U}_q$ be the collection of items generated by solving the min-knapsack instance over the class of items $\mathcal{U}_q$, and let $\hat{B}_q$ be the resulting decomposable assortment specified by $\text{MinKnapsack}(\cdot)$ within this class. Similarly, we let $\hat{S}_T \subseteq \mathcal{N}_{\text{likely}}$ be the subset of likely items constructed in the last iteration of $\text{Greedy}(\cdot)$.

A close inspection of our analysis of the unconstrained setting in Appendix EC.4.4 reveals that an identical performance guarantee holds in the emptiness-constrained setting. That is, using precisely the same line of reasoning as in inequality (EC.11), we obtain $\mathbb{E}_{S \sim \hat{B}}[\mathcal{R}(S)] \geq (1 - 6\epsilon) \cdot \mathcal{R}(S^*)$. Hence, in order to establish Theorem EC.1, it remains to show that the decomposable assortment $\hat{B}$ indeed satisfies the emptiness criterion $\Pr_{S \sim \hat{B}}[C_S = \emptyset] \geq \Phi$, and thus, it is not discarded in step 3 of our modified approximation scheme.

To this end, we relate $\Pr_{S \sim \hat{B}_{\text{unlike}}}[C_S = \emptyset]$ to the analogous probability with respect to $S^* \cap \mathcal{N}_{\text{unlike}}$. The lower bound stated in Lemma EC.2 is the first step towards showing that our final assortment $\hat{B}$ meets the emptiness constraint.

LEMMA EC.2. $\Pr_{S \sim \hat{B}_{\text{unlike}}}[C_S = \emptyset] \geq \Pr[C_{S^* \cap \mathcal{N}_{\text{unlike}}} = \emptyset]$.

Proof. By definition of the decomposable assortment $\hat{B}_{\text{unlike}}$, we have:

$$\Pr_{S \sim \hat{B}_{\text{unlike}}}[C_S = \emptyset] = \prod_{q \in \mathcal{Q}} \Pr_{S \sim \hat{B}_q}[C_S = \emptyset]$$

$$\begin{aligned}
&= \prod_{q \in \mathcal{Q}} \prod_{i \in \hat{U}_q} (1 - (1 - \epsilon) \cdot \lambda_i) \\
&\geq \prod_{q \in \mathcal{Q}} \prod_{i \in \hat{U}_q} e^{-\lambda_i} \\
&= \prod_{q \in \mathcal{Q}} e^{-\sum_{i \in \hat{U}_q} \lambda_i},
\end{aligned}$$

where the second equality proceeds from our construction of the decomposable assortment $\hat{B}^q$, where $\Pr[\hat{B}_i^q = 1] = 1 - \epsilon$ for every $i \in U_q$. The subsequent inequality holds since $1 - (1 - \epsilon) \cdot \lambda_i \geq e^{-\lambda_i}$ for $\lambda_i \in [0, \epsilon]$, noting that we indeed have $\lambda_i \leq \epsilon$ since all items $i \in \hat{U}_q$ are unlikely. Recalling that inequality (EC.16) guarantees $\sum_{i \in \hat{U}_q} \lambda_i \leq \sum_{i \in S^* \cap \mathcal{U}_q} \lambda_i$, we conclude that

$$\begin{aligned}
\Pr_{S \sim \hat{B}_{\text{unlikely}}} [C_S = \emptyset] &\geq \prod_{q \in \mathcal{Q}} e^{-\sum_{i \in S^* \cap \mathcal{U}_q} \lambda_i} \\
&\geq \prod_{q \in \mathcal{Q}} \prod_{i \in S^* \cap \mathcal{U}_q} (1 - \lambda_i) \\
&= \Pr [C_{S^* \cap \mathcal{N}_{\text{unlikely}}} = \emptyset].
\end{aligned}$$

□

To complete the proof of Theorem EC.1, we verify that the decomposable assortment $\hat{B}$ meets the emptiness criterion, as shown in the next lemma.

LEMMA EC.3. $\Pr_{S \sim \hat{B}} [C_S = \emptyset] \geq \Phi$.

Proof. Observe that

$$\begin{aligned}
\Pr_{S \sim \hat{B}} [C_S = \emptyset] &= \Pr_{S \sim \hat{B}_{\text{unlikely}}} [C_S = \emptyset] \cdot \Pr_{S \sim \hat{B}_{\text{likely}}} [C_S = \emptyset] \\
&\geq \Pr [C_{S^* \cap \mathcal{N}_{\text{unlikely}}} = \emptyset] \cdot \Pr [C_{S^* \cap \mathcal{N}_{\text{likely}}} = \emptyset] \\
&= \Pr [C_{S^*} = \emptyset] \\
&\geq \Phi,
\end{aligned}$$

where the first equality holds since the decomposable assortments $\hat{B}_{\text{likely}}$ and $\hat{B}_{\text{unlikely}}$ are independent. The first inequality above follow from Lemmas EC.2 and Claim EC.8. In particular, in order to apply Claim EC.8, we note the assortment S sampled from $\hat{B}_{\text{likely}}$ is deterministically equal to $\hat{S}_T$, thus $S \subseteq S^* \cap \mathcal{N}_{\text{likely}}$ with probability 1. □

EC.5.2. Well-separated weight clusters

We begin by defining a way to partition the collection of weight classes into well-separated clusters, which constitute the elementary units of our decomposition into bounded-ratio instances. The concrete meaning of “well-separated clusters” will become clear once we introduce the necessary definitions and constructions.

Creating weight clusters. We begin by reminding the reader that the entire set of items is partitioned into weights classes $\mathcal{W}_0, \mathcal{W}_1, \dots$, where $\mathcal{W}_q = \{i \in [n] : w_i = (1 + \epsilon)^q \cdot \frac{\epsilon}{n}\}$. For ease of notation, let $Q_{\text{skip}} = 2 \cdot \lceil \log_{1+\epsilon}(\frac{n}{\epsilon}) \rceil$ and $Q_{\text{pick}} = \lceil \frac{1}{\epsilon} \rceil \cdot Q_{\text{skip}}$. In addition, we fix a non-positive integer $s \in [-Q_{\text{skip}} - Q_{\text{pick}} + 1, 0]$, which is referred to as the shifting index; the value of this index will be determined shortly.

Given these parameters, we initially define a sequence of pairwise-disjoint subsets $I_1^s, I_2^s, \dots$, where each I_ℓ^s corresponds to a consecutive block of non-negative integers. These subsets are iteratively defined as follows:

- First, we define I_1^s as the interval starting at s and consisting of Q_{pick} successive indices, namely, $I_1^s = \{s, s+1, \dots, s+Q_{\text{pick}}-1\}$.
- Next, we skip Q_{skip} indices, and define I_2^s as the interval formed by the next Q_{pick} indices. Namely, $I_2^s = \{s+Q_{\text{pick}}+Q_{\text{skip}}, s+Q_{\text{pick}}+Q_{\text{skip}}+1, \dots, s+2Q_{\text{pick}}+Q_{\text{skip}}-1\}$.
- Similarly, we skip the next Q_{skip} indices and pick the next Q_{pick} ones to define $I_3^s = \{s+2Q_{\text{pick}}+2Q_{\text{skip}}, s+2Q_{\text{pick}}+2Q_{\text{skip}}+1, \dots, s+3Q_{\text{pick}}+2Q_{\text{skip}}-1\}$. So on and so forth, as we alternate between skipping Q_{skip} indices and forming intervals comprised of Q_{pick} consecutive indices.

With respect to this sequence, we introduce its corresponding sequence of clusters $\mathcal{C}_1^s, \dots, \mathcal{C}_L^s$, where each cluster $\mathcal{C}_\ell^s$ is the union of weight classes over all indices in I_ℓ^s , i.e., $\mathcal{C}_\ell^s = \bigcup_{q \in I_\ell^s} \mathcal{W}_q$. Here, L stands for the largest integer ℓ for which $\mathcal{C}_\ell^s$ is non-empty.

Cluster properties. We proceed by highlighting two structural properties of the sequence $\mathcal{C}_1^s, \dots, \mathcal{C}_L^s$, which come as immediate implications of our construction:

1. For every $\ell \in [L]$, the preference weights of any two items within the cluster $\mathcal{C}_\ell^s$ differ by a factor of at most $(2n/\epsilon)^{2\lceil 1/\epsilon \rceil}$. This property holds since $\mathcal{C}_\ell^s$ is comprised of at most $Q_{\text{pick}} = 2 \cdot \lceil \frac{1}{\epsilon} \rceil \cdot \lceil \log_{1+\epsilon}(\frac{n}{\epsilon}) \rceil$ consecutive weight classes, implying that

$$\frac{\max_{i \in \mathcal{C}_\ell^s} w_i}{\min_{i \in \mathcal{C}_\ell^s} w_i} \leq (1 + \epsilon)^{Q_{\text{pick}}} \leq \left(\frac{2n}{\epsilon} \right)^{2\lceil 1/\epsilon \rceil}.$$

Hence, each cluster taken in isolation forms a bounded-ratio instance, where all item weights are polynomially-related.

2. For every $\ell \in [L-1]$, the preference weights of any two items, one in $\mathcal{C}_\ell^s$ and the other in $\mathcal{C}_{\ell+1}^s$, differ by a factor of at least $(n/\epsilon)^2$. This property holds since $\mathcal{C}_\ell^s$ and $\mathcal{C}_{\ell+1}^s$ are separated by $Q_{\text{skip}} = 2 \cdot \lceil \log_{1+\epsilon}(\frac{n}{\epsilon}) \rceil$ consecutive weight classes, implying that

$$\frac{\min_{i \in \mathcal{C}_{\ell+1}^s} w_i}{\max_{i \in \mathcal{C}_\ell^s} w_i} \geq (1 + \epsilon)^{Q_{\text{skip}}} \geq \left(\frac{n}{\epsilon} \right)^2.$$

Consequently, due to properties 1 and 2, we say that the clusters $\mathcal{C}_1^s, \dots, \mathcal{C}_L^s$ are *well-separated*.

The effect of eliminating unclustered weight classes. A careful examination of the sequence of clusters $\mathcal{C}_1^s, \dots, \mathcal{C}_L^s$ reveals that various weight classes do not appear in any of these clusters, corresponding to sequences of Q_{skip} indices over which we skip during the construction of $I_1^s, I_2^s, \dots$. Therefore, since the dynamic programming approach in Appendices EC.5.3 and EC.5.4 will limit assortment decisions to the clustered weight classes $\mathcal{C}^s = \bigcup_{\ell \in [L]} \mathcal{C}_\ell^s$, the crucial question is: Why can we overlook the revenue contributions of unclustered ones? Letting $S^* \subseteq [n]$ be an optimal assortment, the next claim shows that the shifting index s can be chosen such that we incur a negligible loss in optimality when restricting S^* to the clustered weight classes $\mathcal{C}^s$.

LEMMA EC.4. *There exists a shifting index $s^* \in [-Q_{\text{skip}} - Q_{\text{pick}} + 1, 0]$ for which $\mathcal{R}(S^* \cap \mathcal{C}^{s^*}) \geq (1 - \epsilon) \cdot \mathcal{R}(S^*)$.*

Proof. Our proof is based on a simple application of the probabilistic method (Alon and Spencer 2016). Specifically, let σ be a discrete random variable, uniformly distributed over the set of integers in $[-Q_{\text{skip}} - Q_{\text{pick}} + 1, 0]$. In this case, $S^* \cap \mathcal{C}^\sigma$ is clearly a random assortment, whose expected revenue is given by

$$\begin{aligned} \mathbb{E}_\sigma [\mathcal{R}(S^* \cap \mathcal{C}^\sigma)] &= \sum_{i \in S^*} \mathbb{E}_\sigma [\mathbb{I}[i \in \mathcal{C}^\sigma] \cdot r_i \cdot \pi(i, S^* \cap \mathcal{C}^\sigma)] \\ &= \sum_{i \in S^*} \Pr_\sigma [i \in \mathcal{C}^\sigma] \cdot r_i \cdot \mathbb{E}_\sigma [\pi(i, S^* \cap \mathcal{C}^\sigma) | i \in \mathcal{C}^\sigma] \\ &\geq \sum_{i \in S^*} \Pr_\sigma [i \in \mathcal{C}^\sigma] \cdot r_i \cdot \pi(i, S^*) \\ &= \frac{Q_{\text{pick}}}{Q_{\text{pick}} + Q_{\text{skip}}} \cdot \sum_{i \in S^*} r_i \cdot \pi(i, S^*) \\ &\geq (1 - \epsilon) \cdot \mathcal{R}(S^*) . \end{aligned}$$

Here, the first inequality follows from representation (1) of the choice probabilities, which implies that $\pi(i, S^* \cap \mathcal{C}^\sigma) \geq \pi(i, S^*)$ for any item $i \in S^* \cap \mathcal{C}^\sigma$. The third equality can easily be derived by observing that, due to the uniform choice of σ , any given weight class is clustered into $\mathcal{C}^\sigma$ with probability $\frac{Q_{\text{pick}}}{Q_{\text{pick}} + Q_{\text{skip}}}$. The last inequality holds since $Q_{\text{pick}} = \lceil \frac{1}{\epsilon} \rceil \cdot Q_{\text{skip}}$. In summary, since we have just shown that $\mathbb{E}_\sigma [\mathcal{R}(S^* \cap \mathcal{C}^\sigma)] \geq (1 - \epsilon) \cdot \mathcal{R}(S^*)$, there exists at least one realization s^* of σ for which $\mathcal{R}(S^* \cap \mathcal{C}^{s^*}) \geq (1 - \epsilon) \cdot \mathcal{R}(S^*)$. $\square$

In the remainder of this section, we assume that the shifting index s^* mentioned in Lemma EC.4 is known in advance. From an algorithmic standpoint, the value of s^* can be guessed by exhaustively enumerating all $O(\frac{1}{\epsilon^2} \log \frac{n}{\epsilon})$ integers in $[-Q_{\text{skip}} - Q_{\text{pick}} + 1, 0]$. Furthermore, out of the clusters $\mathcal{C}_1^{s^*}, \dots, \mathcal{C}_L^{s^*}$, we focus our attention on the subsequence $\mathcal{C}_{\psi(1)}^{s^*}, \dots, \mathcal{C}_{\psi(L')}^{s^*}$ of non-empty clusters. In what follows, the reference to s^* and ψ will be implicit, meaning that the clusters $\mathcal{C}_{\psi(1)}^{s^*}, \dots, \mathcal{C}_{\psi(L')}^{s^*}$ will simply be denoted by $\mathcal{C}_1, \dots, \mathcal{C}_L$.

EC.5.3. Value approximation

We proceed by describing an approximate decomposition of the expected revenue $\mathbb{E}_{S \sim B}[\mathcal{R}(S)]$ generated by any decomposable assortment B , whose realization is contained within the clusters $\mathcal{C}_1, \dots, \mathcal{C}_L$, based on the individual revenue contribution of each such cluster. This decomposition will motivate our dynamic programming approach for computing near-optimal assortments, formally presented in Appendix EC.5.4.

To describe our value approximation, for every item $i \in \mathcal{C} = \bigcup_{\ell \in [L]} \mathcal{C}_\ell$, we denote by $\varphi(i)$ the index of the unique cluster that contains item i . In addition, let $\mathcal{C}_{>\ell}$ designate the union of all clusters indexed strictly greater than ℓ , i.e., $\mathcal{C}_{>\ell} = \bigcup_{k \geq \ell+1} \mathcal{C}_k$. We remind the reader that representation (1) of the choice probabilities shows that the expected revenue of any decomposable assortment B can be written as:

$$\mathbb{E}_{S \sim B}[\mathcal{R}(S)] = \mathbb{E}_{S \sim B} \left[\sum_{i \in S} r_i \cdot \pi(i, S) \right] = \mathbb{E}_{S \sim B} \left[\sum_{i \in S} r_i \lambda_i w_i \cdot \mathbb{E} \left[\frac{1}{1 + w_i + w(C_S^{-i})} \right] \right]. \quad (\text{EC.18})$$

In Lemma EC.5 below, we provide a lower bound on the inner expectation term $\mathbb{E}[\frac{1}{1 + w_i + w(C_S^{-i})}]$ for any fixed assortment S . The important feature of this lower bound is that it can be computed knowing only the following ingredients:

1. The distribution of the random subset $S \cap \mathcal{C}_{\varphi(i)}$.
2. The probability that the induced consideration set $C_{S \cap \mathcal{C}_{>\varphi(i)}}$ is empty.

Consequently, by recalling that the preference weights within a single cluster are polynomially-related (see property 1 in Appendix EC.5.2), and by observing that item 2 above corresponds to an emptiness constraint, we facilitate an approximate reduction to the bounded-ratio setting studied in Appendix EC.5.1.

LEMMA EC.5. *For every assortment $S \subseteq \mathcal{C}$ and item $i \in S$,*

$$\mathbb{E} \left[\frac{1}{1 + w_i + w(C_S^{-i})} \right] \geq (1 - \epsilon) \cdot \Pr \left[C_{S \cap \mathcal{C}_{>\varphi(i)}} = \emptyset \right] \cdot \mathbb{E} \left[\frac{1}{1 + w_i + w(C_S^{-i} \cap \mathcal{C}_{\varphi(i)})} \right].$$

Proof. The proof proceeds by observing that

$$\begin{aligned} & \mathbb{E} \left[\frac{1}{1 + w_i + w(C_S^{-i})} \right] \\ & \geq \Pr \left[C_{S \cap \mathcal{C}_{>\varphi(i)}} = \emptyset \right] \cdot \mathbb{E} \left[\frac{1}{1 + w_i + w(C_S^{-i})} \middle| C_{S \cap \mathcal{C}_{>\varphi(i)}} = \emptyset \right] \\ & = \Pr \left[C_{S \cap \mathcal{C}_{>\varphi(i)}} = \emptyset \right] \cdot \mathbb{E} \left[\frac{1}{1 + w_i + w(C_S \cap (\bigcup_{\ell=1}^{\varphi(i)-1} \mathcal{C}_\ell)) + w(C_S^{-i} \cap \mathcal{C}_{\varphi(i)})} \right] \\ & \geq (1 - \epsilon) \cdot \Pr \left[C_{S \cap \mathcal{C}_{>\varphi(i)}} = \emptyset \right] \cdot \mathbb{E} \left[\frac{1}{1 + w_i + w(C_S^{-i} \cap \mathcal{C}_{\varphi(i)})} \right]. \end{aligned}$$

Here, the middle equality follows from the independence of $C_S \cap \mathcal{C}_{>\varphi(i)}$ and $C_S^{-i} \cap (\bigcup_{\ell=1}^{\varphi(i)} \mathcal{C}_\ell)$. The last inequality holds since

$$w \left(C_S \cap \left(\bigcup_{\ell=1}^{\varphi(i)-1} \mathcal{C}_\ell \right) \right) \leq \left| C_S \cap \left(\bigcup_{\ell=1}^{\varphi(i)-1} \mathcal{C}_\ell \right) \right| \cdot \left(\frac{\epsilon}{n} \right)^2 \cdot \min_{j \in \mathcal{C}_{\varphi(i)}} w_j \leq \epsilon w_i ,$$

given that the clusters $\mathcal{C}_1, \dots, \mathcal{C}_L$ are well-separated (see property 2 in Appendix EC.5.2). $\square$

In the next claim, we argue that the lower bound stated in Lemma EC.5 is useful in obtaining a tight approximation for the optimal expected revenue. To formalize this notion, we overload our previous notation by letting $S^* \subseteq \mathcal{C}$ be an optimal assortment restricted to weight classes clustered into $\mathcal{C}$. As shown in Lemma EC.4, this restriction has negligible losses in optimality in comparison to arbitrarily-structured assortments, that may offer items from unclustered weight classes as well.

LEMMA EC.6. *For any $\epsilon \in (0, 1/2)$,*

$$\mathcal{R}(S^*) \leq (1 + 2\epsilon) \cdot \sum_{i \in S^*} r_i \lambda_i w_i \cdot \Pr \left[C_{S^* \cap \mathcal{C}_{>\varphi(i)}} = \emptyset \right] \cdot \mathbb{E} \left[\frac{1}{1 + w_i + w(C_{S^*}^{-i} \cap \mathcal{C}_{\varphi(i)})} \right] .$$

Proof. Using arguments analogous to those appearing in the proof of Lemma EC.5, we derive an upper bound on the expectation $\mathbb{E}[\frac{1}{1 + w_i + w(C_{S^*}^{-i})}]$, for every item $i \in S^*$, by observing that

$$\begin{aligned} & \mathbb{E} \left[\frac{1}{1 + w_i + w(C_{S^*}^{-i})} \right] \\ &= \Pr \left[C_{S^* \cap \mathcal{C}_{>\varphi(i)}} \neq \emptyset \right] \cdot \mathbb{E} \left[\frac{1}{1 + w_i + w(C_{S^*}^{-i})} \middle| C_{S^* \cap \mathcal{C}_{>\varphi(i)}} \neq \emptyset \right] \\ & \quad + \Pr \left[C_{S^* \cap \mathcal{C}_{>\varphi(i)}} = \emptyset \right] \cdot \mathbb{E} \left[\frac{1}{1 + w_i + w(C_{S^*}^{-i})} \middle| C_{S^* \cap \mathcal{C}_{>\varphi(i)}} = \emptyset \right] \\ &\leq \frac{1}{1 + w_i + (\frac{n}{\epsilon})^2 \cdot w_i} + \Pr \left[C_{S^* \cap \mathcal{C}_{>\varphi(i)}} = \emptyset \right] \cdot \mathbb{E} \left[\frac{1}{1 + w_i + w(C_{S^* \cap \mathcal{C}_{\varphi(i)}}^{-i})} \right] \\ &\leq \frac{\epsilon}{n} \cdot \frac{1}{1 + w_i} + \Pr \left[C_{S^* \cap \mathcal{C}_{>\varphi(i)}} = \emptyset \right] \cdot \mathbb{E} \left[\frac{1}{1 + w_i + w(C_{S^* \cap \mathcal{C}_{\varphi(i)}}^{-i})} \right] . \end{aligned} \tag{EC.19}$$

Here, the first inequality holds since the clusters $\mathcal{C}_1, \dots, \mathcal{C}_L$ are well-separated, implying that conditional on the random consideration set $C_{S^* \cap \mathcal{C}_{>\varphi(i)}}$ being non-empty, its random total weight is at least $(\frac{n}{\epsilon})^2 \cdot w_i$. The second inequality is obtained by observing that $1 + w_i + (\frac{n}{\epsilon})^2 \cdot w_i \geq \frac{n}{\epsilon} \cdot (1 + w_i)$ whenever $w_i \geq \frac{\epsilon}{n}$. The property that $w_i \geq \frac{\epsilon}{n}$ follows from our assumption that the instance parameters are specified in rounded form, meaning that $w_i = \frac{\epsilon}{n} \cdot (1 + \epsilon)^q$ for some $q \geq 0$.

Now, according to the revenue representation (EC.18), the optimal expected revenue $\mathcal{R}(S^*)$ can be upper bounded as follows:

$$\mathcal{R}(S^*) = \sum_{i \in S^*} r_i \lambda_i w_i \cdot \mathbb{E} \left[\frac{1}{1 + w_i + w(C_{S^*}^{-i})} \right]$$

$$\begin{aligned}
&\leq \frac{\epsilon}{n} \cdot \sum_{i \in S^*} \frac{r_i \lambda_i w_i}{1 + w_i} + \sum_{i \in S^*} r_i \lambda_i w_i \cdot \Pr \left[C_{S^* \cap \mathcal{C}_{>\varphi(i)}} = \emptyset \right] \cdot \mathbb{E} \left[\frac{1}{1 + w_i + w(C_{S^* \cap \mathcal{C}_{\varphi(i)}}^{-i})} \right] \\
&\leq \epsilon \cdot \mathcal{R}(S^*) + \sum_{i \in S^*} r_i \lambda_i w_i \cdot \Pr \left[C_{S^* \cap \mathcal{C}_{>\varphi(i)}} = \emptyset \right] \cdot \mathbb{E} \left[\frac{1}{1 + w_i + w(C_{S^* \cap \mathcal{C}_{\varphi(i)}}^{-i})} \right],
\end{aligned}$$

where the first inequality is obtained by plugging the upper bound (EC.19) and the second inequality holds since $\mathcal{R}(S^*) \geq r_i \lambda_i \cdot \frac{w_i}{1+w_i}$ for any item $i \in \mathcal{C}$. To see this, note that the optimality of S^* ensures that its expected revenue $\mathcal{R}(S^*)$ is at least the expected revenue of the singleton assortment $\{i\}$, which is precisely $r_i \lambda_i \cdot \frac{w_i}{1+w_i}$. The desired claim is now obtained by rearranging the above bound. $\square$

Based on the preceding discussion, Lemmas EC.5 and EC.6 immediately imply that, in order to compute a random assortment whose expected revenue is within factor $1 - O(\epsilon)$ of the optimal revenue $\mathcal{R}(S^*)$, it suffices to optimize the value approximation $\tilde{\mathcal{R}}(\cdot)$, defined over the set of decomposable assortments B as follows:

$$\begin{aligned}
\tilde{\mathcal{R}}(B) &= \mathbb{E}_{S \sim B} \left[\sum_{i \in S \cap \mathcal{C}} \Pr \left[C_{S \cap \mathcal{C}_{>\varphi(i)}} = \emptyset \right] \cdot r_i \lambda_i w_i \cdot \mathbb{E} \left[\frac{1}{1 + w_i + w(C_{S \cap \mathcal{C}_{\varphi(i)}}^{-i})} \right] \right] \\
&= \mathbb{E}_{S \sim B} \left[\sum_{\ell=1}^L \left(\Pr \left[C_{S \cap \mathcal{C}_{>\ell}} = \emptyset \right] \cdot \sum_{i \in S \cap \mathcal{C}_\ell} r_i \lambda_i w_i \cdot \mathbb{E} \left[\frac{1}{1 + w_i + w(C_{S \cap \mathcal{C}_\ell}^{-i})} \right] \right) \right] \\
&= \sum_{\ell=1}^L \Pr_{S \sim B} \left[C_{S \cap \mathcal{C}_{>\ell}} = \emptyset \right] \cdot \mathbb{E}_{S \sim B} [\mathcal{R}(S \cap \mathcal{C}_\ell)], \tag{EC.20}
\end{aligned}$$

where the second equality is simply a decomposition of $S \cap \mathcal{C}$ into the clusters $\mathcal{C}_1, \dots, \mathcal{C}_L$, and the third equality follows by noting that the inner sum is precisely the expected revenue generated by $S \cap \mathcal{C}_\ell$, conditional on S being the realization of the decomposable assortment B .

EC.5.4. Gluing approximate dynamic program

Continuous dynamic program. The important observation is that the problem of maximizing the value approximation of Appendix EC.5.3 can be formulated as a dynamic program whose state space is described by the following parameters:

- An index $\ell \in [L]$, corresponding to the cluster $\mathcal{C}_\ell$ that is currently being considered.
- A continuous variable $\theta \in [0, 1]$, specifying the probability that none of the items picked from higher-index clusters $\mathcal{C}_{>\ell}$ will appear in our random consideration set.

Given these parameters, we define $F(\ell, \theta)$ as the maximum possible value of $\sum_{k=1}^{\ell} \Pr_{S \sim B} [C_{S \cap \mathcal{C}_{>k}} = \emptyset] \cdot \mathbb{E}_{S \sim B} [\mathcal{R}(S \cap \mathcal{C}_k)]$ over all decomposable assortments B , subject to the probabilistic constraint $\Pr_{S \sim B} [C_{S \cap \mathcal{C}_{>\ell}} = \emptyset] = \theta$. An immediate implication of this definition is that, in order to compute an assortment maximizing the value approximation $\tilde{\mathcal{R}}(\cdot)$, we can equivalently compute the one for

which $F(L, 1)$ is attained. Concurrently, the definition of F can be directly used to express this function in recursive form, by observing that for $\ell \geq 2$,

$$F(\ell, \theta) = \max_{B_\ell \in \mathcal{B}_\ell} \left\{ \theta \cdot \mathbb{E}_{S \sim B_\ell} [\mathcal{R}(S)] + F(\ell - 1, \theta \cdot \Pr_{S \sim B_\ell} [C_S = \emptyset]) \right\}. \quad (\text{EC.21})$$

Here, $\mathcal{B}_\ell$ designates the collection of decomposable assortments over the set of items $\mathcal{C}_\ell$. In addition, for $\ell = 1$, we have

$$F(1, \theta) = \theta \cdot \max_{B_1 \in \mathcal{B}_1} \mathbb{E}_{S \sim B_1} [\mathcal{R}(S)]. \quad (\text{EC.22})$$

Due to the continuity of $\theta \in [0, 1]$, the function F should be viewed as a recursive characterization of optimal random assortments for our value approximation $\tilde{\mathcal{R}}(\cdot)$.

Discretization. For this reason, we explain how to approximately solve the recursive equations (EC.21) and (EC.22) by leveraging the bounded-ratio algorithm given in Theorem EC.1 as a subroutine. To simplify the exposition, we assume that $\max_{i \in \mathcal{C}} \lambda_i \leq 1 - \epsilon$. By the sensitivity analysis established in Lemma 1, this assumption can be made without loss of generality.

Now, in order to discretize the state variable θ , we define the finite set $\Theta = \{(1 + \frac{\epsilon}{n})^{-k} : 0 \leq k \leq n \cdot (1 + \lceil \log_{1+\frac{\epsilon}{n}}(\frac{1}{\epsilon}) \rceil)\}$. For any $\Phi \in [0, 1]$, let $B(\ell, \Phi) \in \mathcal{B}_\ell$ be the decomposable assortment returned by the approximation scheme stated in Theorem EC.1, when the underlying set of items is precisely the cluster $\mathcal{C}_\ell$, subject to the emptiness constraint $\Pr_{S \sim B} [C_S = \emptyset] \geq \Phi$.

With these definitions at hand, we introduce a discrete dynamic program $\tilde{F}$ over the state space $[L] \times \Theta$, i.e., a formulation where our second parameter θ is restricted to take values only within Θ . This program is formally specified through the following recursive equations, for $\ell \geq 2$,

$$\tilde{F}(\ell, \theta) = \max_{\substack{\tilde{\kappa} \in \Theta: \\ \tilde{\kappa} \leq \theta}} \left\{ \theta \cdot \mathbb{E}_{S \sim B(\ell, \tilde{\kappa}/\theta)} [\mathcal{R}(S)] + \tilde{F}(\ell - 1, \tilde{\kappa}) \right\}. \quad (\text{EC.23})$$

In addition, for $\ell = 1$, we have

$$\tilde{F}(1, \theta) = \theta \cdot \mathbb{E}_{S \sim B(1, 0)} [\mathcal{R}(S)]. \quad (\text{EC.24})$$

Analysis. By property 1 in Appendix EC.5.2, within each cluster $\mathcal{C}_\ell$, the ratio between the extremal preference weights is upper-bounded by $(\frac{2n}{\epsilon})^{2\lceil 1/\epsilon \rceil}$. Hence, it is easy to verify that the instance of the constrained click-based MNL instance corresponding to each cluster satisfies Assumption 1. By leveraging the bounded-ratio algorithm of Theorem EC.1, we infer that our recursion can be solved in time $O((\frac{n}{\epsilon})^{O(\frac{1}{\epsilon^4} \log \frac{1}{\epsilon})})$. To establish the performance guarantee of our algorithm, let $\tilde{B}^*$ be the optimal decomposable assortment resulting from the solution of equations (EC.23) and (EC.24). Namely, $\tilde{B}^*$ is obtained by concatenating the decomposable assortments picked at each step of the recursion, in the optimal sequence of dynamic programming decisions that attain the function value $\tilde{F}(L, 1)$. In the remainder of this section, we conclude the analysis by relating $\tilde{\mathcal{R}}(\tilde{B}^*)$ to the optimum value $F(L, 1)$ of the continuous dynamic program F , which was shown earlier to capture the assortment maximizing our value approximation $\tilde{\mathcal{R}}(\cdot)$.

LEMMA EC.7. $\tilde{\mathcal{R}}(\tilde{B}^*) \geq (1 - 3\epsilon) \cdot F(L, 1)$.

To prove the desired claim, we begin by considering the optimal sequence of dynamic programming decisions for equations (EC.23) and (EC.24) that yields the random assortment $\tilde{B}^*$. This sequence is denoted by $(\tilde{\theta}_1^*, \dots, \tilde{\theta}_L^*) \in \Theta^L$, where $\tilde{\theta}_L^* = 1$. Namely, $\tilde{B}^* = \bigcup_{\ell=1}^L B(\ell, \tilde{\Phi}_\ell^*)$ where $\tilde{\Phi}_\ell^* = \tilde{\theta}_{\ell-1}^* / \tilde{\theta}_\ell^*$ for every $\ell \in [1, L]$ and $\tilde{\theta}_0^* = 0$. In the next claim, we relate our value approximation for the decomposable assortment $\tilde{B}^*$ to the value computed by the discretized dynamic program (EC.23)-(EC.24) along the sequence of decisions $(\tilde{\theta}_1^*, \dots, \tilde{\theta}_L^*)$.

CLAIM EC.11. $\tilde{\mathcal{R}}(\tilde{B}^*) = \sum_{\ell=1}^L \tilde{\theta}_\ell^* \cdot \mathbb{E}_{S \sim B(\ell, \tilde{\Phi}_\ell^*)} [\mathcal{R}(S)]$.

Proof. By definition of $\tilde{\mathcal{R}}(\cdot)$ in equation (EC.20), we have:

$$\begin{aligned} \tilde{\mathcal{R}}(\tilde{B}^*) &= \mathbb{E}_{S \sim \tilde{B}^*} [\mathcal{R}(S \cap \mathcal{C}_L)] + \sum_{\ell=1}^{L-1} \Pr_{S \sim \tilde{B}^*} [C_{S \cap \mathcal{C}_{>\ell}} = \emptyset] \cdot \mathbb{E}_{S \sim \tilde{B}^*} [\mathcal{R}(S \cap \mathcal{C}_\ell)] \\ &= \mathbb{E}_{S \sim B(L, \tilde{\Phi}_L^*)} [\mathcal{R}(S)] + \sum_{\ell=1}^{L-1} \left(\prod_{k=\ell+1}^L \Pr_{S \sim B(k, \tilde{\Phi}_k^*)} [C_S = \emptyset] \right) \cdot \mathbb{E}_{S \sim B(\ell, \tilde{\Phi}_\ell^*)} [\mathcal{R}(S)] \\ &= \mathbb{E}_{S \sim B(L, \tilde{\Phi}_L^*)} [\mathcal{R}(S)] + \sum_{\ell=1}^{L-1} \left(\prod_{k=\ell+1}^L \frac{\tilde{\theta}_{k-1}^*}{\tilde{\theta}_k^*} \right) \cdot \mathbb{E}_{S \sim B(\ell, \tilde{\Phi}_\ell^*)} [\mathcal{R}(S)] \\ &= \sum_{\ell=1}^L \tilde{\theta}_\ell^* \cdot \mathbb{E}_{S \sim B(\ell, \tilde{\Phi}_\ell^*)} [\mathcal{R}(S)] , \end{aligned}$$

where the second equality holds since the random sets $C_{S(k_1)}$ and $C_{S(k_2)}$ are independent for every $k_1 \neq k_2$, where $S(k)$ is the random realization of $B(k, \tilde{\Phi}_k^*)$. The next equality follows from the definition of $B(k, \tilde{\Phi}_k^*)$ as a decomposable assortment that satisfies the emptiness constraint $\Pr_{S \sim B(k, \tilde{\Phi}_k^*)} [C_S = \emptyset] \geq \tilde{\Phi}_k^*$. $\square$

Conversely, we now consider a decomposable assortment $\mathcal{A}^*$ that maximizes our value approximation $\tilde{\mathcal{R}}(\cdot)$, namely $\tilde{\mathcal{R}}(\mathcal{A}^*) = F(L, 1)$. Consequently, we recursively define $\theta_L^* = 1$, and $\theta_\ell^* = \lfloor \theta_{\ell+1}^* \cdot \Pr_{S \sim \mathcal{A}^*} [C_{S \cap \mathcal{C}_{\ell+1}} = \emptyset] \rfloor_\Theta$ for every $\ell \in [1, L-1]$, where $\lfloor \cdot \rfloor_\Theta$ is an operator that rounds its argument down to the nearest number in Θ . Next, we define $\Phi_\ell^* = \theta_{\ell-1}^* / \theta_\ell^*$ for every $\ell \in [1, L]$, where $\theta_0^* = 0$. It is not difficult to verify that the sequence of dynamic programming decisions $(\theta_1^*, \dots, \theta_L^*)$ is feasible with respect to the discretized dynamic program (EC.23)-(EC.24). To conclude, we relate the value generated by this sequence to $F(L, 1)$.

CLAIM EC.12. $\sum_{\ell=1}^L \theta_\ell^* \cdot \mathbb{E}_{S \sim B(\ell, \Phi_\ell^*)} [\mathcal{R}(S)] \geq (1 - 3\epsilon) \cdot F(L, 1)$.

Before presenting the proof of Claim EC.12, we explain how the desired inequality proceeds from the above two claims. By Claim EC.11, we have:

$$\tilde{\mathcal{R}}(\tilde{B}^*) = \sum_{\ell=1}^L \tilde{\theta}_\ell^* \cdot \mathbb{E}_{S \sim B(\ell, \tilde{\Phi}_\ell^*)} [\mathcal{R}(S)]$$

$$\begin{aligned}
&\geq \sum_{\ell=1}^L \theta_\ell^* \cdot \mathbb{E}_{S \sim B(\ell, \Phi_\ell^*)} [\mathcal{R}(S)] \\
&\geq (1 - 3\epsilon) \cdot F(L, 1) ,
\end{aligned}$$

where the first inequality holds by the optimality of the dynamic programming decisions $(\tilde{\theta}_1^*, \dots, \tilde{\theta}_L^*)$, and the last inequality immediately follows from Claim EC.12.

Proof of Claim EC.12. We begin by bounding the approximation error generated by the rounding operator $\lfloor \cdot \rfloor_\Theta$. By the definition of the set Θ , for every $x \geq \epsilon^n \cdot (1 + \frac{\epsilon}{n})^{-n}$, we necessarily have $(1 - \frac{\epsilon}{n}) \cdot x \leq \lfloor x \rfloor_\Theta \leq x$. Hence, by our earlier assumption that $1 - \lambda_i \geq \epsilon$ for every $i \in [n]$, and by using the relationships $\theta_{\ell-1}^* = \lfloor \theta_\ell^* \cdot \Pr_{S \sim \mathcal{A}^*} [C_{S \cap \mathcal{C}_\ell} = \emptyset] \rfloor_\Theta$ and $\Phi_\ell^* = \theta_{\ell-1}^* / \theta_\ell^*$, a straightforward induction over $\ell \in [1, L]$ yields

$$\left(1 - \frac{\epsilon}{n}\right) \cdot \Pr_{S \sim \mathcal{A}^*} [C_{S \cap \mathcal{C}_\ell} = \emptyset] \leq \Phi_\ell^* \leq \Pr_{S \sim \mathcal{A}^*} [C_{S \cap \mathcal{C}_\ell} = \emptyset] . \quad (\text{EC.25})$$

Thus, by noting that $\theta_\ell^* = \prod_{k=\ell+1}^L \Phi_k^*$, we obtain

$$\theta_\ell^* \geq \left(1 - \frac{\epsilon}{n}\right)^{L-\ell} \cdot \Pr_{S \sim \mathcal{A}^*} [C_{S \cap \mathcal{C}_{>\ell}} = \emptyset] . \quad (\text{EC.26})$$

In addition, for every $\ell \in [1, L]$, by applying Theorem EC.1 to the collection of items contained in $\mathcal{C}_\ell$ with $\Phi = \tilde{\Phi}_\ell^*$, we infer from inequality (EC.25) that

$$\mathbb{E}_{S \sim B(\ell, \Phi_\ell^*)} [\mathcal{R}(S)] \geq (1 - \epsilon) \cdot \mathbb{E}_{S \sim \mathcal{A}^*} [\mathcal{R}(S \cap \mathcal{C}_\ell)] . \quad (\text{EC.27})$$

By combining inequalities (EC.26) and (EC.27), we finally obtain

$$\begin{aligned}
\sum_{\ell=1}^L \theta_\ell^* \cdot \mathbb{E}_{S \sim B(\ell, \Phi_\ell^*)} [\mathcal{R}(S)] &\geq (1 - \epsilon) \cdot \sum_{\ell=1}^L \left(1 - \frac{\epsilon}{n}\right)^{L-\ell} \cdot \Pr_{S \sim \mathcal{A}^*} [C_{S \cap \mathcal{C}_{>\ell}} = \emptyset] \cdot \mathbb{E}_{S \sim \mathcal{A}^*} [\mathcal{R}(S \cap \mathcal{C}_\ell)] \\
&\geq (1 - \epsilon) \cdot \left(1 - \frac{\epsilon}{n}\right)^n \cdot \sum_{\ell=1}^L \Pr_{S \sim \mathcal{A}^*} [C_{S \cap \mathcal{C}_{>\ell}} = \emptyset] \cdot \mathbb{E}_{S \sim \mathcal{A}^*} [\mathcal{R}(S \cap \mathcal{C}_\ell)] \\
&\geq (1 - 3\epsilon) \cdot \tilde{\mathcal{R}}(\mathcal{A}^*) \\
&= (1 - 3\epsilon) \cdot F(L, 1) .
\end{aligned}$$

where the second inequality holds since $L \leq n$, by observing that the number of non-empty clusters, as defined in Section EC.5.2, is at most n . $\square$

EC.6. Leveraging Rare Items

In this section, we formally establish the efficacy of our greedy approach for choosing B_{rare} . More specifically, letting $\mathcal{N}_{\text{rare}} = \{i \in [n] : \lambda_i < \epsilon/n\}$ denote the set of rare items, we prove the following result.

LEMMA EC.8. *Let B_{likely} and B_{unlikely} be the collections of likely and unlikely items computed by the Product Recommendation PTAS. Then, for any set of rare items $S_{\text{rare}} \subseteq \mathcal{N}_{\text{rare}}$ that satisfies $|S_{\text{rare}}| = C - |B_{\text{likely}} \cup B_{\text{unlikely}}|$,*

$$\mathcal{R}(B_{\text{likely}} \cup B_{\text{unlikely}} \cup B_{\text{rare}}) \geq (1 - 2\epsilon) \cdot \mathcal{R}(B_{\text{likely}} \cup B_{\text{unlikely}} \cup S_{\text{rare}}) .$$

Proof. Let S_{rare}^* be an assortment that maximizes $\mathcal{R}(B_{\text{likely}} \cup B_{\text{unlikely}} \cup S_{\text{rare}})$ over all $S_{\text{rare}} \subseteq \mathcal{N}_{\text{rare}}$ with $|S_{\text{rare}}| = C - |B_{\text{likely}} \cup B_{\text{unlikely}}|$, and let $B_{\text{other}} = B_{\text{likely}} \cup B_{\text{unlikely}}$. To prove the desired claim lemma, it suffices to argue that

$$\sum_{i \in B_{\text{rare}}} r_i \cdot \pi(i, B_{\text{other}} \cup B_{\text{rare}}) \geq (1 - 2\epsilon) \cdot \sum_{i \in S_{\text{rare}}^*} r_i \cdot \pi(i, B_{\text{other}} \cup S_{\text{rare}}^*), \quad (\text{EC.28})$$

and that

$$\sum_{i \in B_{\text{other}}} r_i \cdot \pi(i, B_{\text{other}} \cup B_{\text{rare}}) \geq (1 - 2\epsilon) \cdot \sum_{i \in B_{\text{other}}} r_i \cdot \pi(i, B_{\text{other}} \cup S_{\text{rare}}^*). \quad (\text{EC.29})$$

To prove (EC.28), observe that

$$\begin{aligned} \sum_{i \in B_{\text{rare}}} r_i \cdot \pi(i, B_{\text{other}} \cup B_{\text{rare}}) &\geq \sum_{i \in B_{\text{rare}}} \lambda_i r_i w_i \cdot \mathbb{E} \left[\frac{1}{1 + w_i + w(C_{B_{\text{other}}}) + w(C_{B_{\text{rare}}})} \right] \\ &\geq \Pr[C_{B_{\text{rare}}} = \emptyset] \cdot \sum_{i \in B_{\text{rare}}} \lambda_i r_i w_i \cdot \mathbb{E} \left[\frac{1}{1 + w_i + w(C_{B_{\text{other}}})} \right] \\ &\geq \left(1 - \frac{\epsilon}{n}\right)^n \cdot \sum_{i \in B_{\text{rare}}} \lambda_i r_i w_i \cdot \mathbb{E} \left[\frac{1}{1 + w_i + w(C_{B_{\text{other}}})} \right] \\ &\geq (1 - 2\epsilon) \cdot \sum_{i \in B_{\text{rare}}} \lambda_i r_i w_i \cdot \mathbb{E} \left[\frac{1}{1 + w_i + w(C_{B_{\text{other}}})} \right] \\ &= (1 - 2\epsilon) \cdot \sum_{i \in B_{\text{rare}}} \rho_i, \end{aligned}$$

where the last equality follows by recalling that $\rho_i = r_i \cdot \pi(i, B_{\text{other}} \cup \{i\})$. We proceed by observing that

$$\sum_{i \in B_{\text{rare}}} \rho_i \geq \sum_{i \in S_{\text{rare}}^*} \rho_i = \sum_{i \in S_{\text{rare}}^*} r_i \cdot \pi(i, B_{\text{other}} \cup \{i\}) \geq \sum_{i \in S_{\text{rare}}^*} r_i \cdot \pi(i, B_{\text{other}} \cup S_{\text{rare}}^*) .$$

Here, the first inequality holds since B_{rare} is the assortment of rare items $S \subseteq \mathcal{N}_{\text{rare}}$ that maximizes the quantity $\sum_{i \in S} \rho_i$ out of all assortments with cardinality $C - |B_{\text{likely}} \cup B_{\text{unlikely}}|$. The second inequality holds since $(B_{\text{other}} \cup \{i\}) \subseteq (B_{\text{other}} \cup S_{\text{rare}}^*)$ for any $i \in S_{\text{rare}}^*$. Inequality (EC.28) immediately follows from the above sequence of inequalities.

We proceed to prove (EC.29), by noting that

$$\sum_{i \in B_{\text{other}}} r_i \cdot \mathbb{E}[\pi(i, B_{\text{other}} \cup B_{\text{rare}})] \geq \Pr[C_{B_{\text{rare}}} = \emptyset] \cdot \sum_{i \in B_{\text{other}}} \lambda_i r_i w_i \cdot \mathbb{E} \left[\frac{1}{1 + w_i + w(C_{B_{\text{other}}})} \right]$$

$$\begin{aligned}
&\geq (1 - 2\epsilon) \cdot \sum_{i \in B_{\text{other}}} r_i \cdot \pi(i, B_{\text{other}}) \\
&\geq (1 - 2\epsilon) \cdot \sum_{i \in B_{\text{other}}} r_i \cdot \pi(i, B_{\text{other}} \cup S_{\text{rare}}^*),
\end{aligned}$$

where the last inequality holds since $B_{\text{other}} \subseteq B_{\text{other}} \cup S_{\text{rare}}^*$. $\square$

EC.7. Estimation Case Study

EC.7.1. Fitting the click-based MNL model

In this section, we describe the methods used to estimate the parameters of the click-based MNL using maximum likelihood estimation (MLE). To begin, for each seller, we employ a 75/25 train/test split of the sales data. We then combine the training data for each seller and use this aggregated data set to seed the MLE problem. Once the optimal sets of parameters have been derived, we measure the accuracy of the fitted model using the log-likelihood computed on the testing sets of each individual seller. The series of steps described above – 75/25 train/test split, solving the MLE problems, computing the log-likelihood on the test data sets – make up what we refer to as a single trial. In our main table, we report the average out-of-sample log-likelihood for each seller over 20 trials. We report summary statistics for the fitted click-based MNL models in Appendix EC.7.2.

Log-likelihood estimation. Under the MNL model, the random utility that customer t associates with product $i \in S_t$ is given by $U_{it} = V_{it} + \epsilon_{it}$, where V_{it} is the known deterministic component of this utility and ϵ_{it} is an i.i.d. Gumbel random variable that captures heterogeneity in the customer population. To incorporate product and customer features, one can write the deterministic component as a linear combination of the feature values, meaning that $V_{it} = \beta^T X_{it}$, where X_{it} is the vector of feature values that customer t associated with product $i \in S_t$. In this case, the probability that customer t purchases product $i \in S_t$ is given by

$$\pi(i, S_t, X_t) = \frac{e^{\beta^T X_{it}}}{1 + \sum_{j \in S_t} e^{\beta^T X_{jt}}},$$

whereas the no-purchase option is selected with probability

$$\pi(0, S_t, X_t) = \frac{1}{1 + \sum_{j \in S_t} e^{\beta^T X_{jt}}}.$$

Having fully described the second stage of our click-based MNL model, we now formalize the initial consideration set formulation step. Initially, we assume that the click probabilities do not depend on the feature sets of arriving customers. As a result, we let λ_i be the probability $\Pr[c_{it} = 1]$ that any given customer t clicks on product $i \in S_t$. We use $\lambda = \{\lambda_i : i \in \mathcal{N}\}$ to denote the set of all click probabilities.

With this notation, we formulate the log-likelihood, written as a function of PurchaseHistory, under the click-based MNL model as follows:

$$\begin{aligned}
\mathcal{LL}(\beta, \lambda | \text{PurchaseHistory}) &= \log \left(\prod_{t=1}^{\tau} \left(\prod_{i \in C_{S_t}} \lambda_i \prod_{i \in S_t \setminus C_{S_t}} (1 - \lambda_i) \right) \pi(z_t, C_{S_t}, X_t) \right) \\
&= \sum_{t=1}^{\tau} \left(\sum_{i \in C_{S_t}} \log \lambda_i + \sum_{i \in S_t \setminus C_{S_t}} \log (1 - \lambda_i) \right) + \sum_{t=1}^{\tau} \left(\beta^T X_{z_t t} - \log \left(1 + \sum_{i \in C_{S_t}} e^{\beta^T X_{it}} \right) \right). \quad (\text{EC.30})
\end{aligned}$$

Therefore, the MLE problem of interest, used to derive estimates of the feature weights β and the click probabilities λ , is given by:

$$\max_{\lambda \in [0,1]^n, \beta} \mathcal{LL}(\beta, \lambda | \text{PurchaseHistory}). \quad (\text{MLE Click})$$

Clearly, given the log-likelihood function in (EC.30), the MLE problem decomposes into independent optimizations problems, one over the feature weights β and the other over the click probabilities λ . Hence, (MLE Click) can be rewritten as:

$$\begin{aligned}
\max_{\lambda \in [0,1]^n, \beta} \mathcal{LL}(\beta, \lambda | \text{PurchaseHistory}) &= \max_{\lambda \in [0,1]^n} \sum_{t=1}^{\tau} \left(\sum_{i \in C_{S_t}} \log \lambda_i + \sum_{i \in S_t \setminus C_{S_t}} \log (1 - \lambda_i) \right) \\
&\quad + \max_{\beta} \sum_{t=1}^{\tau} \left(\beta^T X_{z_t t} - \log \left(1 + \sum_{i \in C_{S_t}} e^{\beta^T X_{it}} \right) \right).
\end{aligned}$$

Solving the MLE problem. Fortunately, the problem of finding the optimal feature weights β^* is exactly an MLE problem under the standard MNL model, which is a well-known convex optimization problem (see Chapter 3 of Train (2009)). To numerically solve this problem, we use Tensorflow (Abadi et al. 2015), through which we implement a highly optimized version of stochastic gradient descent.

Since the click probabilities follow a Multinomial distribution, it is easy to verify that the optimal click probabilities are given by $\lambda_i^* = \sum_{t=1}^{\tau} c_{it} / \sum_{t=1}^{\tau} \mathbb{1}_{i \in S_t}$. In other words, each optimal click probability λ_i^* is merely the empirical click probability, i.e., the proportion of customers who clicked product i out of those to whom this product was offered. **Although this item-level estimator for λ^* is easy to compute, its predictive performance on out-of-sample data is poor due to overfitting and not controlling for customer attributes. Indeed, most sellers carry more than 100 products, and certain products are rarely clicked on, meaning that our data is insufficient to estimate a single click rate parameter for each item. Instead, we estimate the click rates as the outputs of a machine learning classifier, taking as input the same set of 25 features used to specify the deterministic utilities V_{it} of the MNL choice model. Specifically, we fit a gradient-boosted classification**

tree $\lambda_{it} = g(X_{it})$ to the click data, where $g(\cdot)$ is the resultant mapping from customer-item input features to click rates. The Python code for implementing catboost (Prokhorenkova et al. 2018) and the hyperparameter tuning are detailed below.

As spelled out in Figure EC.1, the following function takes as input a pandas dataframe `sales_data`, which has a row for each product displayed to each customer and whose columns give the various feature values. Furthermore, an additional binary column (`is_click`) indicates whether the given product was clicked or not. The second input to this function is `feature_list`, which is a list of column names corresponding to the features that will be used to fit the catboost model.

In the code below, we optimize over two hyperparameters, `depth` and `l2_leaf_reg`, whose exact nature is formalized next. These two hyperparameters were selected after extensive trial and error to find the hyperparameters that had the greatest effect on prediction accuracy. The `depth` parameter controls the maximum depth allowed for any of the fitted trees and the `l2_leaf_reg` parameter is the coefficient of the L2 regularization term of the cost function that is minimized.

```
def Get_Click_Probs_Catboost(sales_data, feature_list):

    # Get feature values (X) and response (y) for each customer.
    # Convert both to numpy arrays for catboost
    X = np.array(sales_data.loc[:, feature_list])
    y = np.array(sales_data.is_click)

    # Hyperparameter tuning
    params = {'depth': [3,4,5], 'l2_leaf_reg': [1,4,9]}

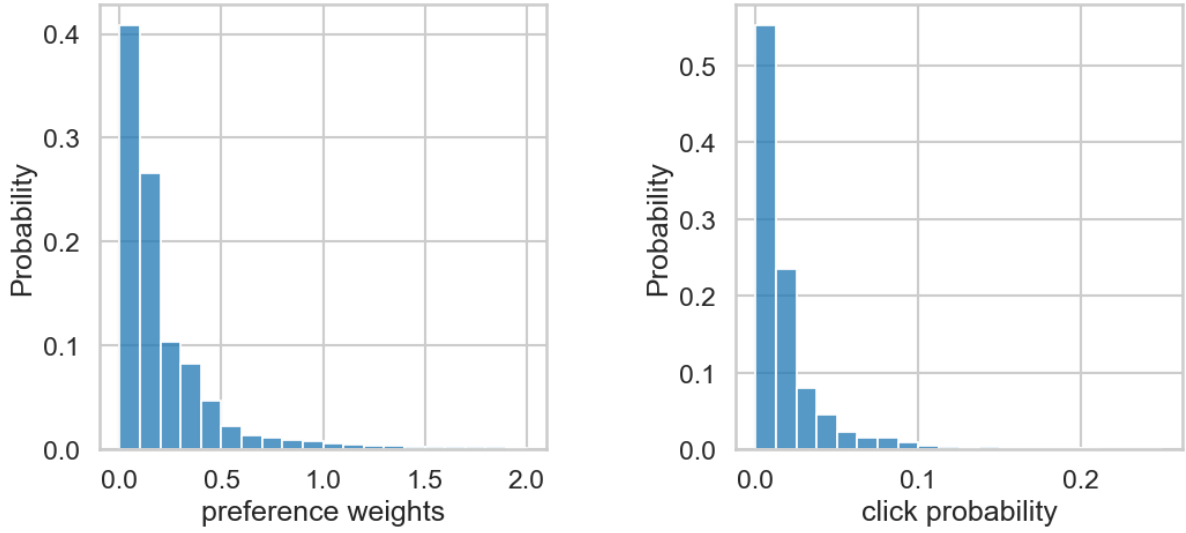
    # Instantiate a catboost model, and fit with X and y.
    # We do 5-fold cross validation (cv=5) to get the best hyperparameters
    cb = CatBoostClassifier(logging_level = "Silent")
    model = GridSearchCV(cb, params, cv = 5).fit(X, y)

    return model
```

Figure EC.1 The catboost implementation

EC.7.2. Descriptive statistics of click-based MNL models

In this section, we report descriptive statistics of the click-based MNL instances fitted to Alibaba's data and conduct a correlational analysis of click data to investigate the extent to which clicks on products are mutually independent.



(a) Preference weights of fitted MNL models

(b) Estimated click rates from Catboost models

Figure EC.2 Distribution of estimated click rates and preference weights over all customers-items-sellers**Table EC.1** Summary statistics of preference weights for the fitted click-based MNL models

Seller ID	Sample #	Mean	Std	Min	25%	50%	75%	Max
1	200 430	0.144	0.275	0.014	0.055	0.071	0.111	3.098
2	1 025 334	0.174	0.144	0.022	0.075	0.115	0.249	1.195
3	433 290	0.363	0.398	0.027	0.124	0.223	0.419	2.694
4	189 474	0.281	0.486	0.025	0.059	0.076	0.154	2.149
5	631 314	0.222	0.244	0.033	0.065	0.109	0.311	1.411
6	304 308	0.320	0.290	0.017	0.126	0.196	0.439	2.364
7	161 148	0.538	0.667	0.054	0.113	0.252	0.622	3.353
8	192 468	0.398	0.588	0.029	0.110	0.155	0.396	3.346
9	670 896	0.123	0.147	0.022	0.051	0.079	0.127	1.019
10	182 022	0.663	0.981	0.122	0.167	0.224	0.561	4.052

Table EC.2 Summary statistics of click rates for the fitted click-based MNL models

Seller	Sample #	Mean	Std	Min	25%	50%	75%	Max
1	200 430	0.033	0.049	1.630×10^{-5}	6.183×10^{-3}	0.012	0.030	0.802
2	1 025 334	0.009	0.009	3.870×10^{-8}	4.148×10^{-3}	0.006	0.010	0.800
3	433 290	0.018	0.017	2.570×10^{-4}	7.772×10^{-3}	0.013	0.022	0.644
4	189 474	0.029	0.030	2.950×10^{-5}	8.639×10^{-3}	0.018	0.038	0.772
5	631 314	0.040	0.056	6.800×10^{-5}	1.141×10^{-2}	0.020	0.040	0.864
6	304 308	0.019	0.025	1.850×10^{-6}	6.415×10^{-3}	0.011	0.020	0.943
7	161 148	0.024	0.029	1.380×10^{-4}	7.297×10^{-3}	0.014	0.024	0.496
8	192 468	0.017	0.027	1.220×10^{-5}	3.910×10^{-3}	0.008	0.018	0.879
9	670 896	0.021	0.016	5.400×10^{-7}	9.118×10^{-3}	0.014	0.031	0.740
10	182 022	0.015	0.031	8.020×10^{-5}	8.470×10^{-4}	0.001	0.007	0.866

Description of click-based MNL instances. In Tables EC.1 and EC.2, we report the summary statistics for the consideration probabilities and preference weights on the Alibaba data, respectively. we report the summary statistics of the fitted consideration probabilities and preference weights from the experiments of Section 5. Specifically, we consider the first of the ten training sets considered (so approximately 75% of the sales data), for which we impute the consideration probability and preference weight of each offered product. These distributions are visualized in Figure EC.2. There is some variability across the 10 sellers, particularly concerning the preference weights. Across all sellers, we find that click probabilities are below 1.1% for nearly 50% of items and below 8% for 95%. The preference weights are below 0.13 for about 51% of all items and below 1 for 95%. Nonetheless, preference weights can be as large as 4 – a purchase rate of nearly 80% for the corresponding item conditional on being the only one clicked and considered – and click rates can be as high as 86% for certain combinations of items-customers-sellers. These statistics indicate that preference weights and click rates tend to be low for most customer-item combinations, although these parameters are large in some cases. The wide range of these parameters suggests that computational methods that only focus on one regime (low click rate or high click rate) may be sub-optimal, e.g., while greedy seems efficacious when all items have small click rates, this may not be the case for instances that mix both high and low click rate items.

Descriptive analysis of click independence assumption. Next, we investigate the extent to which the Alibaba data set is consistent with the assumption that the random clicks on products are mutually independent, conditional on the available information about the customer and offered assortment. Recall that the historical sales data is $\text{PurchaseHistory} = \{(S_t, X_t, C_t, z_t) : t \in [\tau]\}$, where S_t is the set of products offered to customer t , and $C_t \subseteq S_t$ is the set of clicked products. For each customer $t \in [\tau]$, we denote by $\hat{\lambda}_{it}$ our estimate of the click probability for product $i \in S_t$.

To test the assumption of independent clicks, we sample two products uniformly at random in each assortment of the historical dataset. Then, we compare the empirical click rates of the second product in each pair, conditional on whether the first product was clicked on or not. Under the assumption of probabilistic independence, we expect these two quantities to be very close, i.e., whether or not the first product was clicked should have little to no effect on the likelihood that the second product was clicked.

To be more specific, we obtain Figure EC.3 using the following procedure:

1. For each $t \in [\tau]$, we sample two products, i_{1t} and i_{2t} , uniformly at random over S_t . Let $\hat{\lambda}_{\min} = \min_t \hat{\lambda}_{i_{2t}t}$ and $\hat{\lambda}_{\max} = \max_t \hat{\lambda}_{i_{2t}t}$, and let $L = \lceil \frac{\hat{\lambda}_{\max}}{\hat{\lambda}_{\min}} \rceil$.
2. For some $\delta > 0$ that will be specified below, and for each $\ell \in [L]$, let $\mathcal{I}_\ell = [\hat{\lambda}_{\min}\delta\ell, \hat{\lambda}_{\min}\delta \cdot (\ell + 1)]$. Next, we define $T_{0\ell} = \{t \in [\tau] : \hat{\lambda}_{i_{2t}t} \in \mathcal{I}_\ell, C_{i_{1t}t} = 0\}$ and $T_{1\ell} = \{t \in [\tau] : \hat{\lambda}_{i_{2t}t} \in \mathcal{I}_\ell, C_{i_{1t}t} = 1\}$. Namely, $T_{1\ell}$ is the set of observations for which the click rate of product i_{2t} resides in the interval

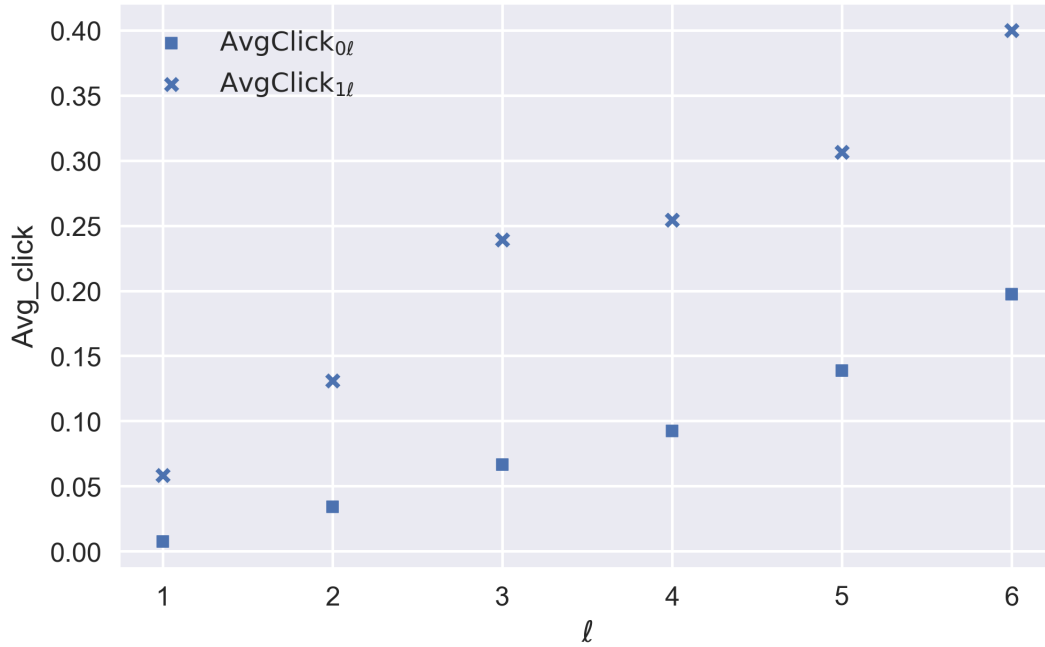


Figure EC.3 AvgClick $_{0\ell}$ and AvgClick $_{1\ell}$ plotted as a function of ℓ

$\mathcal{I}_\ell$ and product i_{1t} is clicked on. Conversely, $T_{0\ell}$ is the set of observations for which the click rate of product i_{2t} resides in the interval $\mathcal{I}_\ell$ and product i_{1t} is *not* clicked on.

- For each pair of buckets $T_{0\ell}$ and $T_{1\ell}$, we compute the corresponding empirical click rates $\text{AvgClick}_{0\ell} = \sum_{t \in T_{0\ell}} C_{i_{2t}} / |T_{0\ell}|$ and $\text{AvgClick}_{1\ell} = \sum_{t \in T_{1\ell}} C_{i_{2t}} / |T_{1\ell}|$. If clicks were indeed mutually independent, it should be the case that $\text{AvgClick}_{0\ell} \approx \text{AvgClick}_{1\ell}$ for each $\ell \in [L]$, since (informally) this proximity implies that the observed empirical click probability of i_{2t} is unaffected by whether or not product i_{1t} is clicked.

Figure EC.3 shows AvgClick $_{0\ell}$ and AvgClick $_{1\ell}$ plotted as a function of ℓ , where via trial and error, we selected $\delta = 0.03$ for better readability. We only plot the results for those buckets that have a sufficiently large sample size, $\min\{|T_{0\ell}|, |T_{1\ell}|\} \geq 10$; otherwise, the empirical click rates are too noisy to obtain meaningful insights. We find that $\text{AvgClick}_{1\ell} > \text{AvgClick}_{0\ell}$, meaning that clicking on one random product i_{1t} increases the probability of clicking on another random product i_{2t} . This positive correlation suggests that clicks begets more clicks. This might be due to latent (unobserved) factors that make customers more likely to search for a product, such as timeliness of the purchase, level of customer engagement, and intent of search query.

Correlational analysis of click independence assumption. We conduct linear regressions to test the correlational insights from Figure EC.3. In the basic specification, we regress the binary variable $is_click_{i_{1t}} = C_{i_{1t}}$, indicating whether or not product i_{1t} is clicked, on the click probability $click_proba(i_{1t})$ predicted by our Catboost model, the preference weight $pref_weight(i_{1t})$ predicted

by our MNL model and the binary variable $is_click(i_{2t}) = C_{i_{2t}}$, indicating whether or not the second randomly sampled product in the assortment S_t is clicked. We wish to test if there is a positive correlation between $click_proba(i_{1t})$ and $click_proba(i_{2t})$ after controlling for the customer-item propensities $click_proba(i_{1t})$ and $pref_weight(i_{1t})$ estimated by our models — notably, because positive correlations observed in the aggregate analysis of Figure EC.3 might be explained by observed features describing the customer-assortment interaction. Based on Model 1 in Table EC.3, we find that the coefficient of $is_click(i_{2t})$ is positive (0.0862) and significantly different from zero. We expand our specification in Models 2 and 3 to understand whether additional clicks on other products different from i_{2t} may also increase the click rate of the focal product i_{2t} . In Model 3, we sample another product $i_{3t} \neq i_{1t}, i_{2t}$ and include $is_click(i_{3t})$ as well as its interaction $is_click(i_{3t}) \times is_click(i_{2t})$ in our specification. In Model 3, we drop the variable $is_click(i_{2t})$ and instead introduce the binary variable $\mathbb{I}[\#clicks_t \geq 1]$, indicating whether any other product $i \in S_t \setminus \{i_{1t}\}$ is clicked or not, and the binary variable $\mathbb{I}[\#clicks_t \geq 2]$, indicating whether any two other products are clicked or not. The results indicate that the relationships between clicks are not merely additive. The positive interaction in Model 2 suggests a synergistic effect, where clicks beget more clicks beyond the separate pairwise correlations.

Table EC.3 Regression of click binary outcomes on other clicks outcomes in the assortment, controlling for the estimated click rate and preference weight of the focal product

	Model 1: $is_click\ i_{1t}$		Model 2: $is_click\ i_{1t}$		Model 3: $is_click\ i_{1t}$	
	Coef. (SE)	<i>p</i> -value	Coef. (SE)	<i>p</i> -value	Coef. (SE)	<i>p</i> -value
<i>Intercept</i>	-0.0028 (0.000)	***	-0.0042 (0.000)	***	-0.0070 (0.000)	***
$click_proba(i_{1t})$	1.0893 (0.006)	***	1.0828 (0.006)	***	1.0700 (0.006)	***
$pref_weight(i_{1t})$	-0.0038 (0.000)	***	-0.0031 (0.000)	***	-0.0017 (0.000)	***
$is_click(i_{2t})$	0.0862 (0.001)	***	0.0711 (0.001)	***		
$is_click(i_{3t})$			0.0722 (0.001)	***		
$is_click(i_{2t}) \times is_click(i_{3t})$			0.0832 (0.004)	***		
$\mathbb{I}[\#clicks_t \geq 1]$					0.0476 (0.002)	***
$\mathbb{I}[\#clicks_t \geq 2]$					0.1267 (0.006)	***
R^2	0.065		0.072		0.085	
Adj. R^2	0.065		0.072		0.085	
Num. obs.	665114		665114		665114	

Note: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. Standard errors in parentheses.

EC.7.3. Fitting the latent class MNL model

The latent class MNL models are also fit to the historical Alibaba sales data via an MLE-based approach. To ensure a fair comparison between the latent class MNL and click-based MNL models, we utilize the same training and testing sets to fit and assess both models. In what follows, we first formalize the latent class MNL model and its corresponding MLE problem. We then describe how we employ a sequential fitting procedure to determine the number of customer segments for each fitted latent class MNL model.

The latent class MNL model. Under the latent class MNL model, the customer population is partitioned into G customer segments, who each make purchasing decisions according to a distinct MNL model. As such, in the Alibaba setting, the probability that customer t purchases product $i \in S_t$ is

$$\pi(i, S_t, X_t) = \sum_{g \in [G]} \theta_g \cdot \frac{e^{\beta_g^T X_{it}}}{1 + \sum_{j \in S_t} e^{\beta_g^T X_{jt}}},$$

whereas the no-purchase option is selected with probability

$$\pi(0, S_t, X_t) = \sum_{g \in [G]} \theta_g \cdot \frac{1}{1 + \sum_{j \in S_t} e^{\beta_g^T X_{jt}}}.$$

In the above choice probability expressions, θ_g and β_g respectively designate the arrival probability and vector of feature weights associated with customer segment g . We use $\theta = \{\theta_g : g \in [G]\}$ to denote the set of all arrival probabilities. Finally, it is important to note that the segment to which each arriving customer belongs is latent, since the sales data does not reveal this information.

Given the specification described above, the log-likelihood function under the latent class MNL model can be written as follows:

$$\begin{aligned} \mathcal{LL}(\beta_1, \dots, \beta_G, \theta | \text{PurchaseHistory}) &= \log \left(\prod_{t=1}^{\tau} \pi(z_t, S_t, X_t) \right) \\ &= \sum_{t=1}^{\tau} \log \left(\sum_{g \in [G]} \theta_g \cdot \frac{e^{\beta_g^T X_{z_t t}}}{1 + \sum_{j \in S_t} e^{\beta_g^T X_{jt}}} \right). \end{aligned}$$

Therefore, the MLE problem of interest, used to derive estimates of the feature weights $\beta_1, \dots, \beta_G$ and the arrival probabilities θ , is given by:

$$\max_{\theta \in [0,1]^n, \beta_1, \dots, \beta_G} \mathcal{LL}(\beta_1, \dots, \beta_G, \theta | \text{PurchaseHistory}), \quad (\text{MLE LC-MNL})$$

with the additional constraint that $\sum_{g \in [G]} \theta_g = 1$.

Solving the MLE problem and choosing G . We solve problem (MLE LC-MNL) by directly maximizing the log-likelihood using MATLAB's constrained non-linear solver *fmincon* (MATLAB Optimization Toolbox). Although this log-likelihood function is non-convex, we found that *fmincon* generally converged to a local optima.

The number of customer segments is determined by solving problem (MLE LC-MNL) in sequential stages, where in each stage k , we set $G = k$. Furthermore, in stage k , we seed the MLE problem with the optimal estimates from stage $k - 1$ so as to ensure that the training likelihood strictly increases as we add more customer segments. To determine when to stop this process, we employ a train/validate approach that is a common machine learning practice. More specifically, we split the training set into a smaller training set and a validation data set that will be used to assess

the benefits of additional customer segments. The smaller training set that seeds the MLE problem now consists of 60% of the data, while the validation set consists of the remaining 15% that made-up the initial training set. The testing set remains unchanged. In each stage, after fitting the latent class MNL model as described above, we compute the fitted model’s likelihood on the validation set. If this validation likelihood is an improvement over the previous stage’s validation likelihood, then we increment G by one and move to the next stage. Otherwise, we stop the fitting procedure and output the fitted model with the highest validation likelihood. We set a time limit of three hours for fitting each latent class MNL model, allowing any stage that has commenced to be carried out to completion before halting the procedure.

It is worth mentioning that, since the click probabilities estimated in Section EC.7.1 depend on customer attributes, the fitted click-based MNL model corresponds to a latent class MNL model with over 2^6 segments (since each assortment has 6 products) whose mixture distribution varies as a function of customer attributes. By contrast, as standard in earlier literature (Berbeglia et al. 2022), the latent class MNL model assumes a fixed mixture distribution, which does not depend on customer attributes. Therefore, in this parametric setting, the family of latent class MNL models is not a generalization of the click-based MNL model.

EC.7.4. Additional estimation experiments

Table EC.4 5x2cv F-test results comparing the catboost click-based MNL model to the MNL model

	Seller 1	Seller 2	Seller 3	Seller 4	Seller 5	Seller 6	Seller 7	Seller 8	Seller 9	Seller 10
LL difference	676.56	245.51	146.81	90.34	270.19	229.63	105.39	8.31	32.14	124.03
f statistic	366.22	178.22	38.44	16.75	16.55	69.26	42.39	1.27	2.14	36.01
p-value	$1.60 \cdot 10^{-6}$	$9.63 \cdot 10^{-6}$	$4.22 \cdot 10^{-4}$	0.0030	0.0031	$9.99 \cdot 10^{-5}$	$3.32 \cdot 10^{-4}$	0.41	0.20	$4.94 \cdot 10^{-4}$

Table EC.5 5x2cv F-test results comparing the catboost click-based MNL model to the LC-MNL model
($G = 3$)

	Seller 1	Seller 2	Seller 3	Seller 4	Seller 5	Seller 6	Seller 7	Seller 8	Seller 9	Seller 10
LL difference	-36.26	75.68	-27.25	86.14	142.29	-107.37	-25.11	-80.50	29.80	26.80
f statistic	1.64	3.59	0.97	8.18	0.80	2.30	2.88	22.83	1.17	1.20
p-value	0.30	0.085	0.54	0.015	0.47	0.18	0.12	0.0014	0.45	0.44

Table EC.6 5x2cv F-test results comparing the logistic-regression click-based MNL model to the MNL model

	Seller 1	Seller 2	Seller 3	Seller 4	Seller 5	Seller 6	Seller 7	Seller 8	Seller 9	Seller 10
LL difference	72.68	-95.38	-74.43	4.09	-52.32	7.18	-70.26	-95.15	-66.78	-44.74
f statistic	1.65	1.54	13.67	0.79	10.21	1.03	4.83	29.47	3.75	1.84
p-value	0.30	0.32	0.0049	0.64	0.0096	0.51	0.048	$8.03 \cdot 10^{-4}$	0.078	0.25

Seller-level statistical comparison. We conduct a more detailed comparison of the predictive performance of the MNL, click-based MNL, and LC-MNL models on each seller’s dataset. In particular, we report the combined 5x2cv F-test (Alpaydin 1999, Raschka 2018) for comparing the predictive errors of two ML classifiers. We repeat 2-fold cross-validations on 5 random data splits of each seller’s data and calculate the test statistic as

$$f = \frac{\sum_{i=1}^5 \sum_{j=1}^2 (\Delta \ell_{i,j})^2}{2 \sum_{i=1}^5 s_i^2},$$

where $\Delta \ell_{i,j}$ is the difference in loglikelihood between the two methods measured on the test set formed by the j -th fold of the i -th random split and $s_i^2 = \sum_{j=1}^2 (\ell_{i,j} - \bar{\ell}_i)^2$ for $\bar{\ell}_i = \frac{1}{2}(\ell_{i,1} + \ell_{i,2})$ is the variance of the difference over the two folds for the i -th split. The statistic f is approximately F -distributed with 10 and 5 degrees of freedom.

In Table EC.4, we find that our click-based MNL model consistently outperforms MNL with significant differences (at 5% level) in out-of-sample log-likelihoods for 8 out of 10 sellers. Now, comparing our click-based MNL model to a 3-segment LC-MNL model, we find that the average differences between the models’ log-likelihoods are not significantly different than zero except for two sellers (note that this is without adjustment to multiple hypothesis testing). Compared to Table 4, here we only use 50% of the data in training the respective models, which may explain why the differences between the models are not as salient.

Additionally, for the click-based MNL model, we test both our Catboost algorithm and a simpler logistic regression model to predict the clicks on products. When we use a simple logistic regression instead of Catboost, the differences in out-of-sample log-likelihoods between the click-based MNL model and the MNL model are reduced, and MNL is even significantly more predictive than click-based MNL for 4 sellers out of 10, as per Table EC.6. This suggests that the predictive edge of our click-based MNL model over the MNL model requires leveraging machine learning methods in predicting customer clicks, to reduce bias in the first-stage estimation.

EC.7.5. The cost of model misspecification

In this section, we evaluate differences between the MNL and click-based MNL choice models from the standpoint of their ability to recommend profitable assortments. Since the click-based MNL model generalizes standard MNL, and since its fit to historical data is significantly more accurate (see Table 4), we wish to quantify the potential loss of revenue incurred by using the standard MNL model in place of the click-based MNL model to make assortment recommendations, when the latter forms the ground truth. Additionally, as a secondary goal, we seek to investigate the extent to which our fitting procedure for estimating the click-based MNL model has the ability to recapture a ground truth click-based MNL through the sales data that it generates. To accomplish these goals, we implement a four step procedure, whose finer level details are fleshed out in the paragraphs that follow.

Step 1 - The ground truth model. Throughout these experiments, we assume that there is a single underlying click-based MNL model, which governs the true purchasing behavior of each arriving customer. We take this ground truth model to be one of the click-based MNL models fit within the experiments of Section 5. Specifically, from the estimation experiments outlined in Section 5, we have ten distinct fitted click-based MNL models to choose from, each of which was derived from a training data set consisting of half of all available sales data. As our ground truth model, we simply select the fitted model from the first of the ten training data sets. The ground truth model is used only to generate semi-synthetic sales data in Step 2, and to ultimately assess the profitability of the assortments recommended by the two fitted models. The fitting procedures employed in Step 3 and the procedure to determine each recommended assortment carried out in Step 4 are blind to the ground truth model.

Step 2 - Generating sales data. In this step, we randomly generate varying levels of sales data by resampling the click and purchasing behavior of each historical customer assuming that their behavior is guided by the ground truth click-based MNL model from Step 1. To do so, we sample $T \in \{1000, 10000, 100000\}$ customers from the Alibaba training data set that was used to derive the ground truth click-based MNL model. For each such customer, recall that the data consists of 25-dimensional feature vectors characterizing each of the six offered products. We take each of these feature vectors and pass it to the ground truth model to derive the MNL-based weight and click probability associated with each displayed product. From here, to generate our fresh collection of sales data, we resample the clicks and purchasing decision of each customer based on the estimated weights and click probabilities derived from the ground truth model. In this way, we are able to exploit the richness of the Alibaba data set to generate new semi-synthetic sales data, which seeds the estimation procedures discussed next.

Step 3 - Generating assortment instances. Next, we fit standard MNL and click-based MNL models to the sales data generated in Step 2 using the MLE-based procedures outlined in Section EC.7.1. Naturally, the standard MNL model is fit without using the clickstream data, i.e., assuming that all offered products are clicked. With the fitted models in-hand, for each seller, we generate 20 instances, jointly describing the click-based MNL and standard MNL assortment optimization problems, using the following procedure. To start, for each seller, we filter the sampled collection of T customers down to only those that pertain to the seller at-hand, and then further downsample this set so as to select n distinct products. We vary $n \in \{10, 20\}$, to ensure that the complete enumeration of all feasible assortments can be employed to discover optimal assortments. Next, we utilize the resulting product features and the estimates for the model parameters β and the function $g(\cdot)$ obtained by solving the MLE problem described in Section EC.7.1 to specify the MNL weights w_i and the consideration probabilities λ_i for the click-based MNL model. Similarly, we

utilize these product features and the MLE estimates obtained by solving problem (MLE LC-MNL) with $G = 1$ to specify the preference weights of the standard MNL model. Finally, to mirror the Alibaba coupon setting, we add a cardinality constraint to the assortment problem, forcing any feasible assortment to contain exactly six products.

Step 4 - Assortment recommendations. We compare the expected revenues of the assortments recommended by the MNL and click-based MNL models. Assuming that the ground truth click-based MNL model governs reality, we compute the revenue loss incurred if Alibaba were to adopt the assortments recommended by the MNL fits instead. The gap between these two recommended assortments can be interpreted as the cost of a model misspecification with respect to these nested families of choice models. In other words, this gap is the loss of revenue from applying a standard MNL choice model in settings where customers follow a more general two-stage choice-making process. Clearly, this performance gap does not realistically measure the extent to which revenue would increase if a platform such as Alibaba utilizes the click-based MNL model vis-a-vis the standard MNL model, since this difference can only be measured by conducting a controlled experiment. However, these simulations do provide a measure of the degree of dissimilarity between the assortment recommendations generated by the two choice models. Indeed, these experiments provide, in some sense, a secondary “operational” measure of fitting accuracy, since they consider the efficacy of the downstream assortment decisions that arise from the two fitted models.

Results. We repeat the above procedure five times, so that for each combination of T and n , there are 100 assortment instances. Let S^{CMNL} be the assortment returned from the fitted click-based MNL model, and let S^{MNL} be the assortment recommended by the fitted MNL model. In Figure EC.4, we depict the average optimality gap of the assortment S^{CMNL} over S^{MNL} for each combination of T and n over the generated instances. More formally, we define the optimality gap of assortment $S \in \{S^{\text{CMNL}}, S^{\text{MNL}}\}$ as $\frac{\mathcal{R}(S^*) - \mathcal{R}(S)}{\mathcal{R}(S^*)}$, where S^* is the optimal assortment under the ground truth model. It is important to note that $\mathcal{R}(S)$ stands for the expected revenue of the assortment S under the ground truth click-based MNL model.

Figure EC.4 reveals that the optimality gaps of the assortments recommended by the click-based MNL model are generally between 10-20% smaller than those observed under the assortments recommended by the MNL fits. This observation suggests that the two choice models lead to significantly distinct assortment decisions. Interestingly, the optimality gaps for the click-based MNL model fits essentially vanish when T exceeds 10,000, showing that our fitting procedure recovers the true click-based MNL model fairly quickly, even in this highly featurized setting.



Figure EC.4 The average optimality gaps of the recommended assortment for each seller. Each tick on the x-axis is a pair (T, n) .